

Lecture 1: Introduction to Generalised Linear Models

MATH3701 Statistical Modelling

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Course notes: Chapter 1 (Introduction)

MATH3701 Statistical Modelling notes, Chapter 1

What we will cover:

- ➊ Revision of linear models with normal errors
- ➋ Introduction to generalised linear models (GLMs)
- ➌ Logistic regression models
- ➍ Loglinear models and contingency tables

Key idea: GLMs extend normal linear models to handle responses that are *not necessarily normal*.

Prerequisites: Comfort with R programming and basic statistical computation.

The Fundamental Question

Goal: Describe how a response variable Y depends on p explanatory variables $\mathbf{x} = (x_1, x_2, \dots, x_p)$.

Normal linear model assumption:

$$Y \mid \mathbf{x} \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2)$$

The problem: Many real-world responses are:

- Binary (success/failure, yes/no)
- Counts (number of events)
- Proportions (bounded between 0 and 1)
- Positive continuous (times, costs)

$\Rightarrow$ The normal distribution is often inappropriate.

Motivating Example: Beetle Mortality

Dose-response experiment: Beetles exposed to carbon disulphide gas.

Dose (x_i)	Beetles (m_i)	Killed (y_i)
1.6907	59	6
1.7242	60	13
1.7552	62	18
1.7842	56	28
1.8113	63	52
1.8369	59	53
1.8610	62	61
1.8839	60	60

Question: How does mortality rate depend on dose?

Why Linear Regression Fails

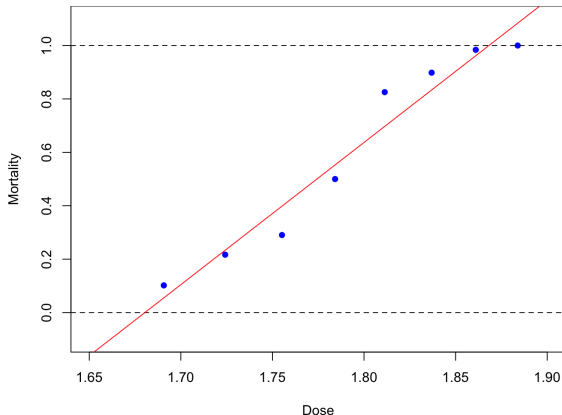
Fitted linear model:

$$\hat{p} = \hat{\alpha} + \hat{\beta}x$$

where $p = y/m$ is the mortality proportion.

Problems:

- Predicts $p > 1$ at high doses
- Predicts $p < 0$ at low doses
- Mortality is *bounded* $[0, 1]$



A Better Approach: The Logistic Model

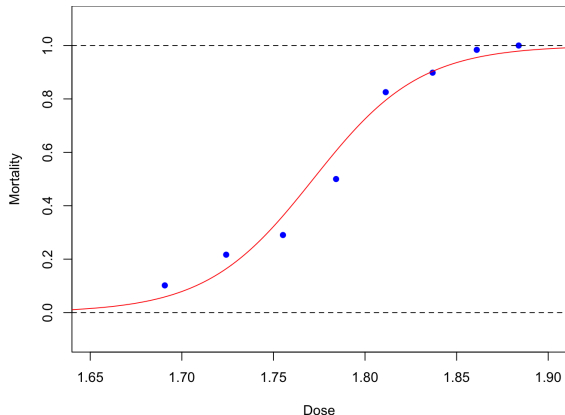
Logistic function:

$$p = \frac{e^{\alpha + \beta x}}{1 + e^{\alpha + \beta x}}$$

Properties:

- S-shaped (sigmoid) curve
- Always between 0 and 1
- Fitted via maximum likelihood

This is an example of a **Generalised Linear Model**.



Revision: The Linear Model

For n paired observations (x_i, y_i) :

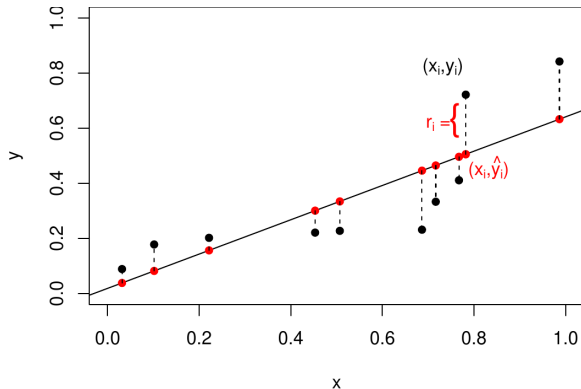
$$y_i = \alpha + \beta x_i + \epsilon_i$$

Assumptions:

- Errors ϵ_i are independent
- $\mathbb{E}[\epsilon_i] = 0$
- $\text{Var}[\epsilon_i] = \sigma^2$ (constant variance)

Goal: Estimate α , β , and σ^2 from data.

Visualizing the Fitting Problem



The data points are fixed; we choose α, β to move and rotate the line. Each point has a vertical residual $r_i = y_i - (\alpha + \beta x_i)$; RSS sums the squared residuals.

Least Squares Estimation

Residual Sum of Squares:

$$\text{RSS}(\alpha, \beta) = \sum_{i=1}^n (y_i - (\alpha + \beta x_i))^2$$

Minimize RSS to obtain the **least squares estimators**:

$$\hat{\beta} = \frac{s_{xy}}{s_x^2}, \quad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

where:

- $\bar{x}, \bar{y}$ are sample means
- $s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ is sample covariance
- $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is sample variance of x

Properties of Least Squares Estimators

Unbiasedness:

$$\mathbb{E}[\hat{\alpha}] = \alpha, \quad \mathbb{E}[\hat{\beta}] = \beta$$

Fitted values and residuals:

- Fitted values: $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$
- Residuals: $r_i = y_i - \hat{y}_i$

Error variance estimation:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n r_i^2$$

This is an unbiased estimator: $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.

Types of Variables

Quantitative variables:

- **Continuous:** Real-valued measurements (height, weight, time)
- **Count (discrete):** Non-negative integers (number of events)

Qualitative (categorical) variables:

- **Ordinal:** Ordered categories (mild/moderate/severe)
- **Nominal:** Unordered categories
 - **Binary/Dichotomous:** Two categories (yes/no, male/female)
 - **Polytomous:** Multiple categories (blood type, eye color)

⇒ Variable type determines the appropriate modeling approach.

Looking Ahead: The GLM Framework

Three components of a GLM:

- ① **Random component:** Distribution of Y (exponential family)

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

- ② **Systematic component:** Linear predictor

$$\eta = \sum_{j=1}^p \beta_j x_j = \mathbf{x}'\boldsymbol{\beta}$$

- ③ **Link function:** Connects mean to linear predictor

$$\eta = g(\mu), \quad \mu = g^{-1}(\eta)$$

Normal linear regression is a special case with $g(\mu) = \mu$ (identity link).

Key points from today:

- Linear regression assumes normal errors, which is often inappropriate
- GLMs extend linear models to non-normal responses
- The beetle example shows why we need bounded response models
- Least squares estimation minimizes RSS
- Variable classification guides model choice

Next lecture: Normal linear models in detail — matrix formulation and model fitting in R.

Lecture 2: Normal Linear Models — Structure, Types and Matrix Form

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Course notes: MATH3701 Statistical Modelling notes, Chapter 2

Note: This is revision material. The matrix formulation will be needed for GLM theory in later lectures.

The Normal Linear Model

For n observations, the **general linear model** is:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

- $\mathbf{Y} = (Y_1, \dots, Y_n)^T$: response vector
- $\mathbf{X}$: $n \times p$ **design matrix** (known)
- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$: parameter vector (unknown)
- $\boldsymbol{\epsilon}$: independent errors with constant variance σ^2

Equivalently:

$$Y_i \mid \mathbf{x}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{x}_i^T \boldsymbol{\beta}, \sigma^2)$$

Types of Normal Linear Models

Parameters	Explanatory variables	Model name
$p = 2$	1 quantitative	Simple linear regression
$p > 2$	> 1 quantitative	Multiple linear regression
$p = 2$	1 binary	Two-sample t -test
$p = k$	1 qualitative (k levels)	One-way ANOVA
$p > k$	Qualitative only	Multi-way ANOVA
$p > 2$	Mixed	ANCOVA

Key insight: These are all *special cases* of $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.

The Design Matrix

Recipe for building $\mathbf{X}$:

- 1 Column of 1s (intercept)
- 2 One column per quantitative variable
- 3 For a qualitative variable with k levels: $k - 1$ dummy indicator columns (one level = reference)
- 4 Interaction columns = element-wise products

Example (simple regression with $n = 4$):

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ 1 & x_4 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$\mathbf{X}$ must have full column rank p for a unique solution to exist.

Least Squares Estimation

Minimise the residual sum of squares (RSS):

$$\text{RSS}(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

Setting $\partial \text{RSS} / \partial \boldsymbol{\beta} = 0$ gives the **normal equations**:

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}$$

Least squares (OLS) estimator:

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}}$$

Properties: $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ (unbiased), $\text{Var}[\hat{\boldsymbol{\beta}}] = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$.

Hat Matrix, Fitted Values, Residuals

Fitted values:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \underbrace{\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T}_{\mathbf{H}} \mathbf{Y}$$

H is the **hat matrix**: symmetric, idempotent ($\mathbf{H}^2 = \mathbf{H}$), $\text{tr}(\mathbf{H}) = p$.

Residuals: $\mathbf{r} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$

Error variance estimate:

$$\hat{\sigma}^2 = \frac{\mathbf{r}^T \mathbf{r}}{n - p} \quad (\text{unbiased: } \mathbb{E}[\hat{\sigma}^2] = \sigma^2)$$

The hat matrix will reappear in Chapter 13 as the *smoothing matrix*.

Fitting in R

```
# Fit the model
fit <- lm(y ~ x1 + x2, data = mydata)

# Key output
summary(fit)      # coefficients, SE, t-tests, R^2
anova(fit)        # analysis of variance table

# Fitted values and residuals
fitted(fit)
residuals(fit)

# Compare nested models
fit0 <- lm(y ~ 1, data = mydata)
anova(fit0, fit)   # F-test

# Access the design matrix
model.matrix(fit)
```

Nested Model Comparison: The F-Test

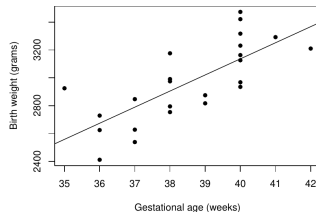
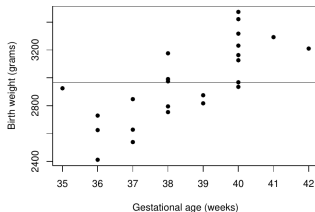
For nested models $M_0 \subset M_1$ with $r_0 > r_1$ residual degrees of freedom:

$$F_{01} = \frac{(\text{RSS}_0 - \text{RSS}_1)/(r_0 - r_1)}{\text{RSS}_1/r_1} \sim F_{r_0-r_1, r_1} \quad \text{under } H_0$$

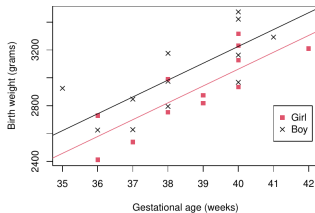
- Large $F \Rightarrow$ small p -value $\Rightarrow$ prefer M_1
- Small $F \Rightarrow$ large p -value $\Rightarrow$ use simpler M_0

Note: The F -test has a direct analogue in GLMs via the deviance ratio (Chapter 4).

Example: Nested Models for Birthweight Data

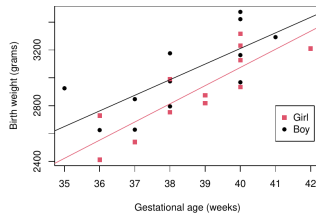


Model 0: intercept only



Model 2: + sex

Model 1: + age



Model 3: + age $\times$ sex

F -tests decide which additional terms are justified by the data.

Check the model assumptions by plotting residuals $r_i = y_i - \hat{y}_i$:

- ➊ **Residuals vs fitted:** Should show no pattern (tests linearity and constant variance)
- ➋ **Normal Q-Q plot:** Residuals should lie on a diagonal line (tests normality)
- ➌ **Scale-location:** Should be flat (tests homoscedasticity)
- ➍ **Residuals vs leverage:** Identifies influential observations

In R: `plot(fit)` produces these four plots automatically.

Key points:

- The normal linear model: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- The design matrix $\mathbf{X}$ encodes all model structure
- OLS estimator: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$
- The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ maps data to fitted values
- Nested models compared via F -test
- Residual plots check model assumptions

Next lecture: Generalised Linear Models — the exponential family and the three-component GLM structure.

Lecture 3: Normal Linear Models — Notation and R Fitting

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The $\sim$ operator: “is modeled by”

Symbol	Meaning
$y \sim x$	y is modeled by x
1	Intercept (included by default)
E	Main effect of variable E
$E + F$	Main effects of E and F
$E : F$	Interaction between E and F
$E * F$	Main effects + interaction ($= E + F + E : F$)
-E	Remove term E from model
-1 or 0	Remove intercept

More Formula Notation

Symbol	Meaning
E / F	Nested factors: $E + E : F$
$\text{poly}(E, \ell)$	Polynomial of degree ℓ in E
$I()$	“As is” — use arithmetic literally
$.$	All other variables in data frame

Examples:

- $y \sim x + I(x^2)$: quadratic regression
- $y \sim \text{poly}(x, 3)$: cubic polynomial (orthogonal)
- $y \sim . - z$: all variables except z

Formula Examples

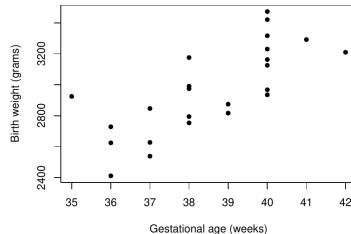
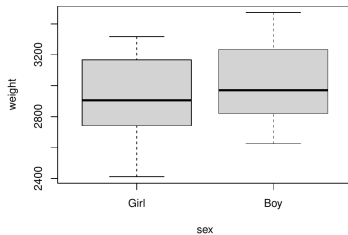
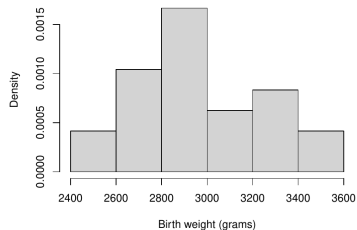
Birth weight example:

- `weight ~ age`: simple regression on age
- `weight ~ age + sex`: parallel lines model
- `weight ~ age * sex`: separate lines (with interaction)
- `weight ~ age + sex + age:sex`: same as above

More examples:

- `yield ~ variety + poly(rainfall, 2)`: ANCOVA with quadratic term
- `score ~ school/class`: students nested in classes within schools
- `profit ~ investment - 1`: regression through the origin

The Birthweight Data



Both age and sex look like useful predictors — exactly what `model2 <- lm(weight ~ age + sex)` will fit.

The `lm()` Function

Basic syntax:

```
my.lm <- lm(formula, data = mydata)
```

Example:

```
# Fit the parallel lines model
model2 <- lm(weight ~ age + sex, data = birthwt)

# View results
summary(model2)
```

Note: R automatically handles:

- Design matrix construction
- Dummy coding for factors
- Parameter estimation

Extracting Information from lm Objects

```
# Coefficient estimates
coefficients(my.lm)    # or coef(my.lm)

# Fitted values
fitted.values(my.lm)   # or fitted(my.lm)

# Residuals
residuals(my.lm)       # or resid(my.lm)

# Residual degrees of freedom
df.residual(my.lm)

# Design matrix
model.matrix(my.lm)
```

The summary() Output

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1773.324	820.906	-2.160	0.0423 *
age	120.894	21.423	5.643	1.26e-05 ***
sex	163.039	72.814	2.239	0.0360 *

Residual standard error: 172.2 on 21 degrees of freedom

Multiple R-squared: 0.6387, Adjusted R-squared: 0.6043

F-statistic: 18.56 on 2 and 21 DF, p-value: 2.237e-05

Key elements:

- **Estimates:** $\hat{\beta}$ values
- **Std. Error:** $\sqrt{\text{diag}(\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1})}$
- **t value:** Estimate / Std. Error
- **p-value:** Test of $H_0 : \beta_j = 0$

The `anova()` Function

Single model: Sequential analysis of variance

```
anova(my.lm)
```

Multiple models: Compare nested models

```
anova(model1, model2, model3)
```

Output includes:

- Sum of squares for each term
- Degrees of freedom
- Mean squares
- F -statistics and p -values

Making Predictions

Predict at new values of the explanatory variables:

```
# Create new data
new_data <- data.frame(age = c(38, 40, 42),
                       sex = c(0, 1, 1))

# Point predictions
predict(model2, newdata = new_data)

# With confidence intervals
predict(model2, newdata = new_data,
        interval = "confidence")

# With prediction intervals
predict(model2, newdata = new_data,
        interval = "prediction")
```

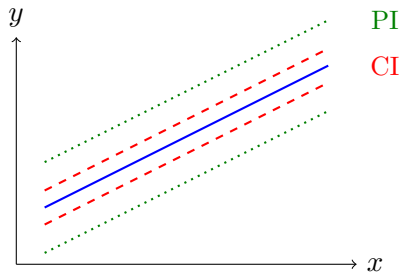
Confidence vs. Prediction Intervals

Confidence interval for $\mathbb{E}[Y \mid \mathbf{x}]$:

- Uncertainty about the *mean* response
- Narrower

Prediction interval for new Y :

- Uncertainty about an *individual* observation
- Wider (includes σ^2)



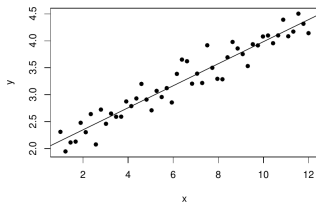
Model Diagnostics: Residual Plots

Key diagnostic plots:

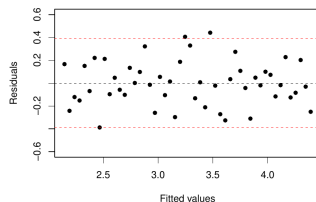
- ① **Residuals vs. fitted:** Check linearity, constant variance
- ② **Q-Q plot:** Check normality of residuals
- ③ **Scale-location:** Check homoscedasticity
- ④ **Residuals vs. leverage:** Identify influential points

```
# All four plots  
par(mfrow = c(2, 2))  
plot(my.lm)
```

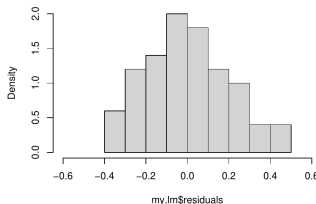
Example: Diagnostics on a Toy Dataset



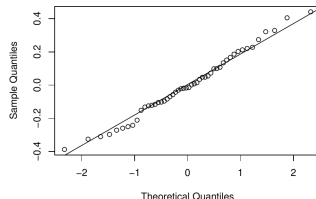
(a) Data and fitted line



(b) Residuals vs. fitted



(c) Histogram of residuals



(d) Normal Q-Q plot

A well-behaved fit: residuals scatter randomly, histogram and Q-Q plot are consistent with normality.

What to Look For

Residuals vs. Fitted:

- **Good:** Random scatter around 0
- **Bad:** Curved pattern $\Rightarrow$ nonlinearity
- **Bad:** Funnel shape $\Rightarrow$ heteroscedasticity

Q-Q Plot:

- **Good:** Points follow diagonal line
- **Bad:** S-shape $\Rightarrow$ heavy/light tails
- **Bad:** Curve $\Rightarrow$ skewness

If assumptions violated:

- Transform response (log, sqrt)
- Add polynomial terms
- Use a different model (e.g., GLM)

Declaring Factors in R

Problem: R may treat numeric codes as continuous.

```
# Sex coded as 0/1
sex <- c(0, 0, 1, 1, 0, 1)

# Wrong: treated as quantitative
lm(y ~ sex) # Only one coefficient for sex

# Right: declare as factor
sex <- as.factor(sex)
lm(y ~ sex) # Now categorical
```

Best practice: Always use `factor()` or meaningful labels:

```
sex <- factor(c("F", "F", "M", "M", "F", "M"))
```

Key points:

- R formula notation provides concise model specification
- `lm()` fits normal linear models
- `summary()` gives coefficient estimates and tests
- `anova()` compares nested models
- `predict()` generates predictions with intervals
- Diagnostic plots check model assumptions
- Declare categorical variables as factors

Next lecture: Introduction to GLM theory — the exponential family.

Lecture 4: GLM Theory — Exponential Family, Moments and Variance Functions

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Course notes: Chapter 3, Sections 3.1–3.5

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Why Generalise?

Normal linear models assume:

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2), \quad \mu_i = \mathbf{x}_i' \boldsymbol{\beta}$$

But many responses are:

- **Counts:** $Y_i \in \{0, 1, 2, \dots\}$ (Poisson)
- **Binary:** $Y_i \in \{0, 1\}$ (Bernoulli)
- **Proportions:** $Y_i/m_i \in [0, 1]$ (Binomial)
- **Positive continuous:** $Y_i > 0$ (Gamma, Exponential)

GLMs: A unified framework for all of these.

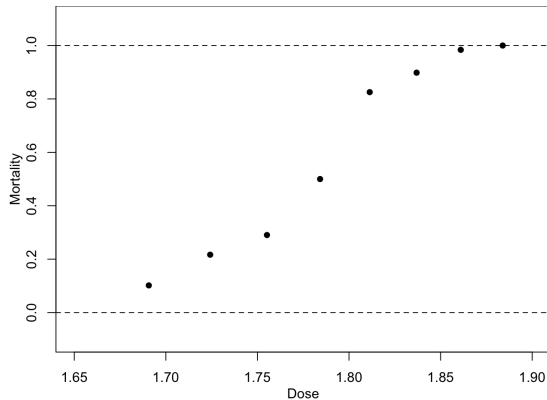
Motivating Example 1: Beetle Mortality

Data: Number killed (Y_i) out of m_i beetles at dose x_i

Model:

$$Y_i \sim \text{Binomial}(m_i, p_i)$$

Goal: Model how p_i depends on dose x_i .



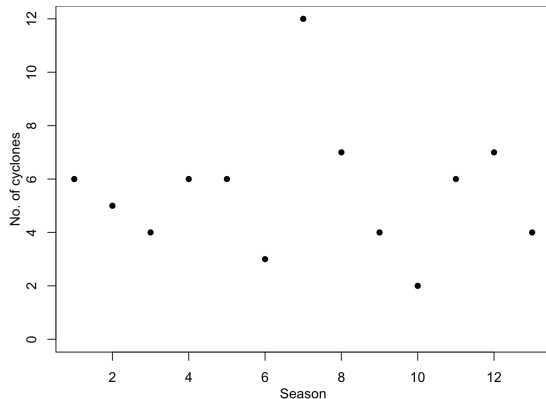
Motivating Example 2: Cyclone Counts

Data: Number of tropical cyclones (Y_i) in season i

Model:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

Goal: Model how λ_i depends on climate variables.



The Three Components of a GLM

A GLM has three parts:

- ① **Random Component:** Distribution of Y from the exponential family
- ② **Systematic Component:** Linear predictor $\eta = \mathbf{X}\beta$
- ③ **Link Function:** Connects mean to linear predictor via $\eta = g(\mu)$

Today: The **random component** (the exponential family and its moments) and the **systematic component**.

Next lecture: The link function.

The Exponential Family

Definition: A distribution belongs to the **exponential family** if its pdf/pmf can be written as:

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

Components:

- θ = **canonical (natural) parameter**
- $\phi > 0$ = **scale (dispersion) parameter**
- $b(\cdot)$ = function determining the distribution
- $c(\cdot, \cdot)$ = normalizing function

Key question: How do we write common distributions in this form?

Poisson Distribution

PMF:

$$f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y = 0, 1, 2, \dots$$

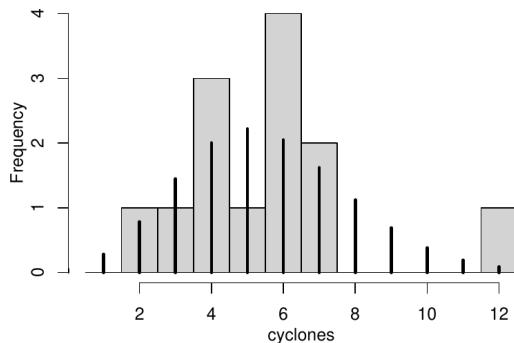
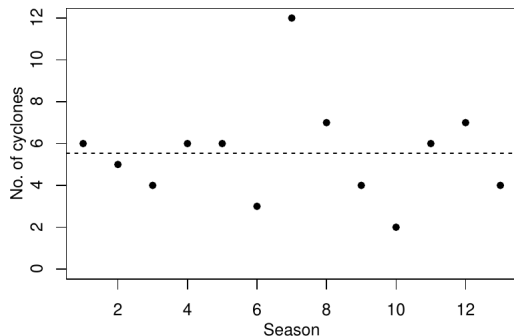
Rewrite using $\lambda^y = e^{y \log \lambda}$:

$$f(y) = \exp\{y \log \lambda - \lambda - \log y!\}$$

Matching to exponential family form:

- $\theta = \log \lambda$ (canonical parameter)
- $b(\theta) = e^\theta = \lambda$
- $\phi = 1$ (no dispersion parameter)
- $c(y, \phi) = -\log y!$

Poisson Fit to the Cyclone Data



Assuming a constant rate $Y_i \sim \text{Poisson}(\lambda)$ across seasons, $\hat{\lambda} = \bar{y}$. The fitted Poisson probabilities (right, bars) match the observed counts (histogram) reasonably well.

Binomial Distribution

PMF:

$$f(y; m, p) = \binom{m}{y} p^y (1-p)^{m-y}, \quad y = 0, 1, \dots, m$$

Rewrite:

$$\begin{aligned} f(y) &= \exp \left\{ y \log p + (m-y) \log(1-p) + \log \binom{m}{y} \right\} \\ &= \exp \left\{ y \log \frac{p}{1-p} + m \log(1-p) + \log \binom{m}{y} \right\} \end{aligned}$$

The key step: factor out to get y times a function of p .

Binomial: Exponential Family Components

From the rewritten form:

$$f(y) = \exp \left\{ y \log \frac{p}{1-p} + m \log(1-p) + \log \binom{m}{y} \right\}$$

Matching to exponential family:

- $\theta = \log \frac{p}{1-p} = \text{logit}(p)$ (canonical parameter)
- $b(\theta) = m \log(1 + e^\theta)$
- $\phi = 1$
- $c(y, \phi) = \log \binom{m}{y}$

Useful identity:

$$p = \frac{e^\theta}{1 + e^\theta}, \quad 1 - p = \frac{1}{1 + e^\theta}$$

Normal Distribution

PDF:

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}$$

Expand and rewrite:

$$\begin{aligned} f(y) &= \exp \left\{ -\frac{y^2 - 2y\mu + \mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\} \\ &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\} \end{aligned}$$

Group terms: $y\theta - b(\theta)$ in the numerator, rest in $c(y, \phi)$.

Normal: Exponential Family Components

Matching to exponential family form:

- $\theta = \mu$ (the canonical parameter equals the mean)
- $b(\theta) = \theta^2/2 = \mu^2/2$
- $\phi = \sigma^2$ (dispersion parameter)
- $c(y, \phi) = -\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi)$

Special property of Normal:

- The canonical parameter θ is just μ
- This is why normal linear models use $\mu = \mathbf{x}'\boldsymbol{\beta}$ directly

Summary: Exponential Family Forms

Distribution	θ	$b(\theta)$	ϕ	$c(y, \phi)$
Poisson(λ)	$\log \lambda$	e^θ	1	$-\log y!$
Binomial(m, p)	$\text{logit}(p)$	$m \log(1 + e^\theta)$	1	$\log \binom{m}{y}$
Normal(μ, σ^2)	μ	$\theta^2/2$	σ^2	$-\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi)$

Key observation: Different distributions share the same unified form.

Why the Exponential Family?

Advantages of the exponential family form:

- ① **Unified theory:** Same formulas work for all members
- ② **Moments from derivatives:** Mean and variance can be found from $b(\theta)$
- ③ **Maximum likelihood:** Score equations have nice form
- ④ **Sufficient statistics:** Data can be summarized efficiently

Members: Normal, Poisson, Binomial, Gamma, Exponential, Geometric, Inverse Gaussian, ...

Not members: Student's t , Uniform (with unknown bounds)

Summary: The Exponential Family

Key points:

- GLMs extend linear models to non-normal responses
- GLMs have three components: random, systematic, link
- The random component uses the exponential family
- Exponential family form: $f(y) = \exp\{(y\theta - b(\theta))/\phi + c(y, \phi)\}$
- Poisson: $\theta = \log \lambda$, $b(\theta) = e^\theta$
- Binomial: $\theta = \text{logit}(p)$, $b(\theta) = m \log(1 + e^\theta)$
- Normal: $\theta = \mu$, $b(\theta) = \theta^2/2$

Moments of Exponential Family Distributions

Proposition

For a distribution in exponential family form:

$$\mathbb{E}[Y] = b'(\theta), \quad \text{Var}[Y] = b''(\theta) \cdot \phi$$

Key insight: The mean and variance are determined entirely by derivatives of $b(\theta)$.

This is why the exponential family is so powerful — once we know $b(\theta)$, we know the moments.

Proof Sketch: Deriving the Mean

Starting point: All probability densities integrate to 1:

$$\int f(y; \theta, \phi) dy = 1$$

Differentiate both sides w.r.t. θ :

$$\frac{\partial}{\partial \theta} \int f(y) dy = 0$$

Move derivative inside:

$$\int \frac{\partial f}{\partial \theta} dy = \int f(y) \cdot \frac{y - b'(\theta)}{\phi} dy = 0$$

Rearranging:

$$\mathbb{E}[Y] = b'(\theta)$$

Proof Sketch: Deriving the Variance

Differentiate again w.r.t. θ :

Taking the second derivative of $\int f(y) dy = 1$ and simplifying gives:

$$\mathbb{E} \left[\left(\frac{Y - b'(\theta)}{\phi} \right)^2 \right] = \frac{b''(\theta)}{\phi}$$

Since $\mathbb{E}[Y] = b'(\theta)$:

$$\frac{\text{Var}[Y]}{\phi^2} = \frac{b''(\theta)}{\phi}$$

Therefore:

$$\text{Var}[Y] = b''(\theta) \cdot \phi$$

Verifying Moments: Poisson

Recall: $\theta = \log \lambda$, $b(\theta) = e^\theta$, $\phi = 1$

Mean:

$$\mathbb{E}[Y] = b'(\theta) = \frac{d}{d\theta} e^\theta = e^\theta = \lambda \quad \checkmark$$

Variance:

$$\text{Var}[Y] = b''(\theta) \cdot \phi = e^\theta \cdot 1 = \lambda \quad \checkmark$$

Note: For Poisson, $\mathbb{E}[Y] = \text{Var}[Y] = \lambda$ (equidispersion).

Verifying Moments: Binomial

Recall: $\theta = \text{logit}(p)$, $b(\theta) = m \log(1 + e^\theta)$, $\phi = 1$

Mean:

$$b'(\theta) = m \cdot \frac{e^\theta}{1 + e^\theta} = mp \quad \checkmark$$

Variance:

$$\begin{aligned} b''(\theta) &= m \cdot \frac{e^\theta \cdot (1 + e^\theta) - e^\theta \cdot e^\theta}{(1 + e^\theta)^2} = m \cdot \frac{e^\theta}{(1 + e^\theta)^2} \\ &= mp(1 - p) \quad \checkmark \end{aligned}$$

Note: Variance depends on mean — this is the *mean-variance relationship*.

Verifying Moments: Normal

Recall: $\theta = \mu$, $b(\theta) = \theta^2/2$, $\phi = \sigma^2$

Mean:

$$\mathbb{E}[Y] = b'(\theta) = \frac{d}{d\theta} \frac{\theta^2}{2} = \theta = \mu \quad \checkmark$$

Variance:

$$\text{Var}[Y] = b''(\theta) \cdot \phi = 1 \cdot \sigma^2 = \sigma^2 \quad \checkmark$$

Note: For Normal, variance is independent of mean.

The Variance Function

Define: The **variance function** $V(\mu)$ by:

$$\text{Var}[Y] = V(\mu) \cdot \phi$$

Distribution	Mean μ	Variance function $V(\mu)$
Poisson	λ	μ
Binomial	mp	$\mu(1 - \mu/m)$
Normal	μ	1
Gamma	α/λ	μ^2

The variance function characterizes how spread changes with the mean.

Summary: Exponential Family Moments

Distribution	θ	$b(\theta)$	ϕ	$\mathbb{E}[Y]$	$\text{Var}[Y]$
Poisson	$\log \lambda$	e^θ	1	λ	λ
Binomial	$\text{logit}(p)$	$m \log(1 + e^\theta)$	1	mp	$mp(1 - p)$
Normal	μ	$\theta^2/2$	σ^2	μ	σ^2

Key formula: $\mathbb{E}[Y] = b'(\theta)$, $\text{Var}[Y] = b''(\theta)\phi$

The Systematic Component

Now we connect the distribution to explanatory variables.

Linear predictor:

$$\eta_i = \sum_{j=1}^p \beta_j x_{ij} = \mathbf{x}_i' \boldsymbol{\beta}$$

In matrix form:

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}$$

where:

- $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)'$ is the $n \times 1$ vector of linear predictors
- $\mathbf{X}$ is the $n \times p$ design matrix
- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)'$ is the parameter vector

Normal Linear Model as a GLM

Random component: $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$

Systematic component: $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$

Link function: Identity link, $g(\mu) = \mu$

So:

$$\mu_i = \eta_i = \mathbf{x}_i' \boldsymbol{\beta}$$

This is exactly the normal linear model we studied earlier.

$\Rightarrow$ Normal regression is a *special case* of GLM.

Summary: Moments and Model Structure

Key points:

- Moments from $b(\theta)$: $\mathbb{E}[Y] = b'(\theta)$, $\text{Var}[Y] = b''(\theta)\phi$
- Variance function $V(\mu)$ describes mean-variance relationship
- Systematic component: linear predictor $\eta = \mathbf{X}\boldsymbol{\beta}$
- Normal regression is a GLM with identity link

Next lecture: The link function — canonical links, then logit, probit and complementary log-log.

Lecture 5: GLM Theory — Link Functions

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Course notes: Chapter 3, Section 3.6

MATH3701 Statistical Modelling notes, Chapter 3

The Link Function

Problem: The mean μ may be constrained:

- Poisson: $\mu > 0$
- Binomial: $0 < \mu/m < 1$

But: The linear predictor $\eta = \mathbf{x}'\boldsymbol{\beta}$ is unconstrained.

Solution: Use a **link function** g such that:

$$\eta = g(\mu) \in \mathbb{R}$$

The inverse $h = g^{-1}$ maps η back to the valid range of μ :

$$\mu = g^{-1}(\eta) = h(\eta)$$

The Canonical Link

Definition: The **canonical link** sets $\eta = \theta$ (the natural parameter).

Since $\mathbb{E}[Y] = \mu = b'(\theta)$, we have:

$$g(\mu) = (b')^{-1}(\mu) = \theta$$

Distribution	Canonical θ	Canonical Link $g(\mu)$	Name
Normal	μ	μ	Identity
Poisson	$\log \lambda$	$\log(\mu)$	Log
Binomial	$\text{logit}(p)$	$\log \frac{\mu/m}{1-\mu/m}$	Logit
Gamma	$-1/\mu$	$-1/\mu$	Reciprocal

Property: with the canonical link, $g'(\mu) = 1/b''(\theta) = 1/V(\mu)$.

Why Use the Canonical Link?

① Sufficient statistics:

$$S_j = \sum_{i=1}^n y_i x_{ij}$$

contain all information about β_j .

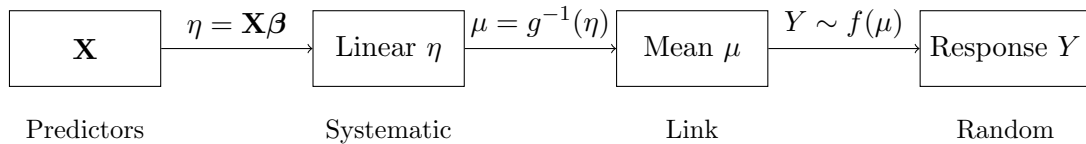
② Simple score equations: MLE satisfies

$$\sum_{i=1}^n (y_i - \hat{\mu}_i) x_{ij} = 0$$

③ Computational efficiency: Algorithms converge reliably.

However: Other links may be more interpretable or provide a better fit.

GLM Summary Diagram



Flow:

$$\text{Predictors} \xrightarrow{\text{linear model}} \eta \xrightarrow{\text{inverse link}} \mu \xrightarrow{\text{distribution}} Y$$

Link Functions for Binomial Data

For proportions/probabilities $p \in (0, 1)$:

Name	Link $g(p)$	Inverse $h(\eta)$
Logit	$\log \frac{p}{1-p}$	$\frac{e^\eta}{1+e^\eta}$
Probit	$\Phi^{-1}(p)$	$\Phi(\eta)$
Cloglog	$\log(-\log(1-p))$	$1 - e^{-e^\eta}$
Cauchit	$\tan(\pi(p - 0.5))$	$0.5 + \frac{1}{\pi} \arctan(\eta)$

where Φ is the standard normal CDF.

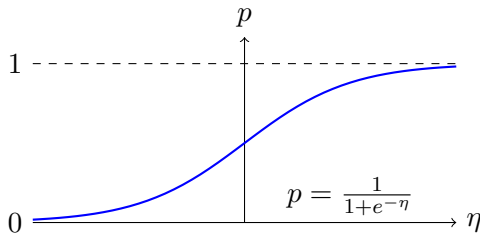
The Logit Link

Definition:

$$\text{logit}(p) = \log \frac{p}{1-p} = \log(\text{odds})$$

Inverse (logistic function):

$$p = \frac{e^{\eta}}{1 + e^{\eta}} = \frac{1}{1 + e^{-\eta}}$$



The Probit Link

Definition:

$$\text{probit}(p) = \Phi^{-1}(p)$$

where Φ is the standard normal CDF.

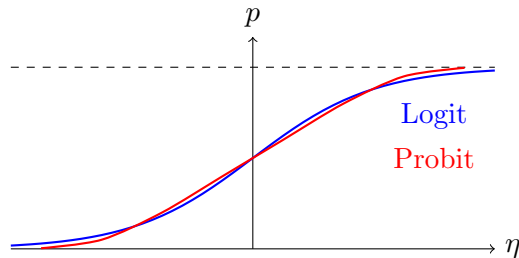
Inverse:

$$p = \Phi(\eta)$$

Interpretation:

- If there's a latent normal variable $Z = \eta + \epsilon$, $\epsilon \sim \mathcal{N}(0, 1)$
- Then $\mathbb{P}(Z > 0) = \Phi(\eta)$
- Probit link arises naturally from latent variable models

Comparing Logit and Probit



Key points:

- Very similar for moderate probabilities
- Logit has slightly heavier tails
- In practice, results are often nearly identical
- Logit preferred for interpretability (odds ratios)

The Complementary Log-Log Link

Definition:

$$\text{cloglog}(p) = \log(-\log(1 - p))$$

Inverse:

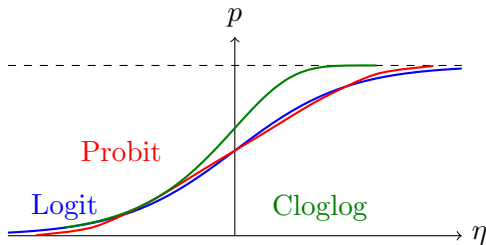
$$p = 1 - \exp(-e^\eta)$$

Key property: Asymmetric — behaves differently near 0 and 1.

Use when:

- Probability of an event increases steeply then levels off
- Complementary to the log-log link: $\log(-\log(p))$
- Arises from extreme value distributions

Comparing Link Functions



Note: Cloglog approaches 1 faster but 0 slower than logit/probit.

Fit logistic regression with different link functions:

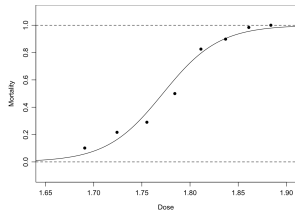
```
# Logit link (default)
glm(y ~ dose, family = binomial(link = "logit"))

# Probit link
glm(y ~ dose, family = binomial(link = "probit"))

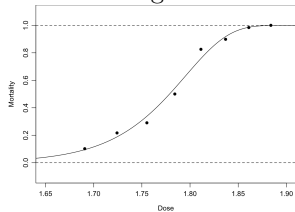
# Complementary log-log link
glm(y ~ dose, family = binomial(link = "cloglog"))
```

Compare using residual deviance and residual plots.

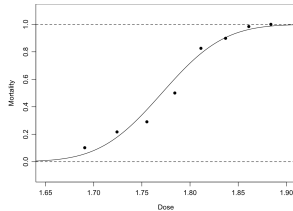
Beetle Data: Fitted Curves with Different Links



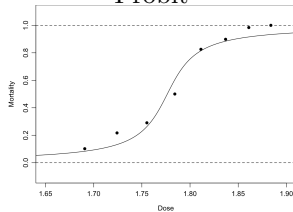
Logit



Cloglog



Probit



Cauchit

Link Functions for Poisson Data

For count data with $\mu > 0$:

Name	Link $g(\mu)$	Ensures $\mu > 0$?
Log (canonical)	$\log(\mu)$	Yes, $\mu = e^\eta > 0$
Square root	$\sqrt{\mu}$	Need $\eta > 0$
Identity	μ	Need $\eta > 0$

Log link is almost always used:

- Ensures positive fitted values
- Coefficients have multiplicative interpretation
- e^{β_j} = rate ratio for unit increase in x_j

Considerations:

- ① **Theoretical:** Does a particular link arise from the science?
 - Probit from latent normal models
 - Cloglog from survival/extreme value theory
- ② **Interpretability:**
 - Logit gives odds ratios
 - Log gives rate ratios
- ③ **Empirical fit:**
 - Compare deviances
 - Check residual patterns

R: Specifying Link Functions

```
# Binomial family with different links
glm(y ~ x, family = binomial(link = "logit"))
glm(y ~ x, family = binomial(link = "probit"))
glm(y ~ x, family = binomial(link = "cloglog"))

# Poisson family with different links
glm(y ~ x, family = poisson(link = "log"))
glm(y ~ x, family = poisson(link = "sqrt"))

# Gaussian family
glm(y ~ x, family = gaussian(link = "identity"))
glm(y ~ x, family = gaussian(link = "log"))
```

Key points:

- Link functions map constrained μ to unconstrained η
- Canonical links have computational advantages
- For binomial: logit, probit, cloglog are common choices
- Logit and probit give similar results; logit is more interpretable
- Cloglog is asymmetric — useful for specific applications
- For Poisson: log link is standard
- Choose based on theory, interpretability, and fit

Next lecture: Maximum likelihood estimation for GLMs.

Lecture 6: GLM Estimation — Maximum Likelihood and IRLS

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Course notes: Chapter 4, Sections 4.1–4.2 and the IRLS material

Supplementary: [Supplementary/proof_IRLS.pdf](#)

The Estimation Problem

Given:

- Observations $y_1, \dots, y_n$
- Explanatory variables $\mathbf{x}_1, \dots, \mathbf{x}_n$
- A GLM specification (distribution, link)

Goal: Estimate the parameter vector $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)'$

Method: Maximum likelihood estimation (MLE)

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} L(\boldsymbol{\beta}; \mathbf{y})$$

The Log-Likelihood Function

For independent observations from exponential family:

$$L(\boldsymbol{\beta}; \mathbf{y}, \phi) = \prod_{i=1}^n f(y_i; \theta_i, \phi)$$

Log-likelihood:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

Key: θ_i depends on $\boldsymbol{\beta}$ through:

$$\theta_i \leftarrow \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i' \boldsymbol{\beta})$$

The Simple Case: i.i.d. Observations

If all $\theta_i = \theta$ (same for all observations):

$$\ell(\theta; \mathbf{y}, \phi) = \frac{n(\bar{y}\theta - b(\theta))}{\phi} + \text{const}$$

Score equation:

$$\frac{\partial \ell}{\partial \theta} = \frac{n(\bar{y} - b'(\theta))}{\phi} = 0$$

Solution:

$$\boxed{b'(\hat{\theta}) = \bar{y}}$$

Since $\mathbb{E}[Y] = b'(\theta)$, the MLE satisfies $\hat{\mu} = \bar{y}$.

Example: Poisson i.i.d. Case

For $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$:

- $b(\theta) = e^\theta$, so $b'(\theta) = e^\theta$
- Score equation: $e^{\hat{\theta}} = \bar{y}$
- Solution: $\hat{\theta} = \log \bar{y}$
- Therefore: $\hat{\lambda} = \bar{y}$

The MLE is the sample mean!

This makes intuitive sense: best estimate of the rate is the observed average.

The General Case: Non-identical θ_i

When θ_i varies with explanatory variables:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \frac{y_i \theta_i(\boldsymbol{\beta}) - b(\theta_i(\boldsymbol{\beta}))}{\phi} + \text{const}$$

Problem: There is no closed-form solution in general.

Exception: Normal linear model has

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

For other GLMs: Need iterative methods (Newton-Raphson, Fisher Scoring).

The Score Function

Definition: The score function is the gradient of the log-likelihood:

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \begin{pmatrix} \frac{\partial \ell}{\partial \beta_1} \\ \vdots \\ \frac{\partial \ell}{\partial \beta_p} \end{pmatrix}$$

Properties:

- $\mathbb{E}[\mathbf{U}(\boldsymbol{\beta})] = \mathbf{0}$ at true parameter value
- MLE satisfies $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$

Observed Fisher information:

$$\mathbf{I}(\boldsymbol{\beta}) = -\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \quad (p \times p \text{ matrix})$$

Expected Fisher information:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{I}(\boldsymbol{\beta})] = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right]$$

Key result:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{U}(\boldsymbol{\beta})\mathbf{U}(\boldsymbol{\beta})'] = \text{Var}[\mathbf{U}(\boldsymbol{\beta})]$$

Newton-Raphson Method

Iterative algorithm to find $\hat{\beta}$:

Starting from $\beta^{(0)}$, update:

$$\beta^{(t+1)} = \beta^{(t)} + \mathbf{I}^{-1}(\beta^{(t)})\mathbf{U}(\beta^{(t)})$$

Intuition:

- $\mathbf{U}(\beta)$: direction of steepest ascent
- $\mathbf{I}^{-1}(\beta)$: adjusts step size based on curvature

Convergence: Stop when $\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$.

Replace observed with expected information:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathbf{J}^{-1}(\boldsymbol{\beta}^{(t)})\mathbf{U}(\boldsymbol{\beta}^{(t)})$$

Advantages over Newton-Raphson:

- $\mathbf{J}$ is guaranteed positive definite
- More stable convergence
- Equivalent to iteratively reweighted least squares (IRLS)

For canonical links: Newton-Raphson and Fisher Scoring are identical.

Key points:

- MLEs maximize the log-likelihood $\ell(\boldsymbol{\beta}; \mathbf{y})$
- For i.i.d. case: $\hat{\mu} = \bar{y}$
- General case requires iterative methods
- Newton-Raphson uses observed information
- Fisher Scoring uses expected information (more stable)

Score for One Observation

For a single observation from the exponential family:

$$\log f(y; \theta, \phi) = \frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$$

Score (first derivative of log-likelihood):

$$\frac{\partial \log f}{\partial \theta} = \frac{y - \mu}{\phi}, \quad \mu = b'(\theta)$$

Information:

$$\mathcal{I}(\theta) = \mathbb{E} \left[\left(\frac{\partial \log f}{\partial \theta} \right)^2 \right] = \frac{b''(\theta)}{\phi} = \frac{V(\mu)}{\phi}$$

For a GLM, we need the score with respect to β , not θ .

Score Equations for a GLM

For independent observations $y_1, \dots, y_n$, the log-likelihood is:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi)$$

By the chain rule $\partial/\partial\beta_j = \sum_i (\partial\theta_i/\partial\mu_i)(\partial\mu_i/\partial\eta_i)(\partial\eta_i/\partial\beta_j)$:

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \frac{(y_i - \mu_i)}{\phi V(\mu_i)} \cdot \dot{\mu}_i \cdot x_{ij},$$

where $\dot{\mu}_i = \partial\mu_i/\partial\eta_i = 1/g'(\mu_i)$.

In matrix form: $\mathbf{X}^T \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$.

Working Weights and Working Response

Define the $n \times n$ diagonal **weight matrix**:

$$\mathbf{W} = \text{diag}(w_1, \dots, w_n), \quad w_i = \frac{\dot{\mu}_i^2}{\phi V(\mu_i)}$$

and the **working (adjusted dependent) variable**:

$$z_i = \eta_i + (y_i - \mu_i) g'(\mu_i) = \eta_i + \frac{y_i - \mu_i}{\dot{\mu}_i}$$

Then the score equations $\partial \ell / \partial \boldsymbol{\beta} = \mathbf{0}$ become:

$$\mathbf{X}^T \mathbf{W} (\mathbf{z} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{0}$$

This is a **weighted least squares** normal equation — but $\mathbf{W}$ and $\mathbf{z}$ depend on $\boldsymbol{\beta}$.

The IRLS Algorithm

Iteratively Reweighted Least Squares:

- 1 **Initialise:** choose starting values $\hat{\mu}_i^{(0)}$
- 2 **Compute** working quantities at current estimate:

$$\eta_i^{(t)} = g(\hat{\mu}_i^{(t)}), \quad w_i^{(t)} = \frac{(\dot{\mu}_i^{(t)})^2}{\phi V(\hat{\mu}_i^{(t)})}, \quad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \hat{\mu}_i^{(t)}}{\dot{\mu}_i^{(t)}}$$

- 3 **Update** via weighted least squares:

$$\hat{\beta}^{(t+1)} = \left(\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)}$$

- 4 **Update** $\hat{\mu}_i^{(t+1)} = g^{-1}(\mathbf{x}_i^T \hat{\beta}^{(t+1)})$
- 5 **Repeat** until $\|\hat{\beta}^{(t+1)} - \hat{\beta}^{(t)}\| < \varepsilon$

R: `glm()` runs IRLS internally; `summary(fit)$iter` shows the count.

IRLS for the Normal Linear Model

For $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$ with identity link $g(\mu) = \mu$:

- $\dot{\mu}_i = 1$, $V(\mu_i) = 1$
- Weights: $w_i = 1/\phi = 1/\sigma^2$ (constant)
- Working response: $z_i = \eta_i + (y_i - \mu_i) = y_i$

Weighted LS reduces to ordinary LS:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$\Rightarrow$ **IRLS converges in one step** for the normal model.

For other families (Binomial, Poisson), IRLS typically converges in < 10 iterations.

Asymptotic Distribution of $\hat{\beta}$

Under regularity conditions, as $n \rightarrow \infty$:

$$\hat{\beta} \sim \mathcal{N}_p(\beta, \mathcal{J}(\beta)^{-1})$$

where the Fisher information matrix is:

$$\mathcal{J}(\beta) = \frac{1}{\phi} \mathbf{X}^T \mathbf{W} \mathbf{X}$$

evaluated at the true β .

In practice: estimate $\mathcal{J}(\beta)$ by substituting $\hat{\beta}$:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\phi} (\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}$$

Standard errors = square roots of diagonal elements.

The Saturated Model

The **saturated model** has one parameter per observation: $\tilde{\mu}_i = y_i$ for each i .

- Achieves the *maximum possible* log-likelihood $\tilde{\ell}$
- Acts as a reference point
- The fitted model M has log-likelihood $\hat{\ell}$

Deviance measures how far M falls short of the saturated model:

$$D = 2\phi(\tilde{\ell} - \hat{\ell})$$

This is the GLM analogue of the residual sum of squares. The deviance is covered in detail in the next lecture.

Summary: IRLS and Asymptotics

Key points:

- GLM score equations $\Rightarrow$ weighted normal equations
- Working weights w_i and working response z_i depend on $\hat{\beta}$
- IRLS iterates weighted LS until convergence
- Reduces to ordinary LS for the normal model (one step)
- MLEs are asymptotically $\mathcal{N}_p(\beta, \mathcal{I}^{-1})$
- Saturated model = maximum achievable likelihood

Next lecture: Deviance, Pearson/deviance residuals, and fitting GLMs in R.

Lecture 7: GLM Estimation — Deviance and Residuals

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Course notes: Chapter 4, Sections 4.3–4.5

MATH3701 Statistical Modelling notes, Chapter 4

Model Comparison: The Key Question

Given:

- A fitted model M with r parameters
- The saturated model $\tilde{M}$ with n parameters

Questions:

- ① How well does model M fit the data?
- ② How do we compare two nested models?

Answer: The **deviance**.

Definition of Deviance

Deviance:

$$D = 2\phi \left\{ \ell(\tilde{\boldsymbol{\theta}}; \mathbf{y}) - \ell(\hat{\boldsymbol{\theta}}; \mathbf{y}) \right\}$$

where:

- $\tilde{\boldsymbol{\theta}}$ = MLE under saturated model ($\tilde{\mu}_i = y_i$)
- $\hat{\boldsymbol{\theta}}$ = MLE under fitted model M
- ϕ = scale parameter

Interpretation:

- Measures “distance” from saturated model
- Larger D = worse fit
- $D = 0$ only for the saturated model

Deviance for Specific Distributions

Normal distribution:

$$D = \sum_{i=1}^n (y_i - \hat{\mu}_i)^2 = \text{RSS}$$

Poisson distribution:

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right\}$$

Binomial distribution:

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - \hat{\mu}_i} \right\}$$

Under the null hypothesis that model M is correct:

$$D \dot{\sim} \chi^2_{n-r}$$

where r = number of parameters in M .

Goodness-of-fit test:

- Large D relative to $n - r \Rightarrow$ poor fit
- p -value: $\mathbb{P}(\chi^2_{n-r} > D)$
- Small p -value $\Rightarrow$ reject model M

Caution: χ^2 approximation may be poor for small samples or sparse data.

Comparing Nested Models

For nested models $M_1 \subset M_2$:

Deviance difference:

$$D_1 - D_2 \sim \chi^2_{r_2 - r_1}$$

under H_0 : M_1 is adequate.

Test procedure:

- 1 Fit both models, obtain D_1 and D_2
- 2 Calculate $\Delta D = D_1 - D_2$
- 3 Compare to $\chi^2_{r_2 - r_1}$ distribution
- 4 Reject H_0 if p -value is small

When ϕ is Unknown

For Normal distribution: $\phi = \sigma^2$ is unknown.

Estimate from saturated model:

$$\hat{\phi} = \frac{D_{\text{sat}}}{n - r_{\text{sat}}}$$

Use F-test instead of χ^2 :

$$F = \frac{(D_1 - D_2)/(r_2 - r_1)}{\hat{\phi}} \sim F_{r_2 - r_1, n - r_{\text{sat}}}$$

Note: For Poisson and Binomial, $\phi = 1$ is known, so χ^2 test applies.

Types of Residuals

1. Raw (response) residuals:

$$e_i^{\text{raw}} = y_i - \hat{\mu}_i$$

2. Pearson (standardized) residuals:

$$e_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

3. Deviance residuals:

$$e_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

where d_i is the contribution of observation i to deviance.

Pearson Residuals

$$e_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

Properties:

- Standardized to have approximately unit variance
- Sum of squared Pearson residuals:

$$X^2 = \sum_{i=1}^n (e_i^P)^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$$

- $X^2 \approx D$ for large samples (both $\sim \chi_{n-r}^2$)

For Poisson: $e_i^P = (y_i - \hat{\mu}_i) / \sqrt{\hat{\mu}_i}$

For Binomial: $e_i^P = (y_i - m_i \hat{p}_i) / \sqrt{m_i \hat{p}_i (1 - \hat{p}_i)}$

$$e_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

where $D = \sum_{i=1}^n d_i$.

Properties:

- $\sum_{i=1}^n (e_i^D)^2 = D$ (deviance)
- Often more normally distributed than Pearson residuals
- Preferred for model diagnostics

For Poisson:

$$d_i = 2 \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right\}$$

Plot residuals to check:

① Residuals vs. fitted values:

- Should show random scatter
- Patterns suggest model misspecification

② Residuals vs. explanatory variables:

- Check for missed nonlinear effects

③ Q-Q plot of residuals:

- Check approximate normality
- (Less critical for GLMs than for normal models)

Fitting GLMs in R

```
# Fit a Poisson GLM
model <- glm(y ~ x1 + x2, family = poisson)

# Fit a binomial GLM (logistic regression)
model <- glm(cbind(success, failure) ~ x,
             family = binomial)

# Summary with coefficients and deviance
summary(model)

# Analysis of deviance
anova(model, test = "Chisq")

# Compare nested models
anova(model1, model2, test = "Chisq")
```

Extracting Residuals in R

```
# Raw (response) residuals
residuals(model, type = "response")

# Pearson residuals
residuals(model, type = "pearson")

# Deviance residuals (default)
residuals(model, type = "deviance")

# Deviance
deviance(model)

# Fitted values
fitted(model)
```

Worked Example: Fitting a Real Poisson GLM

Data: 12 counts classified by two factors **a** (4 levels) and **b** (3 levels).

```
y.glm <- glm(y ~ a + b, family = poisson)
summary(y.glm)
```

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.1408     0.3351   3.404 0.000663 ***
a2            -0.3054     0.3522  -0.867 0.385934
a3             0.1466     0.3132   0.468 0.639712
a4             0.4568     0.2932   1.558 0.119269
b2             0.9163     0.3162   2.898 0.003761 **
b3             0.9445     0.3150   2.999 0.002712 **

Null deviance: 31.725  on 11  degrees of freedom
Residual deviance: 13.150  on 6  degrees of freedom
AIC: 68.227
```

Key output:

- Residual deviance 13.15 on 6 df (ratio ≈ 2.2): against χ^2_6 , $p \approx 0.04$ — some evidence of lack of fit or overdispersion
- Null vs. fitted: $31.73 - 13.15 = 18.58$ on 5 df, $p \approx 0.002$ — the factors clearly improve the fit

Worked Example: Sequential Analysis of Deviance

```
anova(y.glm)
```

```
Terms added sequentially (first to last)
```

	Df	Deviance	Resid.	Df	Resid.	Dev	Pr(>Chi)
NULL				11		31.725	
a	3	6.2793		8		25.445	0.098784 .
b	2	12.2947		6		13.150	0.002139 **

This is the nested-model test from earlier: adding **b** after **a** reduces the deviance by 12.29 on 2 df (χ^2_2 , $p = 0.002$) — a significant improvement in fit.

Key points:

- Deviance measures distance from saturated model
- $D \sim \chi^2_{n-r}$ under correct model specification
- Compare nested models via deviance differences
- Pearson residuals: standardized by variance function
- Deviance residuals: sum of squares equals deviance
- Use residual plots for model diagnostics
- R's `glm()` function handles all major GLM families

Next lecture: Logistic regression for binary and binomial data.

Lecture 8: Modelling Proportions — Logistic Regression, Overdispersion and Odds Ratios

MATH3701 Statistical Modelling

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MATH3701 Statistical Modelling

Course notes: Chapter 5, Sections 5.1–5.4

MATH3701 Statistical Modelling notes, Chapter 5

Binary and Binomial Responses

Bernoulli trials:

$$B = \begin{cases} 1 & \text{if event occurs ("success")} \\ 0 & \text{otherwise ("failure")} \end{cases}$$

with $\mathbb{P}(B = 1) = p$.

Binomial distribution:

Sum of m independent Bernoulli trials with same p :

$$Y = \sum_{j=1}^m B_j \sim \text{Binomial}(m, p)$$

Special case: $m = 1$ gives Bernoulli (binary) data.

The Logistic Regression Model

Model specification:

$$Y_i \sim \text{Binomial}(m_i, p_i)$$

with the logit link:

$$\text{logit}(p_i) = \log \frac{p_i}{1 - p_i} = \mathbf{x}_i' \boldsymbol{\beta}$$

Equivalently:

$$p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})} = \frac{1}{1 + \exp(-\mathbf{x}_i' \boldsymbol{\beta})}$$

This ensures $0 < p_i < 1$ for all $\boldsymbol{\beta}$.

Interpreting Coefficients: Odds

Odds of success:

$$\text{Odds} = \frac{p}{1-p}$$

Example:

- $p = 0.8$: Odds = $0.8/0.2 = 4$ (“4 to 1”)
- $p = 0.5$: Odds = 1 (“even odds”)
- $p = 0.2$: Odds = 0.25 (“1 to 4”)

The logit is the log-odds:

$$\text{logit}(p) = \log(\text{Odds})$$

Interpreting Coefficients: Odds Ratios

Model: $\text{logit}(p) = \alpha + \beta x$

For a unit increase in x :

$$\text{logit}(p_{x+1}) - \text{logit}(p_x) = \beta$$

$$\log \frac{\text{Odds}_{x+1}}{\text{Odds}_x} = \beta$$

$$\frac{\text{Odds}_{x+1}}{\text{Odds}_x} = e^\beta$$

Interpretation:

$e^\beta = \text{Odds Ratio}$

A unit increase in x multiplies the odds by e^β .

Odds Ratio Examples

β	e^{β}	Interpretation
0	1.00	No effect
0.5	1.65	Odds increase by 65%
1.0	2.72	Odds nearly triple
-0.5	0.61	Odds decrease by 39%
-1.0	0.37	Odds reduced to 37%

Key point:

- $e^{\beta} > 1$: Higher $x \Rightarrow$ higher probability of success
- $e^{\beta} < 1$: Higher $x \Rightarrow$ lower probability of success
- $e^{\beta} = 1$ ($\beta = 0$): x has no effect

Log-likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left\{ y_i \log p_i + (m_i - y_i) \log(1 - p_i) + \log \binom{m_i}{y_i} \right\}$$

where $p_i = \text{logit}^{-1}(\mathbf{x}_i' \boldsymbol{\beta})$.

No closed form solution!

Solved iteratively using Fisher Scoring / IRLS.

Fitted values:

$$\hat{p}_i = \frac{\exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}})}{1 + \exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}})}, \quad \hat{y}_i = m_i \hat{p}_i$$

Pearson residuals:

$$e_i^P = \frac{y_i - m_i \hat{p}_i}{\sqrt{m_i \hat{p}_i (1 - \hat{p}_i)}}$$

Deviance residuals:

$$e_i^D = \text{sign}(y_i - m_i \hat{p}_i) \sqrt{d_i}$$

where

$$d_i = 2 \left\{ y_i \log \frac{y_i}{m_i \hat{p}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i (1 - \hat{p}_i)} \right\}$$

For large m_i , residuals should be approximately $\mathcal{N}(0, 1)$.

Deviance for Logistic Regression

Model deviance:

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{y}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - \hat{y}_i} \right\}$$

Goodness of fit: Under correct model, $D \sim \chi^2_{n-r}$.

Alternative: Pearson χ^2 statistic

$$X^2 = \sum_{i=1}^n \frac{(y_i - m_i \hat{p}_i)^2}{m_i \hat{p}_i (1 - \hat{p}_i)}$$

Both D and X^2 are asymptotically χ^2_{n-r} .

Fitting Logistic Regression in R

For grouped binomial data:

```
# y = number of successes, m = number of trials
y <- cbind(successes, failures)
model <- glm(y ~ x1 + x2, family = binomial)

# Or equivalently
model <- glm(cbind(successes, total - successes) ~ x1 + x2,
             family = binomial)
```

For binary (0/1) data:

```
# y is 0 or 1 for each observation
model <- glm(y ~ x1 + x2, family = binomial)
```

Example: R Output

```
Coefficients:
      Estimate Std. Error z value Pr(>|z|)
(Intercept) -60.7175     5.1805  -11.72  <2e-16 ***
dose         34.2703     2.9122   11.77  <2e-16 ***

Null deviance: 284.202  on 7  degrees of freedom
Residual deviance:  11.232  on 6  degrees of freedom
AIC: 41.43
```

Interpretation:

- $\hat{\beta} = 34.27$: log-odds ratio for unit dose increase
- $e^{34.27}$: odds ratio (very large)
- Residual deviance: 11.23 on 6 df — reasonable fit

Analysis of Deviance for the Beetle Data

```
anova(glm.fit)
```

```
Terms added sequentially (first to last)
```

	Df	Deviance	Resid.	Df	Resid. Dev	Pr(>Chi)
NULL				7	284.202	
dose	1	272.97		6	11.232	< 2.2e-16 ***

Adding **dose** reduces the deviance by 272.97 on 1 df — an overwhelming likelihood-ratio test for the effect of dose.

Three types of tests:

1. Wald test (from summary output):

$$z = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim \mathcal{N}(0, 1) \text{ under } H_0 : \beta_j = 0$$

2. Likelihood ratio test (deviance difference):

$$D_0 - D_1 \sim \chi^2_{r_1 - r_0} \text{ under } H_0$$

3. Goodness-of-fit test:

$$D \sim \chi^2_{n-r} \text{ under correct model}$$

Summary: Logistic Regression

Key points:

- Logistic regression models binomial/binary responses
- Logit link: $\text{logit}(p) = \log \frac{p}{1-p} = \mathbf{x}'\boldsymbol{\beta}$
- Coefficients are log-odds ratios
- e^{β_j} = multiplicative effect on odds per unit increase in x_j
- MLE via iterative methods (Fisher Scoring)
- Deviance and Pearson χ^2 for goodness of fit
- Use `family = binomial` in R

Overdispersion: The Problem

Binomial assumption:

$$\text{Var}[Y_i] = m_i p_i (1 - p_i)$$

In practice: The observed variance often exceeds this.

Detection:

- Residual deviance $\gg$ residual degrees of freedom
- Ratio $D/(n - r) \gg 1$

Example: $D = 45.2$ on 10 df suggests overdispersion.

Possible sources:

① Missing explanatory variables:

- Important variables not in the model
- Unmeasured heterogeneity

② Incorrect link function:

- Logit may not be appropriate

③ Lack of independence:

- Trials within groups may be correlated
- Clustering effects

Modelling Overdispersion

Introduce a dispersion parameter $\tau > 1$:

$$\text{Var}[Y_i] = \tau \cdot m_i p_i (1 - p_i)$$

Estimation:

$$\hat{\tau} = \frac{D}{n - r} \quad \text{or} \quad \hat{\tau} = \frac{X^2}{n - r}$$

Effect:

- Parameter estimates $\hat{\beta}$ unchanged
- Standard errors multiplied by $\sqrt{\hat{\tau}}$
- Confidence intervals become wider
- p -values increase (more conservative)

Quasi-Binomial in R

```
# Standard binomial (assumes  $\phi = 1$ )
model1 <- glm(y ~ x, family = binomial)

# Quasi-binomial (estimates  $\phi$ )
model2 <- glm(y ~ x, family = quasibinomial)

# Compare summaries
summary(model1)  # SE assuming no overdispersion
summary(model2)  # SE adjusted for overdispersion
```

Key difference:

- binomial: Fixed $\phi = 1$
- quasibinomial: Estimates ϕ from data

2×2 Contingency Tables

Setup:

	Success	Failure	Total
Group 1	y_1	$m_1 - y_1$	m_1
Group 2	y_2	$m_2 - y_2$	m_2

Probabilities:

- Group 1: $\mathbb{P}(\text{success}) = \pi_1$
- Group 2: $\mathbb{P}(\text{success}) = \pi_2$

Question: Is there an association between group and outcome?

Odds and Odds Ratio

Odds of success in each group:

$$O_1 = \frac{\pi_1}{1 - \pi_1}, \quad O_2 = \frac{\pi_2}{1 - \pi_2}$$

Odds ratio:

$$\psi = \frac{O_1}{O_2} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}$$

Interpretation:

- $\psi = 1$: No association (same odds in both groups)
- $\psi > 1$: Group 1 has higher odds of success
- $\psi < 1$: Group 1 has lower odds of success

Estimating the Odds Ratio

Sample estimates:

$$\hat{\pi}_1 = \frac{y_1}{m_1}, \quad \hat{\pi}_2 = \frac{y_2}{m_2}$$

Estimated odds ratio:

$$\hat{\psi} = \frac{y_1(m_2 - y_2)}{y_2(m_1 - y_1)}$$

Log odds ratio:

$$\log \hat{\psi} = \log y_1 - \log(m_1 - y_1) - \log y_2 + \log(m_2 - y_2)$$

Confidence Interval for Odds Ratio

Standard error of log odds ratio:

$$\text{SE}(\log \hat{\psi}) = \sqrt{\frac{1}{y_1} + \frac{1}{m_1 - y_1} + \frac{1}{y_2} + \frac{1}{m_2 - y_2}}$$

95% CI for $\log \psi$:

$$\log \hat{\psi} \pm 1.96 \cdot \text{SE}(\log \hat{\psi})$$

95% CI for ψ :

$$\left(\hat{\psi} \cdot e^{-1.96 \cdot \text{SE}}, \quad \hat{\psi} \cdot e^{+1.96 \cdot \text{SE}} \right)$$

If CI includes 1: No significant association.

Connection to Logistic Regression

Model: $\text{logit}(\pi_i) = \alpha + \beta \cdot \text{Group}_i$

where $\text{Group}_i = 0$ for group 2 (reference), $\text{Group}_i = 1$ for group 1.

Then:

$$\text{logit}(\pi_2) = \alpha$$

$$\text{logit}(\pi_1) = \alpha + \beta$$

Therefore:

$$\beta = \text{logit}(\pi_1) - \text{logit}(\pi_2) = \log \psi$$

And:

$$e^{\beta} = \psi = \text{Odds Ratio}$$

Example: Odds Ratio Calculation

Data:

	Disease	No Disease	Total
Exposed	30	70	100
Not Exposed	10	90	100

Odds ratio:

$$\hat{\psi} = \frac{30 \times 90}{10 \times 70} = \frac{2700}{700} = 3.86$$

Interpretation: Exposed individuals have 3.86 times the odds of disease compared to unexposed.

Relative Risk vs. Odds Ratio

Relative Risk (Risk Ratio):

$$RR = \frac{\pi_1}{\pi_2}$$

Odds Ratio:

$$OR = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)}$$

Relationship:

- When π_1, π_2 are small: $OR \approx RR$
- OR always more extreme than RR (farther from 1)
- OR has nicer mathematical properties
- OR is what logistic regression estimates

Summary: Overdispersion and Odds Ratios

Key points:

- Overdispersion: variance $>$ binomial assumption
- Detect via $D/(n - r) \gg 1$
- Use `quasibinomial` to adjust standard errors
- Odds ratio: $\psi = O_1/O_2$
- $\psi = 1$ means no association
- From logistic regression: $\psi = e^\beta$
- CI for ψ via log transformation
- OR $\approx$ RR when probabilities are small

Next lecture: Dose-response experiments and applications.

Lecture 9: Modelling Proportions — Dose-Response Experiments

MATH3701 Statistical Modelling

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MATH3701 Statistical Modelling

Course notes: Chapter 5, Section 5.5

MATH3701 Statistical Modelling notes, Chapter 5

Dose-Response Studies

Setting:

- Subjects exposed to varying doses of a treatment/substance
- Binary response: success/failure, survival/death
- Goal: Model how response probability depends on dose

Examples:

- Drug efficacy studies
- Toxicology (beetle mortality!)
- Clinical trials
- Pesticide effectiveness

The Beetle Mortality Data (Revisited)

Dose (x_i)	Beetles (m_i)	Killed (y_i)	Proportion
1.6907	59	6	0.10
1.7242	60	13	0.22
1.7552	62	18	0.29
1.7842	56	28	0.50
1.8113	63	52	0.83
1.8369	59	53	0.90
1.8610	62	61	0.98
1.8839	60	60	1.00

Note: Dose is $\log_{10}(\text{concentration})$.

Logistic Regression Model

Model:

$$Y_i \sim \text{Binomial}(m_i, p_i), \quad i = 1, \dots, 8$$

$$\text{logit}(p_i) = \alpha + \beta x_i$$

Probability of death:

$$p_i = \frac{\exp(\alpha + \beta x_i)}{1 + \exp(\alpha + \beta x_i)}$$

Interpretation:

- $\beta > 0$: Higher dose $\Rightarrow$ higher mortality
- e^β : Odds ratio per unit increase in dose

R Implementation

```
# Load data
beetle <- data.frame(
  dose = c(1.6907, 1.7242, 1.7552, 1.7842,
           1.8113, 1.8369, 1.8610, 1.8839),
  died = c(6, 13, 18, 28, 52, 53, 61, 60),
  total = c(59, 60, 62, 56, 63, 59, 62, 60)
)

# Fit logistic regression
y <- cbind(beetle$died, beetle$total - beetle$died)
model <- glm(y ~ dose, data = beetle,
             family = binomial)

summary(model)
```

Model Output

```
Coefficients:
      Estimate Std. Error z value Pr(>|z|)
(Intercept)  -60.717      5.181  -11.72  <2e-16 ***
dose          34.270      2.912   11.77  <2e-16 ***

Null deviance: 284.202  on 7  degrees of freedom
Residual deviance:  11.232  on 6  degrees of freedom
AIC: 41.43
```

Fitted model:

$$\text{logit}(\hat{p}) = -60.72 + 34.27 \times \text{dose}$$

Deviance: 11.23 on 6 df — reasonable fit.

Sequential Hypothesis Tests

Test 1: Null model adequacy

- H_0 : All beetles have same mortality rate
- Null deviance: 284.2 on 7 df
- $p < 0.001 \Rightarrow$ Reject (mortality varies)

Test 2: Dose effect

- H_0 : $\beta = 0$ (no dose effect)
- $\Delta D = 284.20 - 11.23 = 272.97$ on 1 df
- $p < 0.001 \Rightarrow$ Strong dose effect

Test 3: Model adequacy

- H_0 : Logistic dose model is correct
- Residual deviance: 11.23 on 6 df
- $p \approx 0.082$ — marginally adequate

Predictions

Probability at a given dose:

$$\hat{p}(x) = \frac{\exp(-60.72 + 34.27x)}{1 + \exp(-60.72 + 34.27x)}$$

```
# Predict at new doses
new_dose <- data.frame(dose = c(1.7, 1.75, 1.8))
predict(model, newdata = new_dose,
        type = "response")
```

Example predictions:

- Dose 1.70: $\hat{p} = 0.14$
- Dose 1.75: $\hat{p} = 0.36$
- Dose 1.80: $\hat{p} = 0.70$

Finding the Lethal Dose

LD50: Dose at which 50% mortality is expected.

For logistic model:

$$\text{logit}(0.5) = 0 = \alpha + \beta x_{50}$$

Solving:

$$x_{50} = -\frac{\alpha}{\beta} = -\frac{-60.72}{34.27} = 1.772$$

General formula for LD p (dose for mortality p):

$$x_p = \frac{\text{logit}(p) - \alpha}{\beta} = \frac{1}{\beta} \left(\log \frac{p}{1-p} - \alpha \right)$$

LD Values in R

```
# Extract coefficients
alpha <- coef(model)[1]
beta  <- coef(model)[2]

# LD50
LD50 <- -alpha / beta
LD50  # 1.772

# LD90 (90% mortality)
LD90 <- (log(0.9/0.1) - alpha) / beta
LD90  # 1.836

# LD10 (10% mortality)
LD10 <- (log(0.1/0.9) - alpha) / beta
LD10  # 1.708
```

Alternative Link Functions

Try different links for beetle data:

```
# Logit (canonical)
m1 <- glm(y ~ dose, family = binomial(link = "logit"))

# Probit
m2 <- glm(y ~ dose, family = binomial(link = "probit"))

# Complementary log-log
m3 <- glm(y ~ dose, family = binomial(link = "cloglog"))
```

Compare deviances:

Link	Deviance
Logit	11.23
Probit	10.12
Cloglog	3.45

Residual Analysis

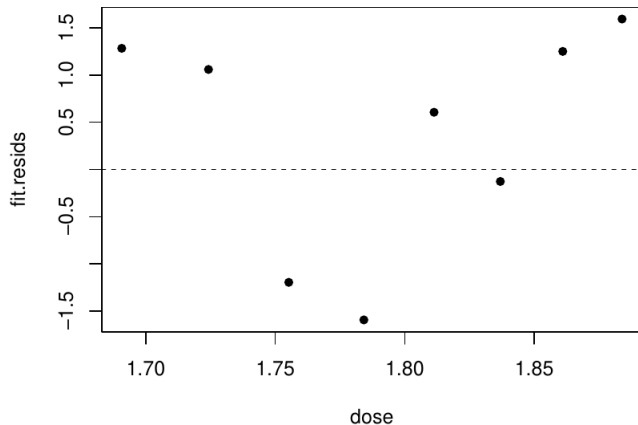
```
# Deviance residuals
residuals(model, type = "deviance")

# Plot residuals vs fitted
plot(fitted(model), residuals(model, type = "deviance"),
     xlab = "Fitted values", ylab = "Deviance residuals")
abline(h = 0, lty = 2)
```

What to look for:

- Random scatter around 0
- No systematic patterns
- U-shaped pattern $\Rightarrow$ try different link

Residuals for the Beetle Data



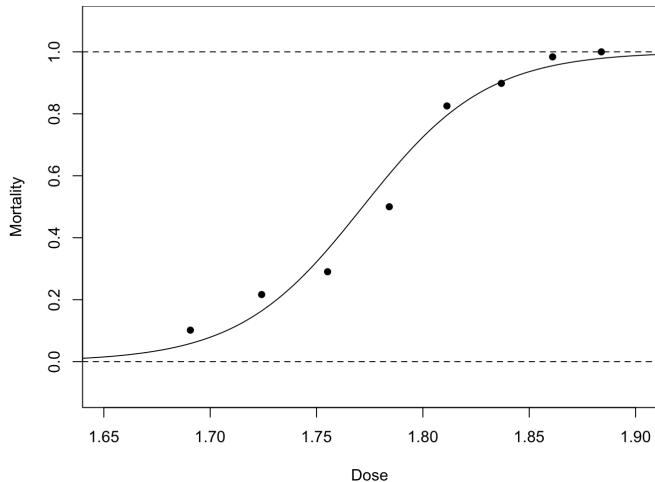
The deviance residuals show a slight u-shaped pattern against dose — a hint that the logit link may not be quite right. Trying the cloglog link (previous slide) improves the fit.

Visualizing the Fitted Model

```
# Plot observed proportions
plot(beetle$dose, beetle$died/beetle$total,
     xlab = "Dose", ylab = "Mortality",
     pch = 16, ylim = c(0, 1))

# Add fitted curve
dose_seq <- seq(1.68, 1.90, length = 100)
pred <- predict(model,
               newdata = data.frame(dose = dose_seq),
               type = "response")
lines(dose_seq, pred, col = "blue", lwd = 2)
```

The Fitted Dose-Response Curve



The LD50 (dose for 50% mortality) is found at $x_{50} = -\alpha/\beta = 1.772$.

Key points:

- Dose-response: model $p(\text{response})$ vs. dose
- Logistic regression: $\text{logit}(p) = \alpha + \beta x$
- β measures effect of dose on log-odds
- $\text{LD50} = -\alpha/\beta$ (dose for 50% response)
- General: $x_p = (\text{logit}(p) - \alpha)/\beta$
- Compare link functions via deviance
- Check residuals for model adequacy

Next lecture: Loglinear models for count data.

Lecture 10: Loglinear Models

MATH3701 Statistical Modelling

Richard P Mann

MATH3701 Statistical Modelling

Course notes: Chapter 6

MATH3701 Statistical Modelling notes, Chapter 6

Count Data and Contingency Tables

Setting:

- Response: counts Y_{ij} (non-negative integers)
- All explanatory variables are categorical (factors)
- Data often displayed in contingency tables

Examples:

- Disease cases by region and age group
- Customer purchases by product and store
- Survey responses by demographic factors

Example: Melanoma Data

400 malignant melanoma patients:

Tumor Type	Head/Neck	Trunk	Extremities	Total
Hutchinson's	22	2	10	34
Superficial	16	54	115	185
Nodular	19	33	73	125
Indeterminate	11	17	28	56
Total	68	106	226	400

Question: Is tumor type associated with body site?

The Poisson Model for Counts

Assume:

$$Y_{ij} \sim \text{Poisson}(\lambda_{ij})$$

independently for each cell.

Loglinear model:

$$\log \lambda_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}$$

Components:

- μ : overall mean (log scale)
- α_i : row effect (tumor type)
- β_j : column effect (body site)
- $(\alpha\beta)_{ij}$: interaction (association)

Problem: The model is overparameterized.

R uses corner constraints:

- $\alpha_1 = 0$ (first row is reference)
- $\beta_1 = 0$ (first column is reference)
- $(\alpha\beta)_{1j} = (\alpha\beta)_{i1} = 0$

Interpretation:

- α_i : log ratio of row i to row 1 (column effects averaged out)
- β_j : log ratio of column j to column 1 (row effects averaged out)
- $(\alpha\beta)_{ij}$: departure from additive model

The Independence Model

No interaction:

$$\log \lambda_{ij} = \mu + \alpha_i + \beta_j$$

Equivalently:

$$\lambda_{ij} = e^{\mu} \cdot e^{\alpha_i} \cdot e^{\beta_j}$$

Key property: Expected counts factor into row and column effects.

Under independence:

$$\hat{\lambda}_{ij} = \frac{y_{i+} \cdot y_{+j}}{y_{++}}$$

(row total $\times$ column total / grand total)

Maximum Likelihood for Independence Model

Fitted marginal totals equal observed marginal totals:

$$\hat{y}_{i+} = y_{i+}, \quad \hat{y}_{+j} = y_{+j}$$

Fitted cell values:

$$\hat{\lambda}_{ij} = \frac{y_{i+} \cdot y_{+j}}{y_{++}}$$

This is the “expected count under independence” familiar from χ^2 tests.

Testing Independence

Deviance (likelihood ratio) test:

$$D = 2 \sum_{i,j} y_{ij} \log \frac{y_{ij}}{\hat{\lambda}_{ij}} \sim \chi^2_{(r-1)(c-1)}$$

Pearson chi-squared:

$$X^2 = \sum_{i,j} \frac{(y_{ij} - \hat{\lambda}_{ij})^2}{\hat{\lambda}_{ij}} \sim \chi^2_{(r-1)(c-1)}$$

Both test H_0 : Row and column variables are independent.

Degrees of freedom: $(r-1)(c-1)$ for $r \times c$ table.

Fitting in R

```
# Create data frame
melanoma <- data.frame(
  Type = factor(rep(c("Hutchinson", "Superficial",
                      "Nodular", "Indeterminate"),
                    each = 3)),
  Site = factor(rep(c("Head", "Trunk", "Extrem"), 4)),
  Count = c(22, 2, 10, 16, 54, 115,
            19, 33, 73, 11, 17, 28)
)

# Independence model
model_ind <- glm(Count ~ Type + Site,
                 family = poisson, data = melanoma)

# Saturated model (with interaction)
model_sat <- glm(Count ~ Type * Site,
                 family = poisson, data = melanoma)
```

R Output: Independence Model

```
Coefficients:
      Estimate Std. Error z value Pr(>|z|)
(Intercept)    2.1517    0.1878  11.459 < 2e-16 ***
TypeIndeterminate -0.4855    0.1731  -2.805  0.00503 **
TypeNodular     0.2993    0.1445   2.071  0.03837 *
TypeSuperficial  0.6946    0.1318   5.271 1.36e-07 ***
SiteHead       -1.2004    0.1421  -8.449 < 2e-16 ***
SiteTrunk       -0.7568    0.1173  -6.450 1.12e-10 ***

Null deviance: 216.130 on 11 degrees of freedom
Residual deviance: 51.795 on 6 degrees of freedom
```

Test of independence:

- Deviance: 51.8 on 6 df
- $p\text{-value} < 0.001 \Rightarrow$ Strong evidence against independence

Examining Residuals

Large Pearson residuals indicate departure from model:

```
# Pearson residuals  
residuals(model_ind, type = "pearson")
```

For melanoma data:

Type	Head	Trunk	Extrem
Hutchinson's	+4.06	-2.02	-1.53
Superficial	-1.55	+0.49	+0.73
Nodular	+0.09	+0.31	-0.26
Indeterminate	+0.39	+0.64	-0.71

Hutchinson's melanoma on head/neck is over-represented.

Three-way contingency table: Factors A , B , C

Saturated model:

$$\log \lambda_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk}$$

Hierarchy of models:

- Mutual independence: $\mu + \alpha_i + \beta_j + \gamma_k$
- Conditional independence: Add some two-way interactions
- Full model: Include three-way interaction

Principle: If a higher-order term is included, all lower-order terms involving those factors must also be included.

Example: If $(\alpha\beta\gamma)_{ijk}$ is in the model, then must include:

- $(\alpha\beta)_{ij}, (\alpha\gamma)_{ik}, (\beta\gamma)_{jk}$
- $\alpha_i, \beta_j, \gamma_k$
- μ

Why? Interpretability — effects should be measured relative to properly defined baselines.

Compare nested models via deviance:

$$\Delta D = D_{\text{simpler}} - D_{\text{complex}} \sim \chi^2_{\Delta df}$$

Strategy:

- ① Start with independence model
- ② Add interactions one at a time
- ③ Test each addition using ΔD
- ④ Choose simplest adequate model

Key points:

- Loglinear models: Poisson GLM with categorical predictors
- $\log \lambda_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}$
- Independence: No interaction term
- Test independence via deviance or Pearson χ^2
- Residuals identify cells with poor fit
- Multi-way tables: hierarchy of interactions
- Hierarchical principle: include lower-order terms

Next lecture: Extensions — fixed marginals and product-multinomial models.

Lecture 11: Extensions to Loglinear Models

MATH3701 Statistical Modelling

Richard P Mann

MATH3701 Statistical Modelling

Course notes: Chapter 7

MATH3701 Statistical Modelling notes, Chapter 7

The Problem: Fixed Marginals

In many studies, some totals are fixed by design:

- Clinical trial: Fixed number per treatment group
- Case-control study: Fixed cases and controls
- Survey: Fixed sample size

When totals are fixed:

- Cell counts are *not* independent Poisson
- The standard Poisson model is theoretically incorrect
- But we can still use Poisson GLM.

Example: Fixed Total

Melanoma study: Total of 400 patients (fixed).

Questions:

- Can we still use Poisson loglinear models?
- What distribution do the counts actually follow?
- How do we interpret the results?

Answer: The counts follow a *multinomial* distribution, but Poisson gives the same parameter estimates.

Proposition

If $Y_1, \dots, Y_k$ are independent $\text{Poisson}(\lambda_i)$, then conditional on $Y_+ = n$:

$$(Y_1, \dots, Y_k) \mid Y_+ = n \sim \text{Multinomial} \left(n; \frac{\lambda_1}{\lambda_+}, \dots, \frac{\lambda_k}{\lambda_+} \right)$$

Key insight: Conditioning independent Poissons on their sum gives a multinomial.

Proposition

If $Y_1 \sim \text{Poisson}(\lambda_1)$ and $Y_2 \sim \text{Poisson}(\lambda_2)$ independently, then:

$$Y_1 \mid (Y_1 + Y_2 = n) \sim \text{Binomial} \left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2} \right)$$

Application: In a 2×2 table with fixed row totals, the row distributions are binomial.

Product-Multinomial Distribution

When row totals y_{i+} are fixed:

Each row follows an independent multinomial:

$$(Y_{i1}, \dots, Y_{ic}) \sim \text{Multinomial}(y_{i+}; \pi_{i1}, \dots, \pi_{ic})$$

This is called a product-multinomial distribution.

The joint distribution is the product of row-wise multinomials.

The Key Theorem

Theorem

Tables with fixed margin sums can be analyzed using a multinomial or product-multinomial model as though they were independent Poisson models, provided terms corresponding to the fixed margins are included in the model.

What this means:

- Fixed total $y_{++} \Rightarrow$ include μ (intercept)
- Fixed row totals $y_{i+} \Rightarrow$ include row main effects α_i
- Fixed column totals $y_{+j} \Rightarrow$ include column main effects β_j

Why Does This Work?

Mathematical reason:

When we condition on fixed margins, the parameters for those margins “absorb” the constraint. The remaining parameters (e.g., interactions) have the same MLEs whether we use:

- Poisson likelihood
- Multinomial/product-multinomial likelihood

Practical implication:

We can use `glm(..., family = poisson)` even when the Poisson assumption is technically violated.

Application: Flu Vaccine Trial

Data: 73 participants, fixed group sizes

Group	Low	Moderate	High	Total
Placebo	25	8	5	38
Vaccine	6	18	11	35
Total	31	26	16	73

Row totals fixed: 38 placebo, 35 vaccine (by design).

Flu Data: R Analysis

```
# Create data
flu <- data.frame(
  Group = factor(rep(c("Placebo", "Vaccine"),
                    each = 3)),
  Response = factor(rep(c("Low", "Mod", "High"), 2),
                    levels = c("Low", "Mod", "High")),
  Count = c(25, 8, 5, 6, 18, 11)
)

# Independence model (must include Group for fixed rows)
model <- glm(Count ~ Group + Response,
             family = poisson, data = flu)
summary(model)
```


Flu Data: Results

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.6461	0.1796	14.729	< 2e-16 ***
GroupVaccine	-0.0816	0.2399	-0.340	0.734
ResponseMod	-0.1757	0.2721	-0.646	0.518
ResponseHigh	-0.6614	0.3164	-2.091	0.037 *

Residual deviance: 18.643 on 2 degrees of freedom

Test of independence:

- Deviance: 18.64 on 2 df
- p -value < 0.001
- Strong evidence of association between group and response

Model Simplification

Observation: Perhaps “Moderate” and “High” can be combined?

```
# Combine Moderate and High
flu$Response2 <- flu$Response
levels(flu$Response2) <- c("Low", "ModHigh", "ModHigh")

# Refit
model2 <- glm(Count ~ Group + Response2,
              family = poisson, data = flu)
```

Result: Deviance 2.405 on 2 df ($p = 0.30$).

Interpretation: Vaccine affects Low vs. Moderate/High, not Moderate vs. High.

Guidelines for Fixed Margins

Two-way table:

Fixed	Include in model
Total only (y_{++})	Intercept μ
Row totals (y_{i+})	Row effects α_i
Column totals (y_{+j})	Column effects β_j
Both margins	Both α_i and β_j

Three-way table: Similar principle — include main effects and interactions corresponding to fixed margins.

1. Small expected counts:

- χ^2 approximation may be poor
- Rule of thumb: Expected counts ≥ 5
- Consider exact tests or combining categories

2. Overdispersion:

- If deviance $\gg$ df, consider `quasipoisson`
- Or investigate causes (missing explanatory variables, clustering)

3. Sparse tables:

- Many zeros cause problems
- May need specialized methods

Key points:

- Fixed marginals $\Rightarrow$ counts are not independent Poisson
- Conditioning Poisson on sum gives multinomial
- Product-multinomial when row totals are fixed
- **Key theorem:** Can use Poisson GLM if fixed-margin terms included
- Include row effects if row totals are fixed
- Include column effects if column totals are fixed
- Parameter estimates for interactions are the same.

Next lecture: Course revision and summary.

Lecture 12: GLM Consolidation and Bridge to GAMs

MATH3701 Statistical Modelling

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MATH3701 Statistical Modelling

Course notes: MATH3701 Statistical Modelling notes, Chapter 8

The Three-Component Structure

Every GLM has exactly three components:

① **Random component:**

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

Mean: $\mu_i = b'(\theta_i)$; Variance: $\phi b''(\theta_i)$

② **Systematic component:** linear predictor

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

③ **Link function:** $g(\mu_i) = \eta_i$

Summary of Standard Families

Distribution	Canonical link	Variance function	Use
Normal	Identity	$V = 1$	Continuous
Binomial	Logit	$V = p(1 - p)$	Proportions
Poisson	Log	$V = \lambda$	Counts

Estimation: IRLS converges to the MLE.

Inference: $\hat{\beta} \sim \mathcal{N}_p(\beta, \mathcal{J}^{-1})$.

Comparison: Deviance difference $\sim \chi^2$ or F -ratio.

Case Study: CHD in South Africa

Data: 4×4 table — men cross-classified by cholesterol group and blood-pressure group;
 y_{ij} = number with CHD out of m_{ij} men.

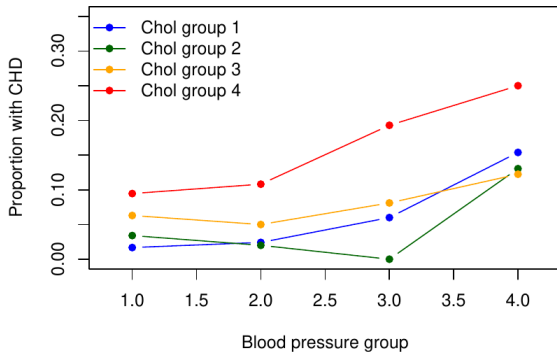
GLM formulation:

$$Y_{ij} \sim \text{Bin}(m_{ij}, p_{ij}), \quad \text{logit}(p_{ij}) = \mu + \alpha_i + \beta_j$$

Models compared:

Model	Parameters	Deviance	df
Null	1	large	15
Main effects	7	moderate	9
Saturated	16	0	0

Exploring the CHD Data



Proportion with CHD by cholesterol and blood pressure group; lines connect groups at the same cholesterol level. Both variables appear associated with CHD; roughly parallel lines suggest no strong interaction.

Test main effects vs null:

$$D_{\text{null}} - D_{\text{main}} \sim \chi_6^2$$

$p < 0.001$ — both cholesterol and blood pressure are highly significant.

Test interaction vs main effects:

$$D_{\text{main}} - D_{\text{saturated}} \sim \chi_9^2$$

$p > 0.05$ — no evidence of interaction; main-effects model is adequate.

Goodness of fit:

$$D_{\text{main}} \sim \chi_9^2$$

$p > 0.05$ — main-effects model fits the data well.

Fitted logit:

$$\text{logit}(\hat{p}_{ij}) = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j$$

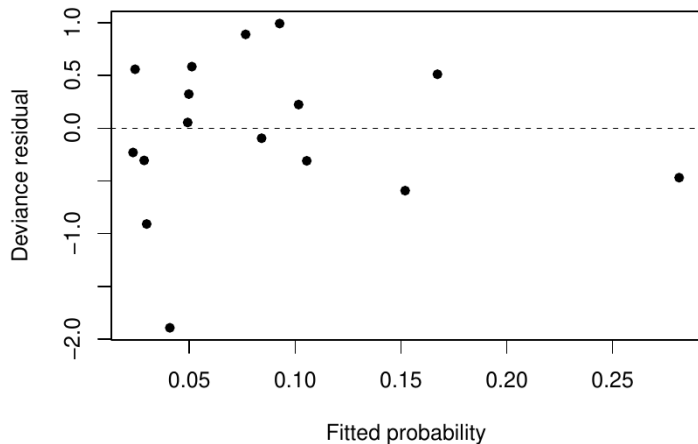
Estimated effects show a clear increasing trend: higher cholesterol and higher blood pressure each raise the log-odds of CHD.

Deviance residuals:

$$e_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

Plotting against fitted values reveals no systematic pattern $\Rightarrow$ well-fitting model.

CHD Model: Deviance Residuals



Deviance residuals from the main-effects logistic regression, plotted against fitted values — no systematic pattern, confirming the goodness-of-fit test.

Limitations of the Linear Predictor

The GLM constraint $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ can be restrictive when a **continuous** explanatory variable has a smooth but *nonlinear* effect.

Options:

- ① **Polynomial regression:** add $x^2, x^3, \dots$
 - Simple, but unstable; degree must be chosen in advance
- ② **Non-parametric smoothing:** replace βx with a smooth function $f(x)$ estimated from the data
 - Flexible, data-driven
 - Leads to **Generalised Additive Models** (Part II)

The GAM replaces the linear predictor with a sum of smooth functions:

$$g(\mu_i) = \alpha + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip})$$

- Each f_j is a **spline** estimated from the data
- f_j replaces the parametric term $\beta_j x_{ij}$
- Penalised fitting controls wiggleness (smoothing parameter λ)
- The random component and link function are the same as in GLMs

Part II: interpolating splines $\rightarrow$ smoothing splines $\rightarrow$ cross-validation $\rightarrow$ GAMs.

Summary of Part I

Theoretical results:

- Exponential family: $\mathbb{E}[Y] = b'(\theta)$,
 $\text{Var}[Y] = \phi b''(\theta)$
- MLE via IRLS
- Asymptotic normality of $\hat{\beta}$
- Deviance as generalised RSS
- LRT: $\Delta D \sim \chi^2$ or F

Applications:

- Logistic regression: binary / proportional responses
- Overdispersion: quasi-binomial / quasi-Poisson
- Loglinear models: contingency tables
- Product-multinomial: fixed marginals

Next lecture: Introduction to non-parametric modelling and splines (Part II begins).

Lecture 13: Introduction to Non-parametric Modelling

MATH3701 Statistical Modelling

Richard P Mann

MATH3701 Statistical Modelling

Course notes: Chapter 9

MATH3701 Statistical Modelling notes

Building on this module: This module extends GLMs in two directions:

- 1 Allow the response to depend on a *smooth function* of the predictor, rather than a linear function
- 2 Combine smooth functions with GLM structure $\Rightarrow$ **Generalised Additive Models**

New topics:

- Interpolating and smoothing splines
- Cross-validation for spline fitting
- Generalised additive models (GAMs)

Motivation: The Coal Seam Data

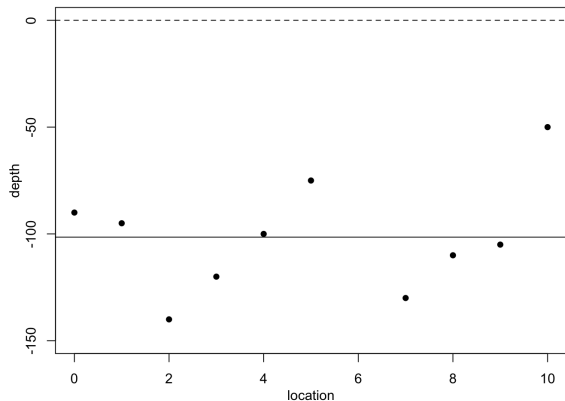
Data: Depths of a coal seam measured at 11 locations along a 10 km transect.

Location (km)	0	1	2	3	4	5	6	7	8	9	10
Depth	35	41	48	47	65	67	?	52	51	68	75

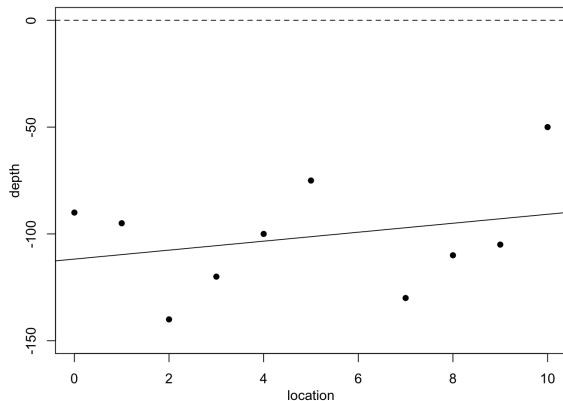
Question: Can we predict the missing depth at $x = 6$?

Different models give different predictions — which is best?

Polynomial Regression: Constant and Linear



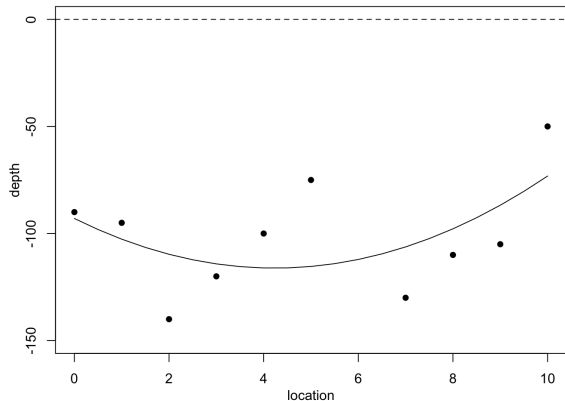
(a) Constant model



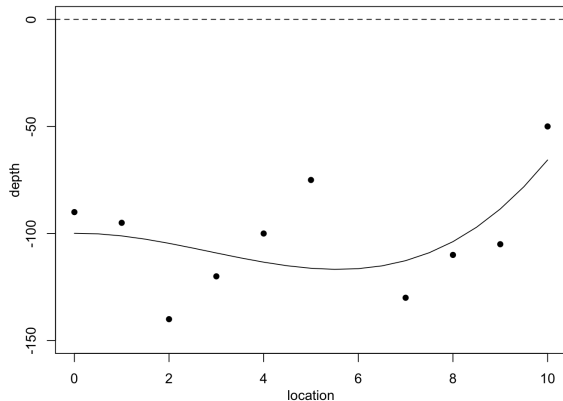
(b) Linear model

Neither captures the local fluctuations in the data.

Polynomial Regression: Quadratic and Cubic



(c) Quadratic model

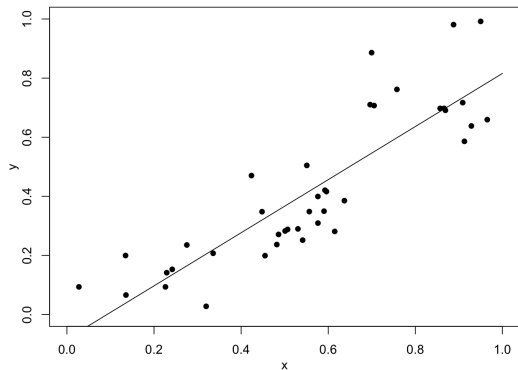


(d) Cubic model

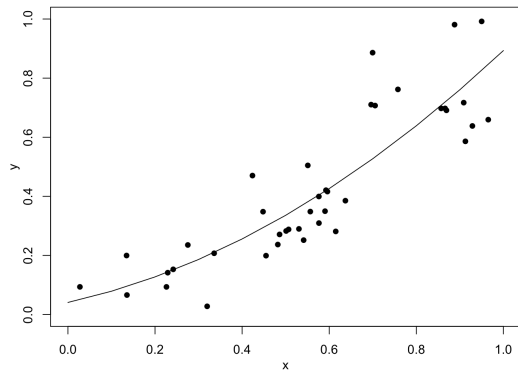
RSS decreases with model complexity (constant: 6252, linear: 5773, quadratic: 4490, cubic: 4243), but no polynomial fits convincingly.

Simulated Change-Point Data: Polynomial Fits

Data: Simulated from a piecewise model with a discontinuity at $x = 0.67$.



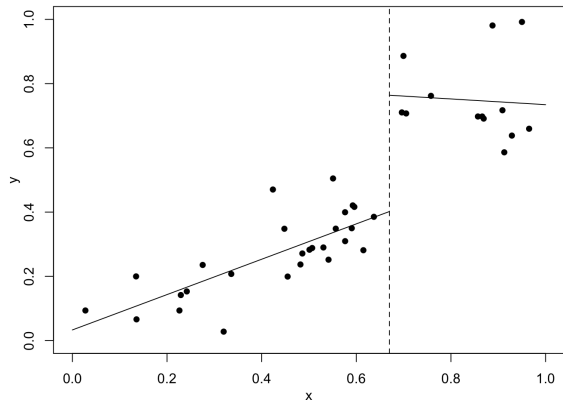
(a) Linear fit



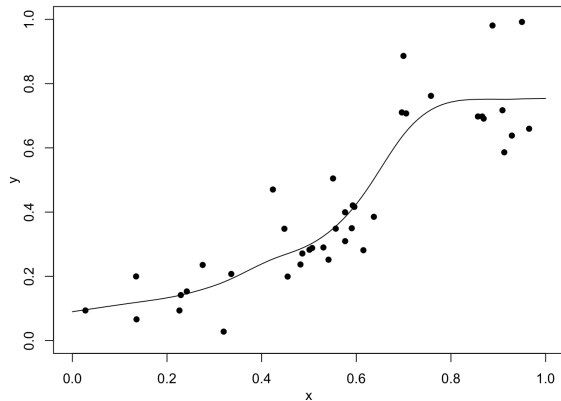
(b) Quadratic fit

Simple polynomial models fail to capture the change-point.

Change-Point Data: Better Approaches



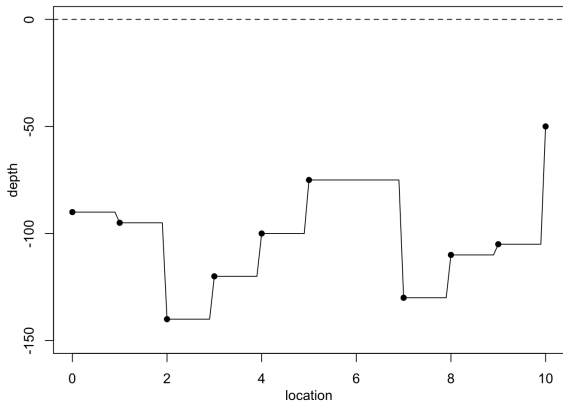
(c) Piecewise linear (known change-point)



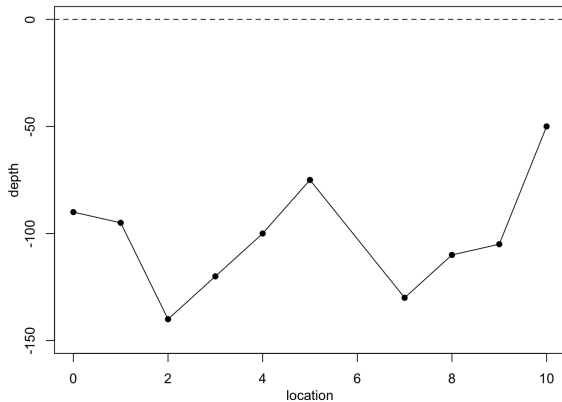
(d) Cubic smoothing spline

The cubic smoothing spline reveals the pattern *without* assuming the change-point location is known. $\Rightarrow$ We need methods that adapt locally to the data.

Spline Approaches: Constant and Linear



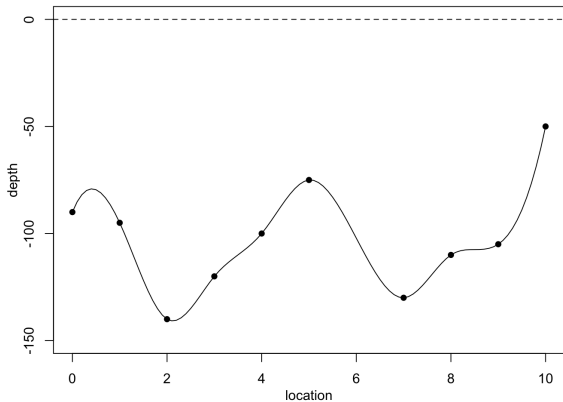
(a) Constant interpolating spline



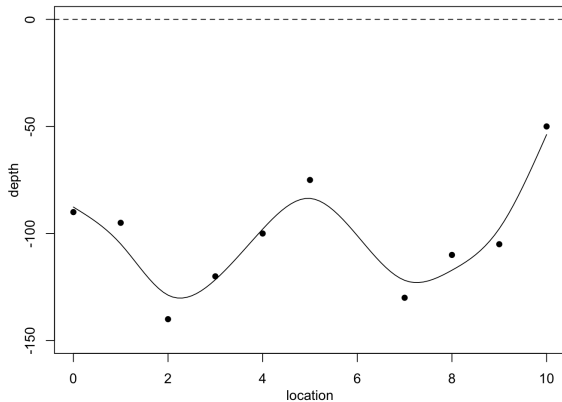
(b) Linear interpolating spline

- Constant: discontinuities at observation locations
- Linear: continuous, but gradient is discontinuous (visible kinks)

Spline Approaches: Cubic Interpolating and Smoothing



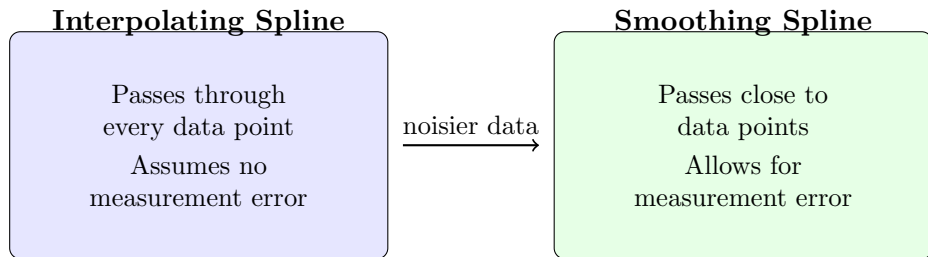
(c) Cubic interpolating spline



(d) Cubic smoothing spline

- Cubic interpolating: passes through all points, continuous gradient and curvature
- Cubic smoothing: allows measurement error, smoother curve

Interpolating vs. Smoothing



Both are **piecewise polynomial** functions with smoothness constraints at join points (**knots**).

Key points:

- Parametric models (polynomials) may not capture local features of data
- Splines are piecewise polynomials joined smoothly at **knots**
- Interpolating splines pass through all data points
- Smoothing splines allow for measurement error
- Different spline orders (constant, linear, cubic) offer different smoothness guarantees

Next lecture: General modelling approaches and the formal definition of splines.

Lecture 14: Introducing Splines — Spline Basics, Smoothness and Degrees of Freedom

MATH3701 Statistical Modelling

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MATH3701 Statistical Modelling

Course notes: Chapter 10

MATH3701 Statistical Modelling notes

The General Model

We observe n pairs (x_i, y_i) at ordered locations $x_1 < x_2 < \dots < x_n$:

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Goal: Estimate the unknown function f .

Two broad strategies:

- ❶ **Parametric:** Specify a functional form for f with finitely many parameters
- ❷ **Non-parametric:** Assume f is “smooth” without specifying its form

Parametric Models

Examples:

- Linear: $f(x) = \alpha + \beta x$
- Polynomial: $f(x) = \alpha + \beta x + \gamma x^2 + \dots + \omega x^p$
- Exponential: $f(x) = \alpha e^{-\beta x}$

Strengths:

- Interpretable parameters
- Well-understood inference (confidence intervals, hypothesis tests)

Limitation: Contains only a small number of parameters, restricting flexibility for arbitrary fluctuations in f .

Non-parametric Models

Assumption: f is *smooth*, meaning $f(x_i)$ is close to $f(x_j)$ when x_i is close to x_j .

No fixed functional form — the data determine the shape.

Approaches:

- **Splines:** Piecewise polynomials with smoothness constraints (our focus)
- Kernel smoothing
- Wavelets
- Local polynomial regression

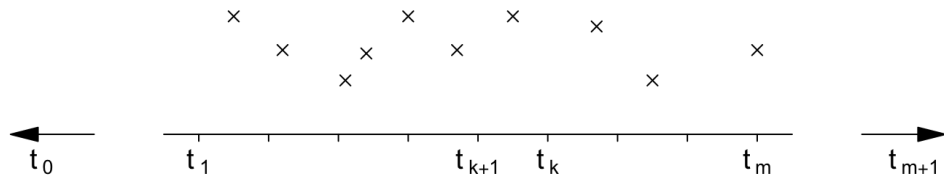
Spline methods fit naturally within the normal linear model and GLM framework.

Problem with global polynomials:

- Low-order polynomials fit poorly across the whole range
- High-order polynomials oscillate wildly (Runge's phenomenon)

Idea: Fit *different* low-order polynomials in different regions.

Requiring continuity and differentiability at the join points leads to **splines**.



Important distinction:

- **Knots** $\{t_k\}$: Where polynomial pieces join
- **Sites** $\{x_i\}$: Where data are observed

Knots and sites need not coincide, though in many applications we place knots at data locations. The knots create $m + 1$ intervals, each with its own polynomial.

Spline Definition

Definition

A **spline of order p** ($p \geq 1$) with knots $t_1 < \dots < t_m$ is a piecewise polynomial of order p that is $(p - 1)$ times differentiable at each knot.

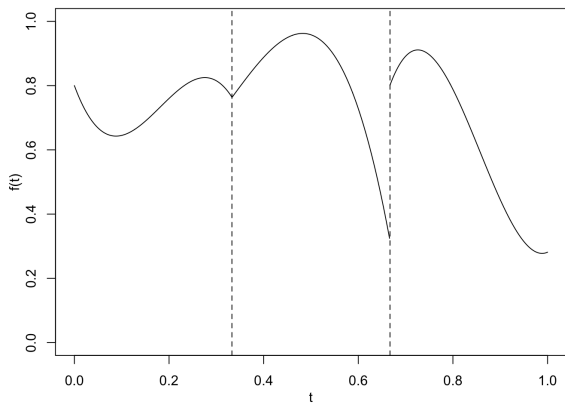
On each interval $[t_k, t_{k+1})$, the spline has the form:

$$f(t) = \sum_{\ell=0}^p a_{k\ell} t^\ell$$

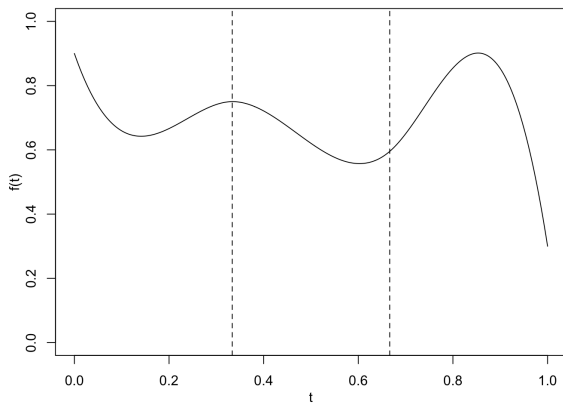
Key property:

- Between knots: infinitely differentiable (it is a polynomial)
- At knots: exactly $(p - 1)$ times differentiable

Piecewise Cubics: Without and With Smoothness



(a) No smoothness constraints



(b) Cubic spline (smooth)

- (a): Piecewise cubic with discontinuities — not a spline
- (b): Piecewise cubic with continuity of f , f' , and f'' at knots — a cubic spline

Common Spline Orders

Order p	Name	Continuity at knots	Typical use
1	Linear	f continuous Slope may jump	Simple interpolation
3	Cubic	f, f', f'' continuous f''' may jump	Standard choice

Why cubic?

- Cubic is the lowest order with continuous curvature (f'')
- Knot locations are “invisible” to the eye
- Higher orders are rarely needed in practice

Key points:

- Non-parametric models allow f to be flexible, guided by the data
- Splines are piecewise polynomials joined smoothly at knots
- A spline of order p is $(p - 1)$ times differentiable at each knot
- Cubic splines ($p = 3$) are the standard choice: continuous f , f' , and f''
- Knots and observation sites are conceptually distinct

Imposing Smoothness

Smoothness constraints: At each knot t_k , require continuity of the function and its first $(p - 1)$ derivatives:

$$\lim_{\varepsilon \rightarrow 0} f^{(\ell)}(t_k - \varepsilon) = \lim_{\varepsilon \rightarrow 0} f^{(\ell)}(t_k + \varepsilon)$$

for $k = 1, \dots, m$ and $\ell = 0, 1, \dots, p - 1$.

Number of constraints: p constraints at each of m knots $= pm$.

Resulting degrees of freedom:

$$\text{df}_{\text{spline}} = (p + 1)(m + 1) - pm = m + p + 1$$

Degrees of Freedom: Examples

Linear spline ($p = 1$) with m knots:

- Unconstrained parameters: $2(m + 1)$
- Continuity constraints: m (one per knot)
- Degrees of freedom: $m + 2$

Cubic spline ($p = 3$) with m knots:

- Unconstrained parameters: $4(m + 1)$
- Smoothness constraints: $3m$ (continuity of f , f' , f'' at each knot)
- Degrees of freedom: $m + 4$

Note: Degrees of freedom grow linearly with the number of knots, regardless of order p .

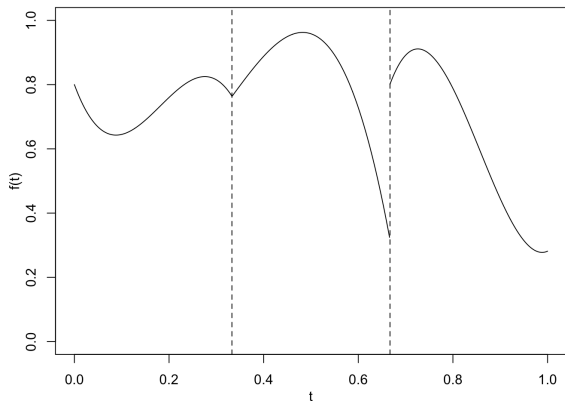
Cubic Spline: Detailed Accounting

Example: Cubic spline with $m = 2$ knots (as in Figure 2.2).

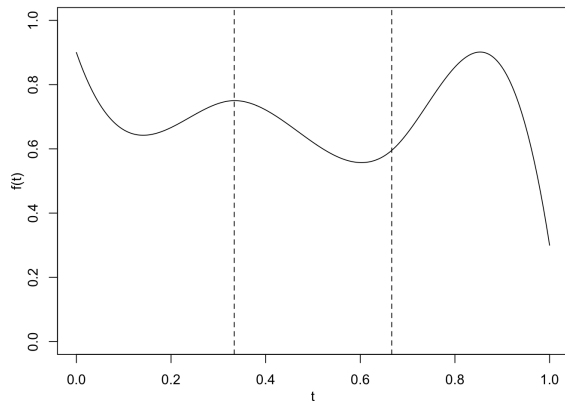
Component	Count
Intervals	3
Coefficients per interval	4
Total coefficients	12
Continuity of f at knots	-2
Continuity of f' at knots	-2
Continuity of f'' at knots	-2
Degrees of freedom	6

Check: $m + p + 1 = 2 + 3 + 1 = 6$. ✓

Constrained vs. Unconstrained



(a) No constraints: 12 free parameters



(b) Spline: 6 free parameters

Smoothness constraints reduce the piecewise cubic from 12 to 6 degrees of freedom, producing a visually smooth curve with no visible knot locations.

What Smoothness Means in Practice

Property	Between knots	At knots
f continuous	✓	✓
f' continuous	✓	✓ (if $p \geq 2$)
f'' continuous	✓	✓ (if $p \geq 3$)
$f^{(p)}$ continuous	✓	Not required
Infinitely differentiable	✓	No

Between knots, the spline is a polynomial and therefore infinitely differentiable. The constraints only matter at the knots themselves.

Looking Ahead: From Definitions to Fitting

We now know:

- What a spline is (piecewise polynomial with smoothness)
- How many degrees of freedom it has ($m + p + 1$)

Next questions:

- How do we choose the coefficients to fit data?
- What happens at the boundaries (beyond the outermost knots)?
- Is there a “best” interpolating spline?

⇒ These lead us to **natural splines** and **roughness penalties**.

Summary: Smoothness and Degrees of Freedom

Key points:

- Smoothness constraints require continuity of derivatives at knots
- A spline of order p with m knots has $m + p + 1$ degrees of freedom
- Each smoothness constraint removes one degree of freedom
- Cubic splines have continuous f , f' , and f''
- The third derivative f''' may be discontinuous at knots
- Between knots, the spline is a standard polynomial

Next lecture: Natural splines and the interpolation problem.

Lecture 15: Natural Splines

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Course notes: Chapter 11

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The Interpolation Problem

Given: n observations $\{(t_i, y_i)\}_{i=1}^n$ at distinct locations $t_1 < \cdots < t_n$.

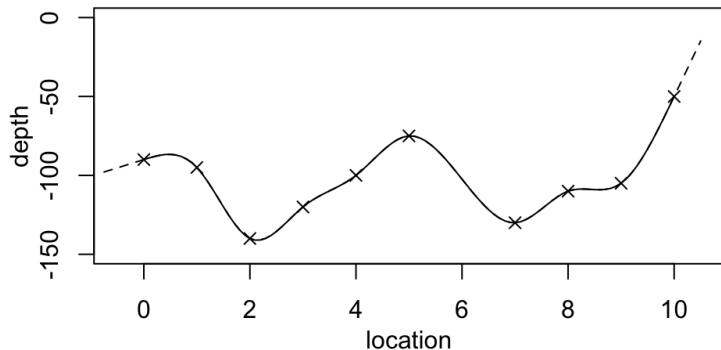
Goal: Find a smooth function f such that $f(t_i) = y_i$ for all i .

A cubic spline with knots at the data locations has $n + 4$ degrees of freedom for n interpolation conditions.

$\Rightarrow$ Infinitely many cubic splines interpolate the data.

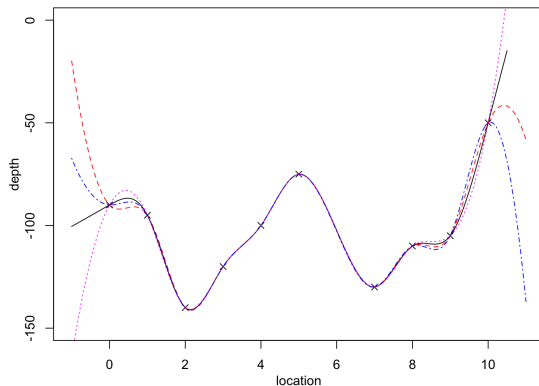
Question: How do we choose among them?

Cubic Interpolating Spline: Coal Seam Data



A cubic interpolating spline fitted to the coal seam data. The dashed line shows extrapolation beyond the data range.

Non-uniqueness of Interpolation



Multiple cubic interpolating splines can pass through the same data. The solid black curve is the **natural spline**; the others differ in their behaviour beyond the boundary knots (dashed lines).

Natural Splines: The Key Idea

Definition

A spline is **natural** if, beyond the boundary knots t_1 and t_n , its higher-order derivatives are zero:

$$f^{(j)}(t) = 0 \quad \text{for } j = \frac{p+1}{2}, \dots, p$$

whenever $t \leq t_1$ or $t \geq t_n$.

Effect: The spline is forced to be a low-order polynomial in the boundary regions.

- **Natural linear spline** ($p = 1$): constant outside $[t_1, t_n]$
- **Natural cubic spline** ($p = 3$): linear outside $[t_1, t_n]$

Natural Cubic Spline: Boundary Conditions

For a natural **cubic** spline, the boundary conditions are:

$$f''(t_1) = f''(t_n) = 0 \quad \text{and} \quad f'''(t_1) = f'''(t_n) = 0$$

More precisely:

$$\lim_{\varepsilon \rightarrow 0} f^{(2)}(t_1 + \varepsilon) = \lim_{\varepsilon \rightarrow 0} f^{(2)}(t_n - \varepsilon) = 0$$

$$\lim_{\varepsilon \rightarrow 0} f^{(3)}(t_1 + \varepsilon) = \lim_{\varepsilon \rightarrow 0} f^{(3)}(t_n - \varepsilon) = 0$$

These impose $p + 1 = 4$ additional constraints.

Degrees of Freedom for Natural Splines

Starting from: $\text{df}_{\text{spline}} = m + p + 1$

Natural boundary conditions: $p + 1$ additional constraints

Therefore:

$$\text{df}_{\text{natural spline}} = (m + p + 1) - (p + 1) = m$$

The number of degrees of freedom equals the number of knots m , **regardless of the spline order p .**

For interpolation: With knots at data locations ($m = n$), we have exactly n degrees of freedom matching n interpolation conditions.

$\Rightarrow$ The natural cubic interpolating spline is **unique**.

Alternative Representations

Proposition (3.1)

Natural splines have convenient closed-form representations.

Linear natural spline ($p = 1$):

$$f(t) = a_0 + \sum_{i=1}^m b_i |t - t_i|, \quad \sum_{i=1}^m b_i = 0$$

Cubic natural spline ($p = 3$):

$$f(t) = a_0 + a_1 t + \sum_{i=1}^m b_i |t - t_i|^3, \quad \sum_{i=1}^m b_i = 0, \quad \sum_{i=1}^m b_i t_i = 0$$

These representations build in the natural boundary conditions automatically via the side constraints on $\{b_i\}$.

Worked Example: Why Does the Linear Representation Work?

Building block: $|t - t_i|$ is itself a linear spline with one knot at t_i .

- Slope -1 for $t < t_i$, slope $+1$ for $t > t_i$

So $f(t) = a_0 + \sum b_i |t - t_i|$ is a sum of linear splines $\Rightarrow$ piecewise linear and continuous $\Rightarrow$ a linear spline.

Why $\sum b_i = 0$? Differentiate on the outer intervals:

- For $t < t_1$: every $|t - t_i| = t_i - t$, so $f'(t) = -\sum b_i$
- For $t > t_m$: every $|t - t_i| = t - t_i$, so $f'(t) = +\sum b_i$

Both are zero (constant outside boundary) $\iff \sum b_i = 0$. ✓

However: can *every* natural linear spline be written this way?

Worked Example: $m = 2$ Knots at $t_1 = 0$, $t_2 = 1$

A natural linear spline here is constant for $t < 0$, linear on $[0, 1]$, constant for $t > 1$. It is determined by two values: $f(0) = y_1$ and $f(1) = y_2$.

Can we solve $f(t) = a_0 + b_1|t| + b_2|t - 1|$ with $b_1 + b_2 = 0$?

Write $b_2 = -b_1$ and evaluate at the knots:

$$f(0) = a_0 + b_1 \cdot 0 + (-b_1) \cdot 1 = a_0 - b_1 = y_1$$

$$f(1) = a_0 + b_1 \cdot 1 + (-b_1) \cdot 0 = a_0 + b_1 = y_2$$

Adding: $2a_0 = y_1 + y_2$. Subtracting: $2b_1 = y_2 - y_1$. So:

$$a_0 = \frac{y_1 + y_2}{2}, \quad b_1 = \frac{y_2 - y_1}{2}, \quad b_2 = \frac{y_1 - y_2}{2}$$

This always has a **unique solution** for any y_1, y_2 .

Worked Example: Numerical Check

Take $y_1 = 3$, $y_2 = 5$. Then $a_0 = 4$, $b_1 = 1$, $b_2 = -1$:

$$f(t) = 4 + |t| - |t - 1|$$

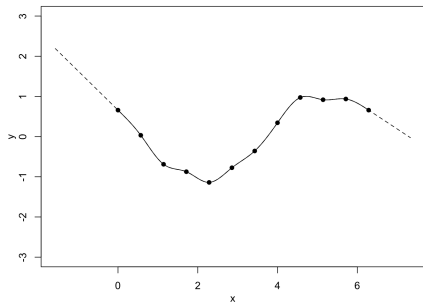
t	$ t $	$ t - 1 $	$f(t)$	Check
-1	1	2	$4 + 1 - 2 = 3$	Constant = y_1 ✓
0	0	1	$4 + 0 - 1 = 3$	$f(0) = y_1$ ✓
0.5	0.5	0.5	$4 + 0.5 - 0.5 = 4$	Midpoint
1	1	0	$4 + 1 - 0 = 5$	$f(1) = y_2$ ✓
2	2	1	$4 + 2 - 1 = 5$	Constant = y_2 ✓

Flat outside $[0, 1]$, linear between — exactly the natural linear spline through $(0, 3)$ and $(1, 5)$. The same approach works for general m : evaluate at each knot to obtain m equations in m unknowns.

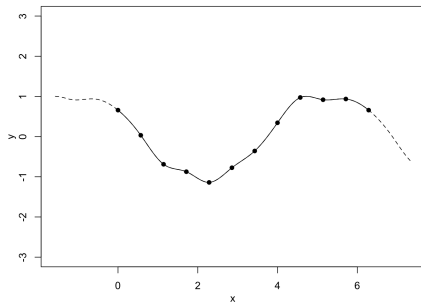
Periodic Splines

For periodic data (e.g., seasonal patterns), an alternative boundary condition:

$$f^{(\ell)}(t_1) = f^{(\ell)}(t_m) \quad \text{for } \ell = 0, 1, \dots, p-1$$



Natural spline



Periodic spline

Caution: Imposing additional restrictions can lead to modelling errors when the full range of x values is not observed.

Key points:

- Without boundary conditions, infinitely many splines interpolate the data
- Natural splines impose linearity (cubic) or constancy (linear) beyond the boundary knots
- A natural spline with m knots has exactly m degrees of freedom
- Natural cubic interpolating splines are unique when knots are at data locations
- Alternative representations express natural splines using $|t - t_i|^p$ basis functions
- Periodic splines are an alternative for cyclic data

Next lecture: Roughness penalties and fitting interpolating splines in R.

Lecture 16: Roughness Penalties and Fitting Interpolating Splines

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Course notes: Chapter 12

MATH3701 Statistical Modelling notes

Why Natural Splines?

We showed that natural splines give a unique interpolant with m degrees of freedom.

There is a deeper justification: natural splines are **optimal** in a precise sense.

Idea: Among all smooth functions that interpolate the data, the natural spline is the *least wiggly*.

The Roughness Penalty

Roughness of a function f is measured by:

$$J_\nu(f) = \int_{-\infty}^{\infty} \left[f^{(\nu)}(t) \right]^2 dt$$

where $f^{(\nu)}$ is the ν th derivative and $\nu \geq 1$.

Interpretation:

- $J_1(f)$: penalises variation in slope (total squared first derivative)
- $J_2(f)$: penalises curvature (total squared second derivative)

A “smooth” function has small $J_\nu(f)$; a “wiggly” function has large $J_\nu(f)$.

The Optimality Result

Proposition (3.2)

Among all functions f satisfying the interpolation conditions $f(t_i) = y_i$, the unique minimiser of $J_\nu(f)$ is a p th-order **natural spline**, where $p = 2\nu - 1$.

Key cases:

ν	$p = 2\nu - 1$	Spline type
1	1	Natural linear spline
2	3	Natural cubic spline

The cubic interpolating spline minimises integrated squared curvature — it is the smoothest interpolant in this sense.

Consequences of the Optimality Result

- **Linear interpolating splines** are shortest-path (piecewise linear) solutions — they minimise $J_1(f)$
- **Cubic interpolating splines** produce curves where knot locations are “invisible” — they minimise $J_2(f)$
- Higher-order splines (quintic, etc.) are rarely motivated in practice

Computational note: Solving the interpolation problem requires inverting an $n \times n$ matrix, which is $O(n^3)$ in general. Specialised algorithms exploit the banded structure to achieve $O(n)$.

Fitting Interpolating Splines in R

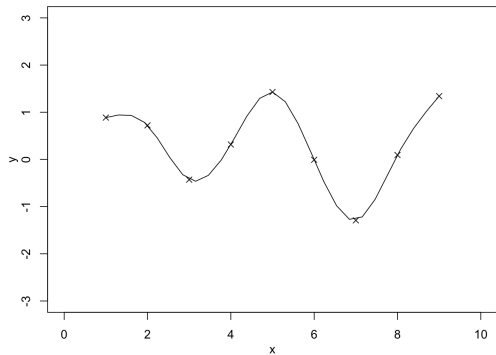
Two functions:

- `spline(x, y, method="natural")`: returns fitted values
- `splinefun(x, y, method="natural")`: returns a *function*

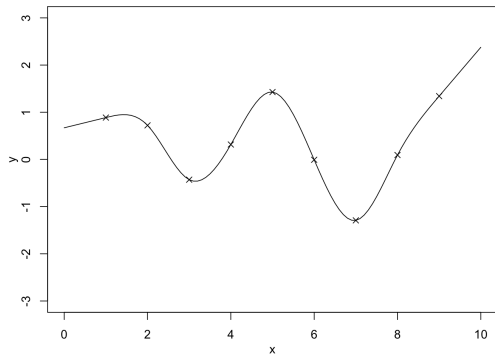
```
# Using spline: returns values at a grid
myfit1 <- spline(x, y, method = "natural")
plot(x, y)
lines(myfit1)

# Using splinefun: returns a function object
myfit2 <- splinefun(x, y, method = "natural")
curve(myfit2, from = 0, to = 10, add = TRUE)
```

R Output: spline vs. splinefun



`spline()` — returns values



`splinefun()` — returns a function

Key difference: `spline()` outputs a list of (x, y) pairs; `splinefun()` outputs a callable function (usable with `curve()`, `integrate()`, etc.).

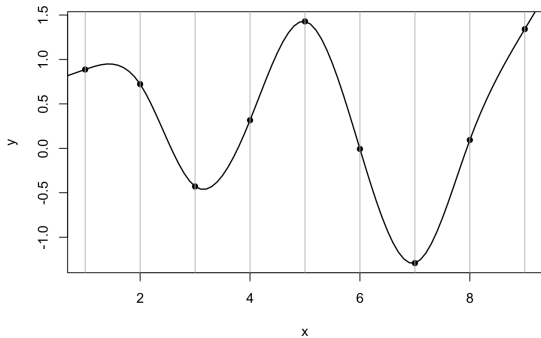
Prediction and Derivatives

```
# Predict at specific locations
spline(x, y, xout = c(2.5, 7.5), method = "natural")

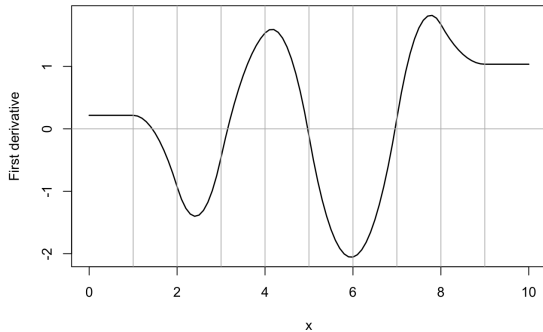
# Or using the function object
myfit2(c(2.5, 7.5))

# Compute derivatives
myfit2(x, deriv = 1) # First derivative
myfit2(x, deriv = 2) # Second derivative
myfit2(x, deriv = 3) # Third derivative
```

Spline Derivatives: Function and First Derivative



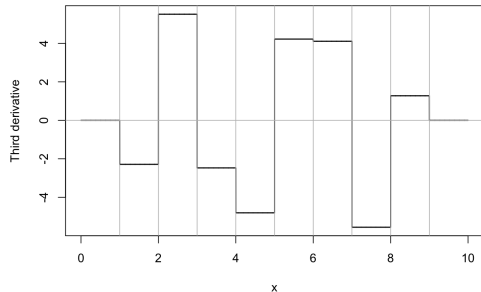
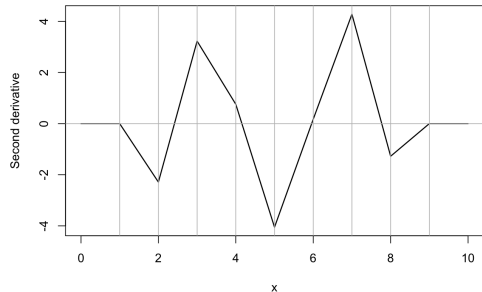
(a) Fitted spline $f(t)$



(b) First derivative $f'(t)$

Both f and f' are continuous everywhere, as expected for a cubic spline.

Spline Derivatives: Second and Third Derivative



(c) Second derivative $f''(t)$ — continuous

(d) Third derivative $f'''(t)$ — jumps at knots

f'' is continuous (piecewise linear); f''' has **jumps at knots**. Beyond the boundary knots, derivatives are constant or zero (natural boundary conditions).

Complete Example: Coal Seam Data

```
# Coal seam data (missing value at x=6)
x <- c(0, 1, 2, 3, 4, 5, 7, 8, 9, 10)
y <- c(35, 41, 48, 47, 65, 67, 52, 51, 68, 75)

# Fit natural cubic interpolating spline
fit <- splinefun(x, y, method = "natural")

# Plot data and fitted curve
plot(x, y, pch = 19, xlab = "Location (km)",
      ylab = "Depth")
curve(fit, from = 0, to = 10, add = TRUE, col = "blue")

# Predict at missing location
fit(6) # Predicted depth at x = 6
```

Key points:

- The roughness penalty $J_\nu(f) = \int [f^{(\nu)}]^2 dt$ measures “wiggleness”
- Natural splines minimise roughness among all interpolating functions
- $\nu = 2$ gives cubic natural splines (minimise curvature)
- In R: `spline()` for values, `splinefun()` for function objects
- `splinefun` can compute derivatives via `deriv` argument
- Cubic spline derivatives are continuous up to order 2; the third derivative jumps at knots

Next lecture: Smoothing splines — what to do when data have measurement error.

Lecture 17: Smoothing Splines

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Course notes: Chapter 13

MATH3701 Statistical Modelling notes

From Interpolation to Smoothing

Interpolation assumes: observations are exact, $y_i = f(t_i)$.

Reality: observations contain noise,

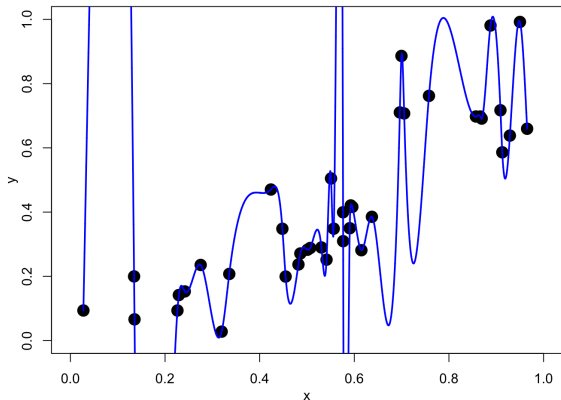
$$y_i = f(t_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Problem with interpolation:

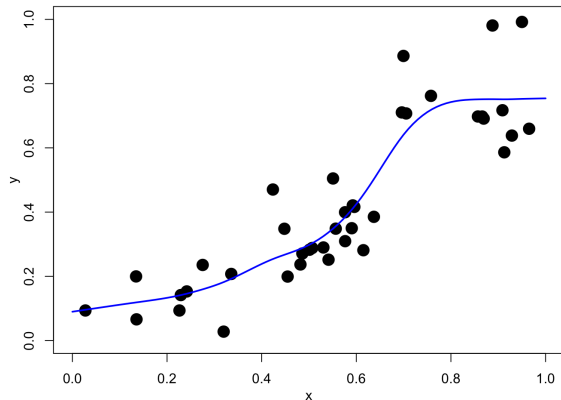
- Fitting every noisy data point exactly produces an overly wiggly curve
- The interpolant tracks noise rather than the underlying signal
- Poor predictions at new locations

⇒ We need to balance **fit to data** against **smoothness**.

Interpolating vs. Smoothing on Noisy Data



(a) Interpolating spline



(b) Smoothing spline

The interpolating spline passes through every point but is overly wiggly. The smoothing spline produces a more realistic estimate of the underlying function.

The Log-Likelihood Perspective

The log-likelihood for the model $y_i = f(t_i) + \epsilon_i$ is:

$$\ell(f; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f(t_i))^2 - n \log \sigma$$

Maximising ℓ without constraints $\Rightarrow$ interpolation ($f(t_i) = y_i$).

This is equivalent to minimising the residual sum of squares:

$$\text{RSS}(f) = \sum_{i=1}^n (y_i - f(t_i))^2$$

Unconstrained minimisation always overfits.

The Penalised Least-Squares Criterion

Idea: Add a roughness penalty to the RSS.

Minimise the **penalised sum of squared residuals**:

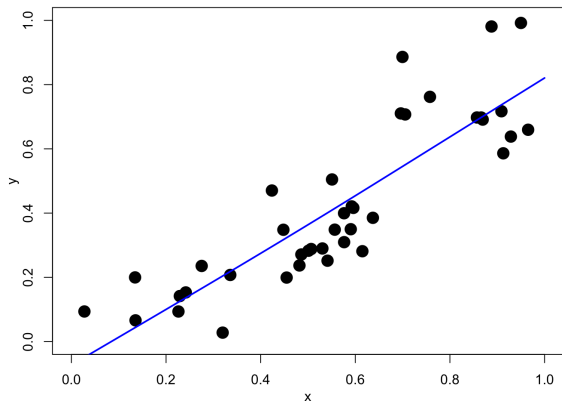
$$R_\nu(f, \lambda) = \underbrace{\sum_{i=1}^n (y_i - f(t_i))^2}_{\text{goodness of fit}} + \underbrace{\lambda J_\nu(f)}_{\text{roughness penalty}}$$

where $J_\nu(f) = \int [f^{(\nu)}(t)]^2 dt$ and $\lambda \geq 0$ is the **smoothing parameter**.

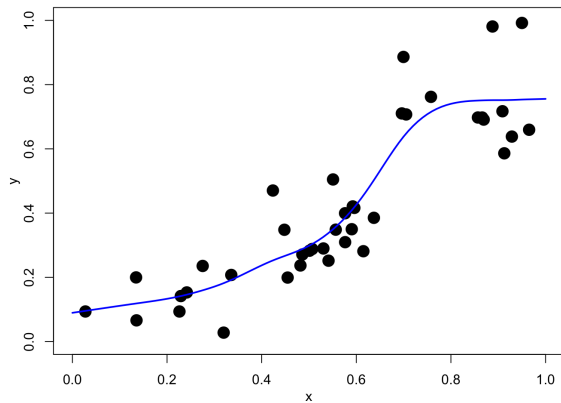
Trade-off controlled by λ :

- $\lambda \rightarrow 0$: no penalty $\Rightarrow$ interpolating spline (overfit)
- $\lambda \rightarrow \infty$: maximal penalty $\Rightarrow$ polynomial regression (underfit)

Effect of λ : Strong and Moderate Smoothing

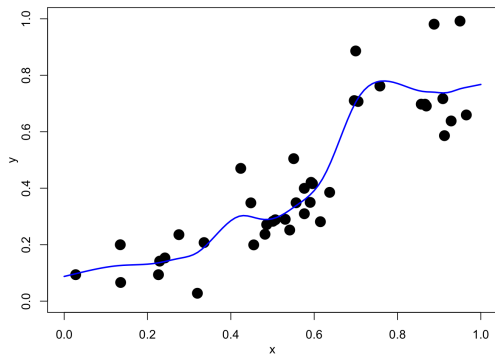


(a) Very large λ — essentially linear

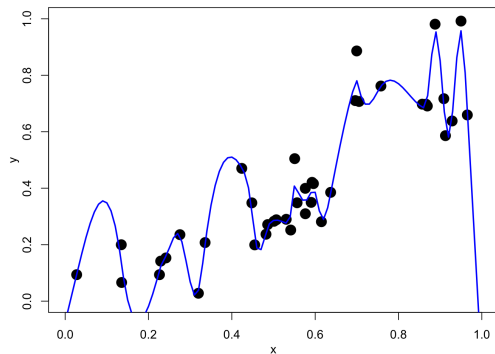


(b) Moderate λ — reasonable fit

Effect of λ : Weak and Very Weak Smoothing



(c) Smaller λ — more flexible



(d) Very small λ — near interpolation

Limiting cases: $\lambda \rightarrow 0$ gives the natural cubic interpolating spline; $\lambda \rightarrow \infty$ gives the OLS line $f(t) = \hat{a}_0 + \hat{a}_1 t$.

The Solution is a Natural Spline

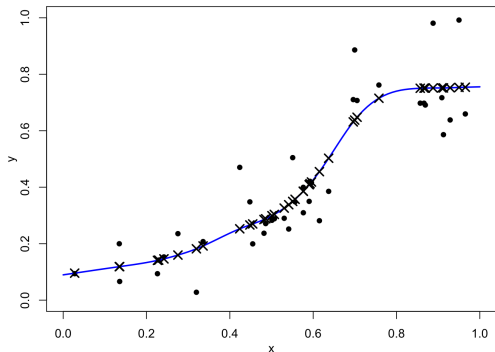
Proposition (4.1)

If $\hat{f}$ minimises $R_\nu(f, \lambda)$ with fitted values $\hat{y}_i = \hat{f}(t_i)$, then $\hat{f}$ is a p th-order natural spline with $p = 2\nu - 1$ and knots at the data locations.

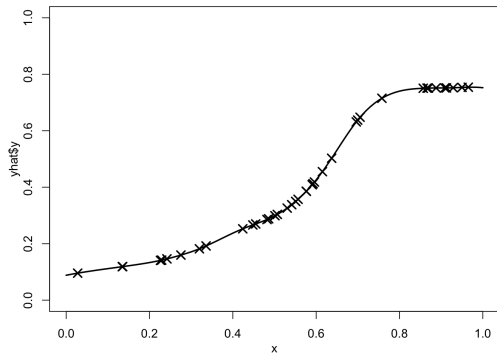
Proof idea: Suppose $\hat{f}^*$ interpolates the same fitted values $(\hat{y}_i)$ but has smaller roughness $J_\nu(\hat{f}^*) < J_\nu(\hat{f})$. Then $R_\nu(\hat{f}^*, \lambda) < R_\nu(\hat{f}, \lambda)$, contradicting optimality.

Consequence: We only need to search over natural splines, which have a finite-dimensional parameterisation.

Proposition 4.1 Illustrated



(a) Smoothing spline: data ($\bullet$) and fitted values ($\times$)



(b) Interpolating spline of the fitted values

The smoothing spline is identical to the interpolating spline of its own fitted values — confirming it is a natural spline with knots at the data locations.

Matrix Formulation

Using the natural spline representation (cubic case, $\nu = 2$):

$$\hat{f}(t) = a_0 + a_1 t + \sum_{i=1}^n b_i |t - t_i|^3, \quad \sum b_i = 0, \quad \sum b_i t_i = 0$$

Fitted values in matrix form:

$$\hat{\mathbf{f}} = \mathbf{K}\mathbf{b} + \mathbf{L}\mathbf{a}$$

where:

- $\mathbf{K}$ is $n \times n$ with (i, k) th element $|t_i - t_k|^3$
- $\mathbf{L}$ is $n \times 2$ with columns $\mathbf{1}$ and $(t_1, \dots, t_n)'$
- $\mathbf{b} = (b_1, \dots, b_n)'$ and $\mathbf{a} = (a_0, a_1)'$

Proposition (4.2)

The roughness penalty can be written as a quadratic form:

Linear ($\nu = 1$): $J_1(f) = -2 \mathbf{b}' \mathbf{K}_1 \mathbf{b}$

Cubic ($\nu = 2$): $J_2(f) = 12 \mathbf{b}' \mathbf{K}_2 \mathbf{b}$

The penalised criterion becomes:

$$R_\nu(f, \lambda) = (\mathbf{y} - \mathbf{Kb} - \mathbf{La})'(\mathbf{y} - \mathbf{Kb} - \mathbf{La}) + \lambda^* \mathbf{b}' \mathbf{Kb}$$

subject to $\mathbf{L}'\mathbf{b} = \mathbf{0}$, where $\lambda^* = c_\nu \lambda$.

The Solution

Proposition (4.3)

The penalised least-squares solution is:

$$\begin{pmatrix} \hat{\mathbf{a}} \\ \hat{\mathbf{b}} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{L}' \\ \mathbf{L} & \mathbf{K} + \lambda^* \mathbf{I}_n \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{0} \\ \mathbf{y} \end{pmatrix}$$

Observations:

- When $\lambda^* = 0$: reduces to interpolation ($\hat{f}(t_i) = y_i$)
- When $\lambda^* \rightarrow \infty$: $\hat{\mathbf{b}} \rightarrow \mathbf{0}$, leaving only $\hat{f}(t) = \hat{a}_0 + \hat{a}_1 t$
- The matrix equation simultaneously enforces the natural spline constraints and minimises the penalised criterion

Key points:

- Smoothing splines balance fit to data against roughness via λ
- The penalised criterion is $R_\nu(f, \lambda) = \text{RSS} + \lambda J_\nu(f)$
- $\lambda = 0$ gives interpolation; $\lambda \rightarrow \infty$ gives polynomial regression
- The minimiser is always a natural spline with knots at data locations
- The solution is obtained by solving a linear system involving $\mathbf{K}$, $\mathbf{L}$, and λ

Next lecture: Fitting smoothing splines in R and choosing the smoothing parameter.

Lecture 18: Choosing the Smoothing Parameter — CV, GCV and Effective Degrees of Freedom

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Smoothing Splines in R

The `smooth.spline()` function fits cubic smoothing splines:

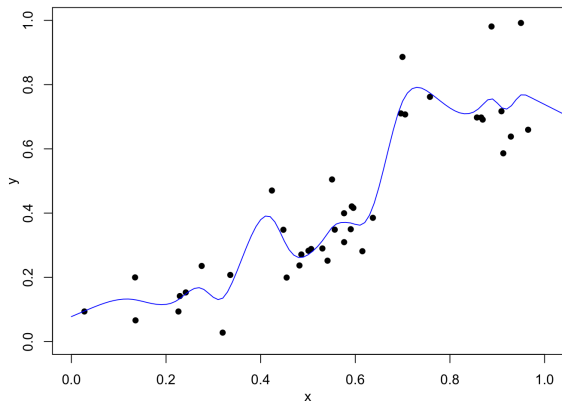
```
# Fit with specified lambda
fit1 <- smooth.spline(x, y, lambda = 0.00001) # Nearly interpolating
fit2 <- smooth.spline(x, y, lambda = 1)       # Very smooth

# Plot
plot(x, y, pch = 19)
lines(fit1, col = "blue")
lines(fit2, col = "red")

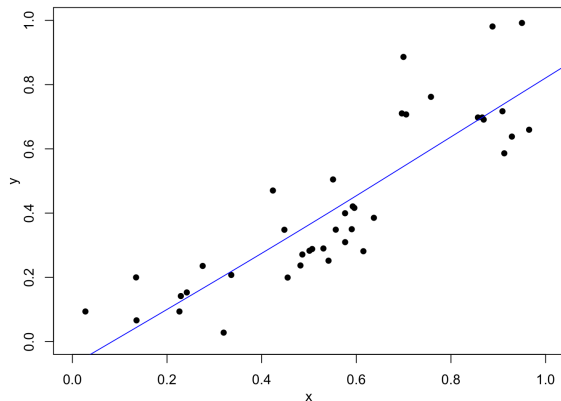
# Predict at new locations
predict(fit1, x = c(2.5, 7.5))
```

Note: If `lambda` is not specified, the optimal value is selected by generalised cross-validation (later in this lecture).

Example: Change-Point Data with Different λ



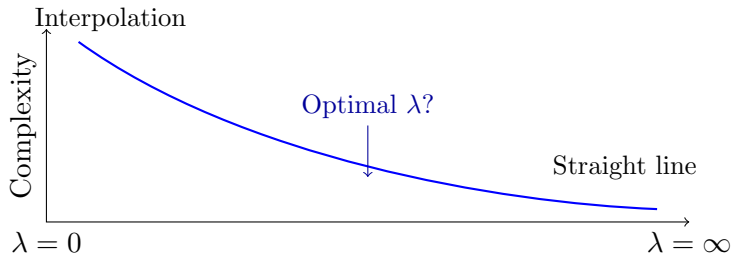
(a) Small $\lambda = 0.00001$



(b) Large $\lambda = 1$

With small λ , the spline captures the change-point but may track noise. With large λ , the spline is too smooth and misses the change-point entirely.

Effect of λ : Visual Summary



The fundamental question: How do we choose λ ?

Strategy 1: Training/Test Split

Idea: Partition data indices into training set I_1 and test set I_2 .

Procedure:

- ① Fit the smoothing spline using only training data $\{(t_i, y_i) : i \in I_1\}$
- ② Evaluate prediction quality on test data:

$$Q_{I_1:I_2}(\lambda) = \sum_{i \in I_2} \left(y_i - \hat{f}_{\lambda, I_1}(t_i) \right)^2$$

- ③ Choose λ minimising $Q_{I_1:I_2}(\lambda)$

Limitation: Wastes data — the test set is not used for fitting.

Strategy 2: Leave-One-Out Cross-Validation

Idea: Use each observation in turn as a single-point test set.

Ordinary Cross-Validation (OCV):

$$Q_{\text{OCV}}(\lambda) = \frac{1}{n} \sum_{j=1}^n \left(y_j - \hat{f}_{\lambda, -j}(t_j) \right)^2$$

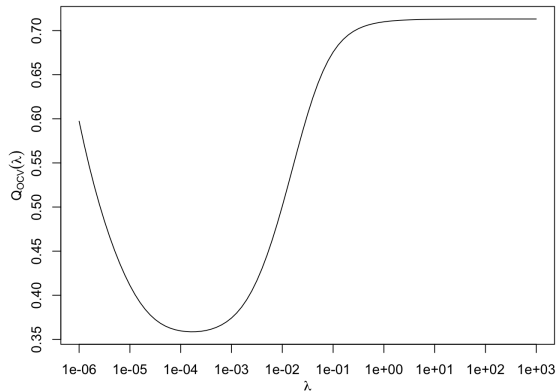
where $\hat{f}_{\lambda, -j}$ is the smoothing spline fitted to all data *except* observation j .

Choose λ to minimise $Q_{\text{OCV}}(\lambda)$.

Apparent problem: Requires fitting n separate smoothing splines.

$\Rightarrow$ A computational trick avoids this (see below).

Example: An OCV Curve



$Q_{OCV}(\lambda)$ vs. λ (log scale) for a simulated change-point dataset. There is a clear minimum here, but no theoretical guarantee of a unique turning point in general.

The Smoothing Matrix

From the solution of the penalised criterion, the fitted values at the data sites are

$$\hat{\mathbf{f}} = \mathbf{K}\hat{\mathbf{b}} + \mathbf{L}\hat{\mathbf{a}} = (\mathbf{K} \quad \mathbf{L}) \begin{pmatrix} \mathbf{M}_{\lambda}^{12} \\ \mathbf{M}_{\lambda}^{22} \end{pmatrix} \mathbf{y}, \quad \mathbf{M}_{\lambda} = \begin{pmatrix} \mathbf{0} & \mathbf{L}' \\ \mathbf{L} & \mathbf{K} + \lambda^* \mathbf{I}_n \end{pmatrix},$$

where $\mathbf{M}_{\lambda}^{12}$, $\mathbf{M}_{\lambda}^{22}$ are the corresponding blocks of $\mathbf{M}_{\lambda}^{-1}$.

Smoothing matrix:

$$\boxed{\hat{\mathbf{f}} = \mathbf{S}_{\lambda} \mathbf{y}}$$

$\mathbf{S}_{\lambda}$ is $n \times n$, symmetric and positive definite for $\lambda > 0$.

The fitted values are a **linear** function of the data.

Effective Degrees of Freedom

Analogy with least squares: $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, and

$$\text{trace}(\mathbf{H}) = \text{trace}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) = \text{trace}(\mathbf{I}_p) = p.$$

$\mathbf{S}_\lambda$ plays the role of the hat matrix, so define

$$\text{edf}_\lambda = \text{trace}(\mathbf{S}_\lambda)$$

Limits:

- $\lambda \rightarrow \infty$: polynomial regression, $\text{edf}_\infty = \nu$
- $\lambda = 0$: interpolating spline, $\text{edf}_0 = n$

More smoothing $\Rightarrow$ fewer effective degrees of freedom.

Computing OCV Quickly, and GCV

OCV shortcut: no need for n separate fits:

$$Q_{\text{OCV}}(\lambda) = \frac{1}{n} \sum_{j=1}^n \left(\frac{y_j - \hat{f}_\lambda(t_j)}{1 - s_{jj}} \right)^2,$$

where $\hat{f}_\lambda$ is fitted to the full data and s_{jj} is the j th diagonal element of $\mathbf{S}_\lambda$.

Generalised cross-validation: replace each s_{jj} by the average $\frac{1}{n}\text{trace}(\mathbf{S}_\lambda) = \frac{1}{n}\text{edf}_\lambda$:

$$Q_{\text{GCV}}(\lambda) = \frac{\frac{1}{n} \sum_{j=1}^n \left(y_j - \hat{f}_\lambda(t_j) \right)^2}{\left(1 - \frac{1}{n}\text{edf}_\lambda \right)^2}$$

GCV is numerically more stable, and is what `smooth.spline()` and `mgcv` use by default.

Key points:

- `smooth.spline()` in R fits cubic smoothing splines
- The smoothing parameter λ controls the trade-off between fit and smoothness
- Small λ : wiggly curve close to data
- Large λ : smooth curve (approaches a straight line)
- Training/test split: simple but wastes data
- Leave-one-out CV: uses all data; shortcut via s_{jj} avoids n fits
- Smoothing matrix $\hat{\mathbf{f}} = \mathbf{S}_\lambda \mathbf{y}$; $\text{edf}_\lambda = \text{trace}(\mathbf{S}_\lambda)$ lies between ν and n
- GCV replaces s_{jj} by edf_λ/n

Next lecture: Generalised Additive Models — extending GLMs with smooth functions.

Lecture 19: Generalised Additive Models

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So far: Single explanatory variable, normal errors.

$$y_i = f(t_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Now: Multiple explanatory variables, possibly non-normal responses.

Generalised Additive Models (GAMs) combine:

- 1 GLM framework (exponential family, link functions)
- 2 Non-parametric smooth functions of predictors

The GAM Framework

A GAM has three components, extending the GLM:

1. **Random component:** Y belongs to the exponential family with parameters θ and ϕ .
2. **Systematic component** (non-linear predictor):

$$\eta = \sum_{j=1}^p f_j(x_j)$$

where each f_j is a *smooth function* (not necessarily linear).

3. **Link function:**

$$\eta = g(\mu), \quad \mu = g^{-1}(\eta)$$

	GLM	GAM
Predictor	$\eta = \sum \beta_j x_j$	$\eta = \sum f_j(x_j)$
Each term	Linear: $\beta_j x_j$	Smooth: $f_j(x_j)$
Parameters	β (finite)	Functions f_j
Estimation	Maximum likelihood	Penalised likelihood
Response	Exponential family	Exponential family

Note: A GLM is a special case of a GAM where each $f_j(x_j) = \beta_j x_j$.

GAMs can also include parametric (linear) terms alongside smooth terms.

Penalised Deviance

For non-Gaussian data: Replace penalised least squares with **penalised deviance**.

With m smooth terms $f_1, \dots, f_m$ and parametric coefficients β :

$$R_\nu = D(\mathbf{y}, f_1, \dots, f_m, \beta) + \sum_{h=1}^m \lambda_h J_\nu(f_h)$$

where:

- $D(\mathbf{y}, \dots)$ is the deviance (from GLM theory)
- λ_h is the smoothing parameter for the h th smooth term
- $J_\nu(f_h)$ is the roughness penalty for f_h

Each smooth term has its *own* smoothing parameter λ_h , chosen by GCV.

GAMs in R: The mgcv Package

Key function: `gam()` from the `mgcv` package.

```
library(mgcv)

# Fit a GAM with smooth terms
fit <- gam(y ~ s(x1, k = 10) + s(x2, k = 10),
           family = "gaussian")

# With specified smoothing parameter
fit <- gam(y ~ s(x1, k = 10, sp = 3.5))

# Without sp: lambda chosen by GCV automatically
fit <- gam(y ~ s(x1, k = 10))
```

`s()` specifies a smooth term (cubic smoothing spline). The argument `k` sets the maximum dimensionality of the spline basis.

Key gam() Output

```
# Model summary
summary.gam(fit)

# Analysis of deviance
anova.gam(fit)

# Useful components
fit$fitted.values  # Fitted values
fit$sp             # Smoothing parameter(s)
fit$gcv.ubre       # GCV criterion value
sum(fit$hat)       # Total effective df
```

Important: The **edf** (effective degrees of freedom) for each smooth term indicates whether the relationship is approximately linear ($\text{edf} \approx 1$) or genuinely non-linear ($\text{edf} \gg 1$).

Example: Coronary Heart Disease

Data: South African CHD case-control study ($n = 462$).

Variables:

- Response: CHD status (binary: 0/1)
- Explanatory: tobacco consumption, age, family history

GLM (linear effects):

$$\text{logit}(\mathbb{P}(\text{CHD})) = \beta_0 + \beta_1 \cdot \text{tobacco} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{famhist}$$

GAM (smooth effects):

$$\text{logit}(\mathbb{P}(\text{CHD})) = \beta_0 + f_1(\text{tobacco}) + f_2(\text{age}) + \beta_3 \cdot \text{famhist}$$

CHD Example: GLM Fit

```
hr <- read.table("SAheart.txt", sep = ",",  
                 header = TRUE, row.names = 1)  
attach(hr)  
  
glm1 <- glm(chd ~ tobacco + age + famhist,  
            family = "binomial")
```

	Estimate	Std. Error	z	p -value
(Intercept)	-3.621	0.445	-8.14	< 0.001
tobacco	0.083	0.026	3.23	0.001
age	0.049	0.009	5.16	< 0.001
famhist (Present)	0.975	0.220	4.43	< 0.001

All variables significant. **However:** assumes logit is *linear* in tobacco and age.

CHD Example: GAM Fit

```
library(mgcv)
gam1 <- gam(chd ~ s(tobacco, k = 20) + s(age, k = 20)
            + famhist, family = "binomial")
summary.gam(gam1)
```

Smooth term results:

Term	edf	Ref.df	χ^2	<i>p</i> -value
s(tobacco)	6.08	7.57	17.89	0.018
s(age)	1.00	1.00	24.11	< 0.001

Key findings:

- **Tobacco:** edf ≈ 6 — genuinely non-linear relationship
- **Age:** edf ≈ 1 — approximately linear (GAM agrees with GLM)

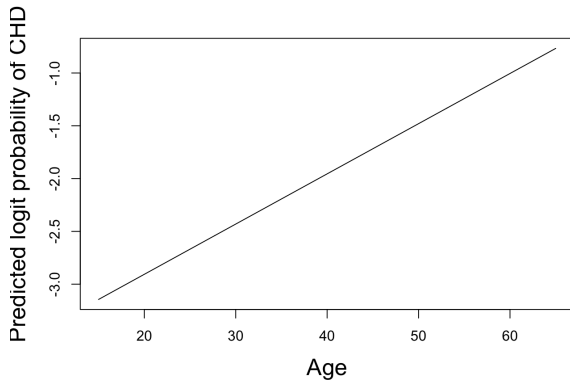
What does edf tell us?

edf	Interpretation
≈ 1	Approximately linear relationship
2–3	Mildly non-linear (e.g., quadratic)
> 5	Substantially non-linear

In the CHD example:

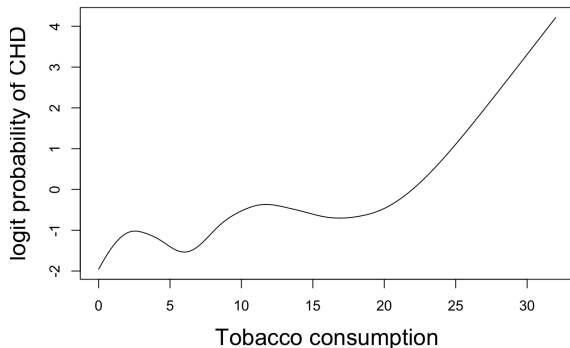
- Age effect is essentially linear — a GLM is adequate for this term
- Tobacco effect is non-linear — the smooth function captures structure that a linear term misses

CHD Example: Predicted Effect of Age



Predicted logit of CHD vs. age (tobacco = 0, no family history). Nearly linear (edf ≈ 1).

CHD Example: Predicted Effect of Tobacco



Predicted logit of CHD vs. tobacco consumption (age = 40, no family history). The non-linear pattern (edf ≈ 6) may reflect unmeasured confounders correlated with tobacco use.

Plotting Smooth Effects in R

```
# Age effect
newdat1 <- data.frame(age = seq(15, 65, by = 0.1),
                      tobacco = 0,
                      famhist = "Absent")
pred1 <- predict.gam(gam1, newdata = newdat1)
plot(newdat1$age, pred1, type = "l",
     xlab = "Age", ylab = "logit P(CHD)")

# Tobacco effect
newdat2 <- data.frame(tobacco = seq(0, 32, by = 0.1),
                      age = 40,
                      famhist = "Absent")
pred2 <- predict.gam(gam1, newdata = newdat2)
plot(newdat2$tobacco, pred2, type = "l",
     xlab = "Tobacco", ylab = "logit P(CHD)")
```

Key points:

- GAMs extend GLMs by replacing linear terms $\beta_j x_j$ with smooth functions $f_j(x_j)$
- Estimation uses penalised deviance with GCV-selected smoothing parameters
- The `mgcv` package in R provides `gam()` for fitting
- Effective degrees of freedom (edf) indicate the degree of non-linearity
- $\text{edf} \approx 1$: linear (GAM reduces to GLM for that term)
- GAMs can mix smooth and parametric terms (e.g., categorical variables)
- Partial effect plots show how each predictor relates to the response

This concludes the module. GAMs provide a flexible framework for modelling non-linear relationships within the GLM paradigm.

Lecture 20: Introduction to Survival Analysis

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What is Survival Analysis?

A **survival time** $T > 0$ is the time until a well-defined event (the **endpoint**).

Applications:

- Time from surgery to death or relapse (medicine)
- Time to failure of mechanical components (reliability)
- Time until criminal reoffends (criminology)
- Time to insurance claim (actuarial science)

Two distinguishing features:

- 1 **Skewness:** $T > 0$, typically right-skewed
- 2 **Censoring:** exact T often unobserved

Motivating Example: AVR

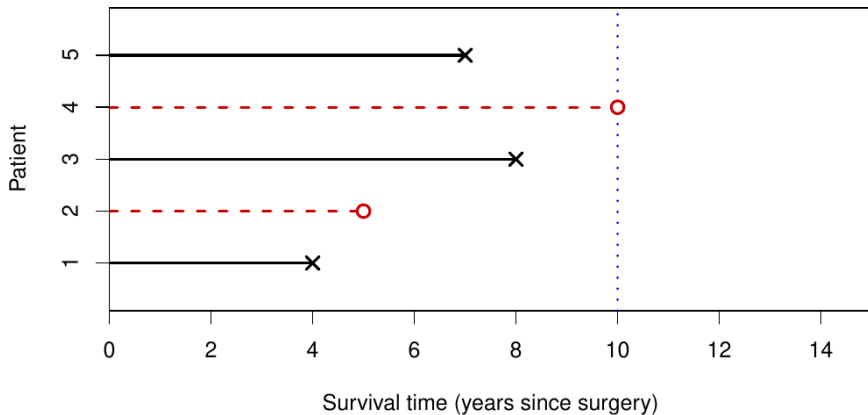
5-patient aortic valve replacement (AVR) study.

Patient	Obs. time	Event	Status
1	4	Yes	Death at year 4
2	5	No	Lost to follow-up
3	8	Yes	Death at year 8
4	10	No	Study ended
5	7	Yes	Death at year 7

Patients 2 and 4 are **right-censored**: we know only $T > t^*$.

Problem: How to estimate $\mathbb{P}(T > 6)$ from this data?

Visualizing the AVR Data



Deaths (x) and censored observations (o) for the 5 AVR patients. The dashed line marks the end of the 10-year study.

Why Naive Approaches Fail

Three naive estimates of $\mathbb{P}(T > 6)$:

- ① Ignore censoring, treat as survivors: $\hat{p} = 4/5 = 0.8$
⇒ Optimistic — **overestimates** survival
- ② Treat censored as deaths: $\hat{p} = 3/5 = 0.6$
⇒ Pessimistic — **underestimates** survival
- ③ Exclude censored observations: $\hat{p} = 2/3 \approx 0.67$
⇒ Discards information; gives $\mathbb{P}(T > 9) \approx 0$ even though patient 4 survived 10 years

All three are unsatisfactory. Part III develops correct methods.

Types of Censoring

Right censoring (most common): observed time t^* is less than the true T :

$$T > t^*, \quad \delta = 0.$$

Left censoring: event occurred *before* t^* ($T < t^*$). Example: disease present at first diagnosis but onset time unknown.

Interval censoring: $T \in (a, b)$. Example: annual screenings.

This course: right censoring throughout.

Observed Data Structure

For each individual $i = 1, \dots, n$, we observe:

$$t_i = \min(T_i, C_i), \quad \delta_i = \mathbf{1}(T_i \leq C_i).$$

- t_i : observed time (event or censoring)
- $\delta_i = 1$: event observed; $\delta_i = 0$: censored
- C_i : censoring time (assumed independent of T_i)

In R:

```
library(survival)
t      <- c(4, 5, 8, 10, 7)
delta  <- c(1, 0, 1, 0, 1)
Surv(t, delta)  # '+' denotes censored obs
```

Core Assumptions

- ① **Uninformative censoring:** $T_i \perp C_i$. No systematic tendency for sicker patients to be censored sooner.
- ② **Eventual event:** $\mathbb{P}(T < \infty) = 1$.
- ③ **i.i.d. sample:** $T_1, \dots, T_n$ are independent and identically distributed.

When explanatory variables are present (regression models), Assumption 3 is replaced by: survival distribution depends on explanatory variables through a specified model.

Key points:

- Survival analysis deals with time-to-event data
- Censoring is the central methodological challenge
- Right censoring: $T > t^*$; event indicator $\delta_i \in \{0, 1\}$
- Uninformative censoring: censoring times independent of survival times
- Naive analyses (ignore/treat as deaths/exclude) all give biased estimates
- The `Surv()` object in R encodes the data structure

Next lecture: The survival function $S(t)$, hazard $h(t)$, and cumulative hazard $H(t)$.

Lecture 21: Key Functions: Survival, Hazard, Cumulative Hazard

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Three Key Functions

For a continuous survival time $T > 0$, three functions completely characterise the distribution:

Function	Definition	Interpretation
Survival	$S(t) = \mathbb{P}(T > t)$	Probability of surviving past t
Hazard	$h(t) = f(t)/S(t)$	Instantaneous risk rate at t
Cumulative hazard	$H(t) = \int_0^t h(s) ds$	Accumulated risk up to t

Key relationship:

$$S(t) = e^{-H(t)}, \quad H(t) = -\log S(t).$$

The Survival Function

$$S(t) = \mathbb{P}(T > t) = 1 - F(t), \quad S(0) = 1, \quad S(\infty) = 0.$$

- Non-increasing step function (when estimated from data)
- $S'(t) = -f(t)$, so $f(t) = -S'(t)$

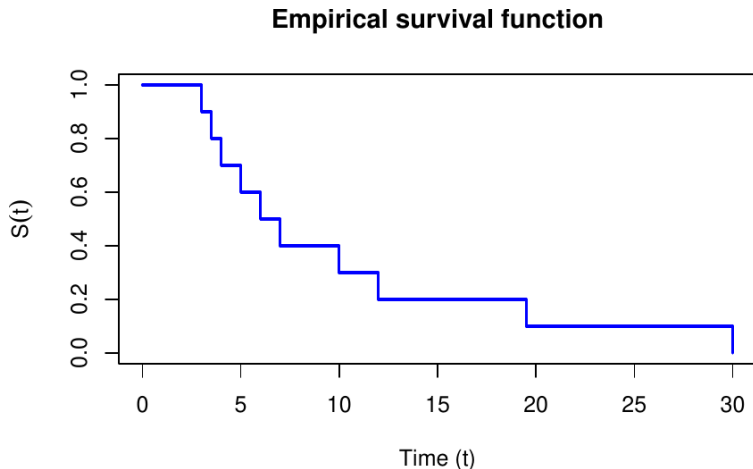
Empirical survival function (no censoring):

$$\hat{S}(t) = \frac{\#\{i : t_i > t\}}{n}.$$

This is unbiased: $\mathbb{E}[\hat{S}(t)] = S(t)$.

With censoring, use the Kaplan–Meier estimator (Lecture 22).

Example: Empirical Survival Function



Empirical survival function for 10 uncensored survival times — a step function dropping by $1/n$ at each observed time.

The Hazard Function

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{\mathbb{P}(T < t + \delta t \mid T \geq t)}{\delta t} = -\frac{d}{dt} \log S(t).$$

- $h(t) \geq 0$; not a probability density (need not integrate to 1)
- $h(t) \delta t \approx \mathbb{P}(\text{event in } [t, t + \delta t) \mid \text{alive at } t)$

Shapes of the hazard:

- **Constant:** exponential distribution
- **Increasing:** ageing / wear-out (Weibull with $\gamma > 1$)
- **Decreasing:** infant mortality (Weibull with $\gamma < 1$)
- **Hump-shaped:** recovery followed by late relapse risk

Cumulative Hazard and the Key Relationship

$$H(t) = \int_0^t h(s) ds \quad \Longleftrightarrow \quad S(t) = e^{-H(t)}.$$

Proof: Since $h(t) = -d \log S(t)/dt$, integrating:

$$H(t) = -\log S(t) \quad \Rightarrow \quad S(t) = e^{-H(t)}.$$

Summary table:

$$f(t) = h(t)e^{-H(t)}, \quad h(t) = -\frac{S'(t)}{S(t)}, \quad S(t) = e^{-H(t)}.$$

Knowing any one of S , h , H determines the others.

Example: Exponential Distribution

$T \sim \text{Exp}(\lambda)$: constant hazard $h(t) = \lambda$.

Function	Formula
Density	$f(t) = \lambda e^{-\lambda t}$
Survival	$S(t) = e^{-\lambda t}$
Hazard	$h(t) = \lambda$ (constant)
Cumulative hazard	$H(t) = \lambda t$ (linear)

Memoryless property:

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t).$$

Past survival gives no information about future risk.

Mean: $\mathbb{E}[T] = 1/\lambda$; Median: $\log(2)/\lambda$.

Mean and Median Survival

Mean:

$$\mathbb{E}[T] = \int_0^\infty t f(t) dt = \int_0^\infty S(t) dt$$

(the second equality follows from integration by parts).

Median $\mu_{1/2}$: the time at which half the population has experienced the event:

$$S(\mu_{1/2}) = 0.5.$$

p th quantile t_p : $S(t_p) = 1 - p$.

The median is often preferred in survival analysis — it is less sensitive to the (unknown) tail behaviour of $S(t)$.

Key points:

- Three equivalent characterisations: $S(t)$, $h(t)$, $H(t)$
- Key relationship: $S(t) = e^{-H(t)}$, $h(t) = -S'(t)/S(t)$
- Exponential: constant hazard — the memoryless distribution
- Mean survival: $\mathbb{E}[T] = \int_0^\infty S(t) dt$
- Median: $S(\mu_{1/2}) = 0.5$ — preferred over mean
- Empirical $\hat{S}(t) = \#\{t_i > t\}/n$ (no censoring)

Next lecture: The Kaplan–Meier estimator — non-parametric survival estimation with censoring.

Lecture 22: Exploratory Analysis — Kaplan–Meier Estimator

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Motivation

The empirical survival function $\hat{S}(t) = \#\{t_i > t\}/n$ is **biased** when data include censored observations.

Solution: The **Kaplan–Meier (KM) estimator** is the non-parametric MLE of $S(t)$ that correctly accounts for right censoring.

Role: KM is primarily an **exploratory / descriptive** tool. It makes no parametric assumptions and does not adjust for explanatory variables. For regression modelling, use Chapters 21–25.

KM is the starting point for any survival analysis.

The Product-Limit Formula

Let $t_{(1)} < \cdots < t_{(k)}$ be the ordered distinct event times. At each $t_{(j)}$:

- n_j = number at risk immediately before $t_{(j)}$
- d_j = number of events at $t_{(j)}$

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \frac{n_j - d_j}{n_j}.$$

- Step function: drops only at observed event times
- Censored observations contribute to risk sets but do not cause drops
- Each factor $(n_j - d_j)/n_j$ = conditional probability of surviving past $t_{(j)}$

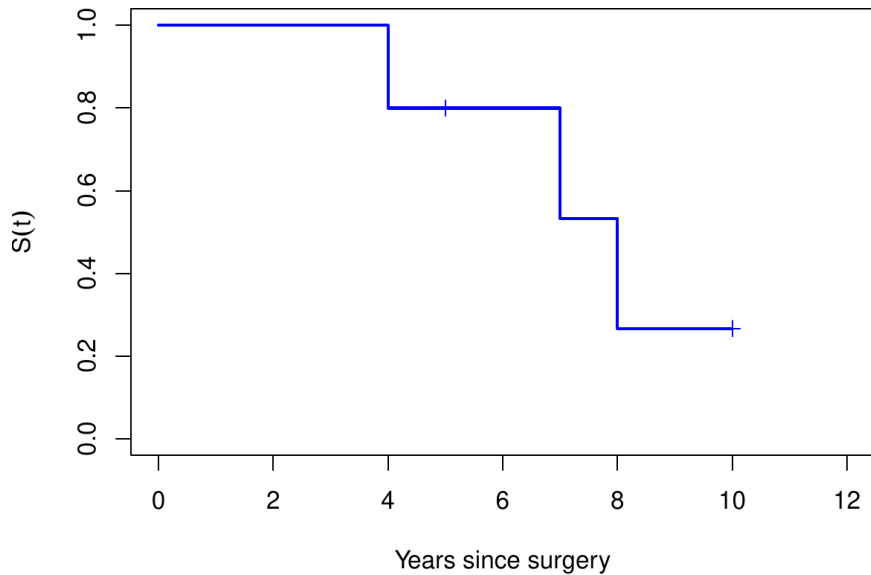
KM Example: AVR Data

Data: (4, 1), (5, 0), (7, 1), (8, 1), (10, 0). Event times: 4, 7, 8.

$t_{(j)}$	n_j	d_j	$(n_j - d_j)/n_j$	$\hat{S}(t_{(j)})$
4	5	1	4/5	0.800
7	3	1	2/3	0.533
8	2	1	1/2	0.267

At $t = 7$: patient 2 was censored at $t = 5$, so only 3 are at risk.
 $\hat{S}(6) = 0.800$; $\hat{S}(9) = 0.267$; Estimated median = 7 years.

The KM Curve for the AVR Data



Greenwood's Formula

The KM estimator has variance estimated by **Greenwood's formula**:

$$\widehat{\text{Var}}(\widehat{S}(t)) = \widehat{S}(t)^2 \sum_{j: t_{(j)} \leq t} \frac{d_j}{n_j(n_j - d_j)}.$$

Pointwise 95% CI (linear scale):

$$\widehat{S}(t) \pm 1.96 \sqrt{\widehat{\text{Var}}(\widehat{S}(t))}.$$

Better: use the **log-log transformation** (automatic in `survfit()`), which always produces CIs in $(0, 1)$.

KM in R

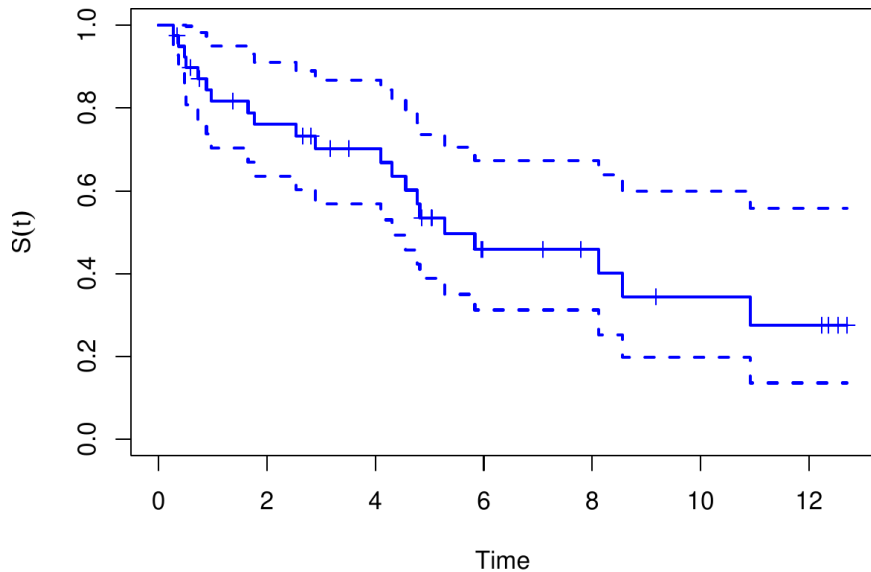
```
library(survival)
t      <- c(4, 5, 8, 10, 7)
delta <- c(1, 0, 1, 0, 1)

# Fit and plot
km <- survfit(Surv(t, delta) ~ 1)
plot(km, conf.int = TRUE,
     mark.time = TRUE,      # tick marks for censored obs
     col = "blue", lwd = 2,
     xlab = "Time", ylab = "S(t)")

# Summary table
summary(km)

# Two groups
km2 <- survfit(Surv(t, delta) ~ group)
plot(km2, col = c("blue", "red"))
```


Output: KM Curve with Confidence Band



Key points:

- KM estimator: non-parametric MLE of $S(t)$ with censoring
- Product-limit formula: $\hat{S}(t) = \prod_{j: t_{(j)} \leq t} (n_j - d_j) / n_j$
- Risk set: individuals still at risk just before each event time
- Censored obs contribute to risk sets but do not cause drops
- Greenwood's formula gives approximate variance
- `survfit()` in R: confidence bands use log-log transform
- KM is descriptive — use regression models to adjust for explanatory variables

Next lecture: Parametric survival distributions — the exponential distribution.

Lecture 23: Parametric Families I — Exponential Distribution

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Course notes: Chapter 18

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Parametric Survival Models

A **parametric** model specifies $S(t; \theta)$ up to a finite-dimensional parameter θ .

Advantages:

- Full inference on $S(t)$, $h(t)$, quantiles
- Maximum likelihood theory applies
- Efficient if model is correctly specified

Disadvantages:

- Biased if model form is wrong

Most important parametric families:

- Exponential (constant hazard)
- Weibull (monotone hazard)
- Gompertz (exponentially growing hazard)

Exponential Distribution

$T \sim \text{Exp}(\lambda), \lambda > 0:$

$$h(t) = \lambda, \quad S(t) = e^{-\lambda t}, \quad f(t) = \lambda e^{-\lambda t}.$$

Properties:

- **Constant hazard:** risk does not change over time
- **Memoryless:** $\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t)$
- Mean: $1/\lambda$; Median: $\log(2)/\lambda \approx 0.693/\lambda$
- Single parameter: easy to fit and interpret

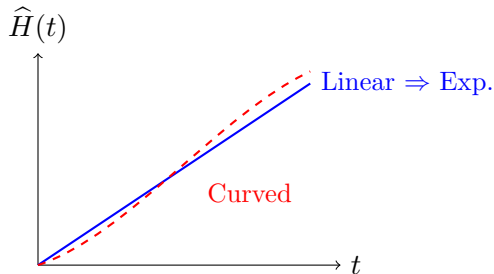
Limitation: constant hazard is rarely realistic. The Weibull generalises this.

Graphical Assessment: Exponential

Exponential check: $H(t) = \lambda t$, so a plot of

$$\hat{H}(t) = -\log \hat{S}(t) \quad \text{vs} \quad t$$

should be **linear through the origin**.



Slope of the linear fit estimates λ .

MLE for the Exponential Model

With r events and total observed time $v = \sum_{i=1}^n t_i$:

$$\ell(\lambda) = r \log \lambda - \lambda v.$$

Setting $\ell'(\lambda) = 0$:

$$\hat{\lambda} = \frac{r}{v} = \frac{\text{events}}{\text{total person-time}}.$$

Variance: $\widehat{\text{Var}}(\hat{\lambda}) \approx \hat{\lambda}^2/r$, so $\widehat{\text{sd}}(\hat{\lambda}) \approx \hat{\lambda}/\sqrt{r}$.

95% CI (log-scale for positivity): $\hat{\lambda} \exp(\pm 1.96/\sqrt{r})$.

Example: AVR Data

Observed times: 4, 5, 8, 10, 7 years; event indicators: 1, 0, 1, 0, 1.

$$r = 3, \quad v = 4 + 5 + 8 + 10 + 7 = 34 \text{ patient-years.}$$

$$\hat{\lambda} = \frac{3}{34} \approx 0.088 \text{ deaths/year.}$$

Estimated mean survival: $1/\hat{\lambda} \approx 11.3$ years.

Estimated median: $\log(2)/\hat{\lambda} \approx 7.9$ years.

```
library(survival)
t      <- c(4, 5, 8, 10, 7)
delta  <- c(1, 0, 1, 0, 1)
fit     <- survreg(Surv(t, delta) ~ 1, dist = "exponential")
lambda_hat <- exp(-coef(fit))
```

Key points:

- Parametric models specify $S(t)$ up to finitely many parameters
- Exponential: $h(t) = \lambda$ (constant), $S(t) = e^{-\lambda t}$
- Memoryless property: risk is the same regardless of time survived
- MLE: $\hat{\lambda} = r/v$ (events per unit of total follow-up)
- Graphical check: $\hat{H}(t)$ vs t should be linear for exponential

Next lecture: Weibull distribution and graphical assessment.

Lecture 24: Parametric Families II — Weibull and Graphical Assessment

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Weibull Distribution

$T \sim \text{Weibull}(\lambda, \gamma)$, $\lambda > 0$ (scale), $\gamma > 0$ (shape):

$$h(t) = \lambda \gamma t^{\gamma-1}, \quad S(t) = e^{-\lambda t^\gamma}.$$

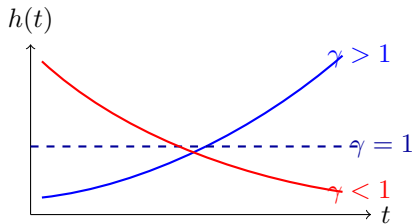
Shape parameter γ :

- $\gamma < 1$: decreasing hazard (infant mortality / early failure)
- $\gamma = 1$: constant hazard (reduces to exponential)
- $\gamma > 1$: increasing hazard (ageing / wear-out)

The Weibull is the most widely used parametric survival distribution because:

- ① Flexible hazard shapes
- ② Satisfies **proportional hazards** property (Lecture 27)

Weibull: Hazard Shapes



All three curves share the same mean (for appropriate λ). The shape γ determines how risk evolves over time.

Graphical Assessment: Weibull

$H(t) = \lambda t^\gamma$, so taking logs:

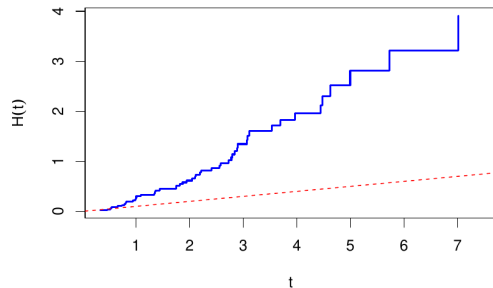
$$\log H(t) = \log \lambda + \gamma \log t.$$

Weibull check: Plot $\log \hat{H}(t) = \log(-\log \hat{S}(t))$ vs $\log t$.

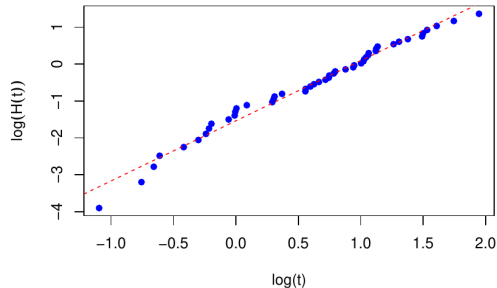
- Should be **linear** if Weibull is appropriate
- Slope $\approx \hat{\gamma}$ (shape parameter)
- Intercept $\approx \log \hat{\lambda}$ (scale parameter)
- Exponential is the special case where slope = 1

This plot is also used to check the **proportional hazards assumption** for two groups (parallel lines $\Leftrightarrow$ PH holds).

Example: Log-Cumulative Hazard Plots



(a) $\hat{H}(t)$ vs. t



(b) $\log \hat{H}(t)$ vs. $\log t$

Simulated Weibull data ($\gamma = 2$). Panel (a) curves away from the dashed exponential line; panel (b) is linear, confirming the Weibull fit and giving $\hat{\gamma}$ as the slope.

Weibull MLE and R Parametrisation

MLEs for Weibull: closed form for $\hat{\lambda}$ given $\hat{\gamma}$; $\hat{\gamma}$ solved numerically.

Note: `survreg()` uses the AFT parametrisation. Convert to PH:

```
library(survival)
fit <- survreg(Surv(t, delta) ~ 1, dist = "weibull")

# Convert AFT -> PH parametrisation:
gamma_hat <- 1 / fit$scale
lambda_hat <- exp(-gamma_hat * coef(fit))

cat("Shape (gamma):", gamma_hat, "\n")
cat("Scale (lambda):", lambda_hat, "\n")

# Median survival
(-log(0.5) / lambda_hat)^(1/gamma_hat)
```

Other Parametric Families

Gompertz: $h(t) = \lambda e^{\theta t}$ ($\lambda > 0$, $\theta \in \mathbb{R}$)

- Hazard grows exponentially with t
- Standard in actuarial models of human mortality
- $S(t) = \exp(-(\lambda/\theta)(e^{\theta t} - 1))$

Gamma: $f(t) = \lambda^\alpha t^{\alpha-1} e^{-\lambda t} / \Gamma(\alpha)$

- Flexible but does **not** satisfy proportional hazards

Log-normal: $\log T \sim N(\mu, \sigma^2)$

- Hump-shaped hazard; common in engineering reliability

Key points:

- Weibull: $h(t) = \lambda\gamma t^{\gamma-1}$, $S(t) = e^{-\lambda t^\gamma}$
- $\gamma < 1$: decreasing; $\gamma = 1$: exponential; $\gamma > 1$: increasing hazard
- Graphical check: $\log \hat{H}(t)$ vs $\log t$ should be linear (Weibull), with slope $\hat{\gamma}$
- `survreg()` uses AFT parametrisation; convert via $\hat{\gamma} = 1/\text{scale}$
- Gompertz and log-normal for other hazard shapes

Next lecture: MLE with censored data — likelihood contributions for censored and uncensored observations.

Lecture 25: MLE with Censored Data

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Likelihood with Censoring

For observed data (t_i, δ_i) , $i = 1, \dots, n$:

Uncensored ($\delta_i = 1$): event at t_i . Contribution: $f(t_i)$.

Censored ($\delta_i = 0$): survived past t_i . Contribution: $S(t_i) = \mathbb{P}(T > t_i)$.

Full likelihood:

$$L(\theta) = \prod_{i=1}^n f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^n h(t_i)^{\delta_i} S(t_i).$$

The second form uses $f(t) = h(t)S(t)$ and is often more convenient.

Exponential MLE: Derivation

For $h(t) = \lambda$, $S(t) = e^{-\lambda t}$:

$$\ell(\lambda) = r \log \lambda - \lambda v,$$

where $r = \sum_i \delta_i$ (events) and $v = \sum_i t_i$ (total time).

Score equation:

$$\frac{d\ell}{d\lambda} = \frac{r}{\lambda} - v = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{r}{v}.$$

Interpretation: estimated rate = events per unit of total follow-up time.

$\widehat{\text{Var}}(\hat{\lambda}) \approx \hat{\lambda}^2/r$, so the 95% CI is $\hat{\lambda} \exp(\pm 1.96/\sqrt{r})$.

Log-likelihood for Weibull(λ, γ):

$$\ell(\lambda, \gamma) = r \log(\lambda\gamma) + (\gamma - 1) \sum_i \delta_i \log t_i - \lambda \sum_i t_i^\gamma.$$

Closed form for $\hat{\lambda}$ given $\hat{\gamma}$:

$$\hat{\lambda} = \frac{r}{\sum_i t_i^{\hat{\gamma}}}.$$

$\hat{\gamma}$ solves numerically:

$$\frac{1}{\gamma} + \frac{1}{r} \sum_i \delta_i \log t_i - \frac{\sum_i t_i^\gamma \log t_i}{\sum_i t_i^\gamma} = 0.$$

In R: `survreg()` handles this automatically.

Fitting in R

```
library(survival)

# Exponential model
fit_exp <- survreg(Surv(t, delta) ~ 1,
                  dist = "exponential")
lambda_hat <- exp(-coef(fit_exp))

# Weibull model
fit_wei <- survreg(Surv(t, delta) ~ 1,
                  dist = "weibull")
gamma_hat <- 1 / fit_wei$scale
lambda_hat <- exp(-gamma_hat * coef(fit_wei)[1])

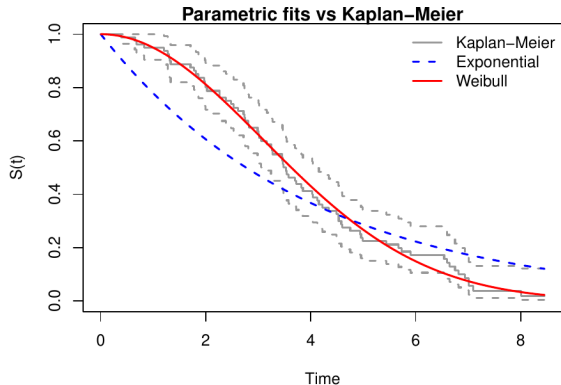
# Log-likelihood for model comparison
logLik(fit_exp)
logLik(fit_wei)
```

Overlay fitted curve on KM estimate:

- 1 Estimate $\hat{\lambda}$ (and $\hat{\gamma}$ for Weibull)
- 2 Compute fitted $\hat{S}(t) = e^{-\hat{\lambda}t}$ (or $e^{-\hat{\lambda}t^{\hat{\gamma}}}$)
- 3 Plot $\hat{S}(t)$ on top of the KM survival curve
- 4 If the fitted curve follows the step function well: good fit

Also use the **log-cumulative hazard plot** (Lecture 24): should be linear in $\log t$ for Weibull.

Example: Fitted Curves vs. KM



KM estimate (step function) with fitted exponential (dashed) and Weibull (solid) curves. The Weibull curve tracks the KM step function more closely.

Key points:

- Censored likelihood: $L(\theta) = \prod_i f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}$
- Uncensored: contributes density; censored: contributes survival function
- Exponential MLE: $\hat{\lambda} = r/v$ (events / total person-time)
- Weibull: $\hat{\lambda}$ in closed form given $\hat{\gamma}$; $\hat{\gamma}$ found numerically
- `survreg()` in R fits both; convert using $\hat{\gamma} = 1/\text{scale}$
- Check fit: overlay fitted $\hat{S}(t)$ on KM estimate

Next lecture: Parametric two-sample comparison using the exponential model.

Lecture 26: Parametric Two-Sample Comparison

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Course notes: Chapter 20

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Setting: Two Groups

Two groups $j = 1, 2$. For group j , observe (t_{ji}, δ_{ji}) .

Sufficient statistics:

- $r_j = \sum_i \delta_{ji}$: events in group j
- $v_j = \sum_i t_{ji}$: total person-time in group j

Under exponential model ($h_j(t) = \lambda_j$):

$$\hat{\lambda}_j = \frac{r_j}{v_j}.$$

Hazard ratio:

$$\psi = \frac{\lambda_2}{\lambda_1}, \quad \hat{\psi} = \frac{\hat{\lambda}_2}{\hat{\lambda}_1} = \frac{r_2 v_1}{r_1 v_2}.$$

Inference for the Hazard Ratio

Asymptotic distribution of log hazard ratio:

$$\log \hat{\psi} \sim N\left(\log \psi, \frac{1}{r_1} + \frac{1}{r_2}\right).$$

95% CI for ψ :

$$\exp\left(\log \hat{\psi} \pm 1.96\sqrt{\frac{1}{r_1} + \frac{1}{r_2}}\right).$$

LRT for $H_0 : \psi = 1$ ($\lambda_1 = \lambda_2$):

$$\Lambda = -2(\hat{\ell}_0 - \hat{\ell}_1) \sim \chi_1^2.$$

Example: Breast Cancer Data

HPA staining: negative (group 1, $n_1 = 13$) vs positive (group 2, $n_2 = 32$).

$$r_1 = 5, v_1 = 1652; \quad r_2 = 21, v_2 = 2679 \text{ (months)}.$$

$$\hat{\lambda}_1 = 5/1652 \approx 0.00303, \quad \hat{\lambda}_2 = 21/2679 \approx 0.00784.$$

$$\hat{\psi} = \frac{21 \times 1652}{5 \times 2679} \approx 2.59.$$

$$95\% \text{ CI} : \exp\left(\log 2.59 \pm 1.96\sqrt{1/5 + 1/21}\right) = [0.98, 6.87].$$

Interpretation: Positive staining nearly triples the hazard of death, though the CI just includes 1.

Weibull Two-Sample Model

Common shape γ , different scales λ_1, λ_2 :

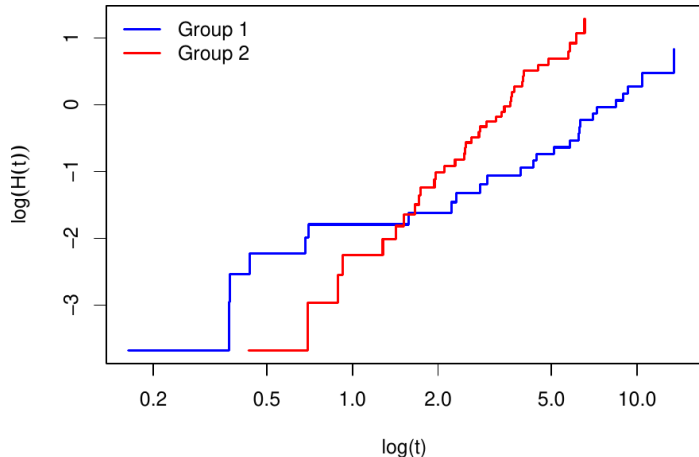
$$h_j(t) = \lambda_j \gamma t^{\gamma-1}.$$

- Hazard ratio $\psi = \lambda_2/\lambda_1$ does not depend on t
- This is the **proportional hazards property** (Lecture 27)
- MLEs: $\hat{\lambda}_j = r_j / \sum_i t_{ji}^{\hat{\gamma}}$; $\hat{\gamma}$ found numerically

Graphical check: log-cumulative hazard plots for each group should be:

- Linear in $\log t$ (Weibull assumption)
- Parallel (PH assumption)

Example: Checking PH for Two Groups



Log-cumulative hazard plots for two simulated groups. Parallel lines with similar slope support both the Weibull assumption and proportional hazards.

Fitting in R

```
library(survival)
d <- data.frame(
  t      = c(t1, t2),
  delta = c(d1, d2),
  group  = c(rep(0, n1), rep(1, n2))
)

# Exponential two-sample model
fit_exp <- survreg(Surv(t, delta) ~ group,
  data = d, dist = "exponential")

# coefficient on group = -log(psi)

# Log-cumulative hazard plot
km_g <- survfit(Surv(t, delta) ~ group, data = d)
plot(km_g, fun = "cloglog",
  col = c("blue", "red"), lwd = 2,
  xlab = "log(t)", ylab = "log H(t)")
```

Key points:

- Sufficient statistics: r_j (events) and v_j (total time) per group
- Exponential MLE: $\hat{\lambda}_j = r_j/v_j$; hazard ratio $\hat{\psi} = r_2v_1/(r_1v_2)$
- 95% CI for ψ : based on asymptotic normality of $\log \hat{\psi}$
- Weibull two-sample: common shape γ , different rates λ_j
- PH property: $\psi = \lambda_2/\lambda_1$ constant in t
- Graphical checks: parallel linear lines in log-cumulative hazard plot

Next lecture: The proportional hazards property and graphical assessment.

Lecture 27: Proportional Hazards Property

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Course notes: Chapter 21

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Proportional Hazards

Definition: Hazards $h_1(t)$ and $h_2(t)$ are **proportional** if

$$h_2(t) = \psi h_1(t), \quad \psi > 0 \text{ constant.}$$

$\psi = h_2(t)/h_1(t)$ is the **hazard ratio**: constant over time.

Equivalent conditions:

$$H_2(t) = \psi H_1(t) \quad \Leftrightarrow \quad S_2(t) = S_1(t)^\psi.$$

Weibull and exponential satisfy PH (common shape γ):

$$\frac{h_2(t)}{h_1(t)} = \frac{\lambda_2 \gamma t^{\gamma-1}}{\lambda_1 \gamma t^{\gamma-1}} = \frac{\lambda_2}{\lambda_1} = \psi.$$

PH Model with Explanatory Variables

With vector of explanatory variables $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^T$:

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x}_i),$$

where $h_0(t)$ is the **baseline hazard**.

- $\exp(\beta_k)$: hazard ratio for a one-unit increase in x_k
- $\boldsymbol{\beta}^T \mathbf{x}$: **prognostic index** / linear predictor

Hazard ratio for individuals with $\mathbf{x}$ vs $\mathbf{x}^*$:

$$\frac{h(t \mid \mathbf{x})}{h(t \mid \mathbf{x}^*)} = \exp(\boldsymbol{\beta}^T (\mathbf{x} - \mathbf{x}^*)),$$

which is **constant in t** — the PH property holds for any $\mathbf{x}$.

Graphical Assessment of PH

Taking log of $H_j(t) = \lambda_j t^\gamma$ (Weibull):

$$\log H_j(t) = \log \lambda_j + \gamma \log t.$$

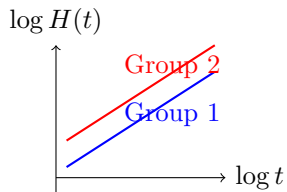
Log-cumulative hazard plot: $\log \hat{H}(t)$ vs $\log t$ for each group.

Pattern	Interpretation
Parallel lines	PH holds
Non-parallel lines	PH violated
Linear (slope = 1)	Exponential
Linear (slope $\neq 1$)	Weibull
Crossing lines	Crossing hazards

Example: PH Check

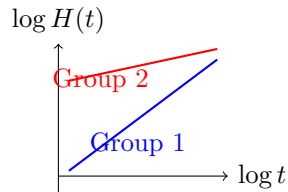
PH holds:

Parallel lines: PH satisfied

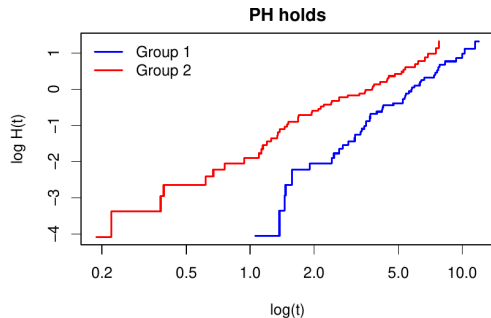


PH violated:

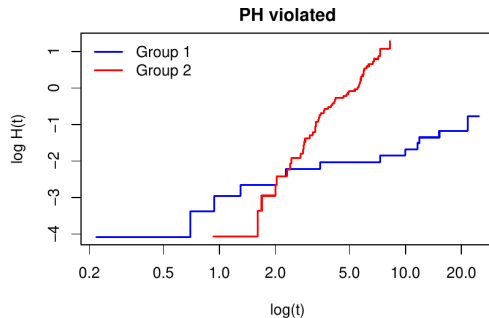
Converging lines: PH violated



The Same Check on Real (Simulated) Data



PH satisfied



PH violated

The same idea with realistic noise: the left panel shows two approximately parallel lines; the right panel shows lines that converge, indicating the hazard ratio changes over time.

PH Check in R

```
library(survival)

# Log-cumulative hazard plot
km <- survfit(Surv(t, delta) ~ group, data = mydata)
plot(km,
      fun    = "cloglog",      # plots  $\log(-\log(S(t)))$  vs  $\log(t)$ 
      col    = c("blue", "red"),
      lwd    = 2,
      xlab   = "log(t)",
      ylab   = "log H(t)")

# Parallel => PH satisfied
# Non-parallel => PH violated
```

Key points:

- PH: $h_2(t) = \psi h_1(t)$ for constant ψ (hazard ratio)
- Equivalent: $S_2(t) = S_1(t)^\psi$
- Weibull/exponential satisfy PH when γ is common
- PH model: $h(t \mid \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x})$
- Hazard ratio $\exp(\beta_k)$ for one-unit increase in x_k
- Graphical check: log-cumulative hazard curves should be parallel
- Crossing or converging curves indicate PH violation

Next lecture: Parametric PH regression — single and multiple explanatory variables.

Lecture 28: Parametric PH Regression — Simple Case and Multiple Explanatory Variables

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Course notes: Chapter 22

Parametric PH Regression: Setup

Model:

$$h(t \mid x_i) = h_0(t) \exp(\beta x_i), \quad h_0(t) = \lambda \gamma t^{\gamma-1}$$

Single binary explanatory variable ($x_i \in \{0, 1\}$):

- Group 0 (baseline): $h(t \mid 0) = \lambda \gamma t^{\gamma-1}$
- Group 1: $h(t \mid 1) = \lambda \gamma t^{\gamma-1} e^{\beta} = \psi \cdot h(t \mid 0)$

Hazard ratio: $\psi = e^{\beta}$.

Survival functions:

$$S(t \mid 0) = e^{-\lambda t^{\gamma}}, \quad S(t \mid 1) = e^{-\lambda e^{\beta} t^{\gamma}}.$$

From the censored likelihood of Lecture 25:

$$\ell(\lambda, \gamma, \beta) = \sum_{i=1}^n \left[\delta_i \{ \beta x_i + \log(\lambda \gamma) + (\gamma - 1) \log t_i \} - \lambda t_i^\gamma e^{\beta x_i} \right].$$

MLEs $\hat{\lambda}, \hat{\gamma}, \hat{\beta}$ obtained numerically.

Note on survreg() output:

$$\hat{\gamma} = 1/\text{scale}$$

$$\hat{\beta}^{\text{PH}} = -\hat{\gamma} \times \text{coefficient on } x$$

$$\hat{\psi} = e^{\hat{\beta}^{\text{PH}}}$$

Signs are reversed in the AFT parametrisation!

Example: Breast Cancer

Positive staining ($x = 1$) vs negative ($x = 0$); exponential model ($\gamma = 1$):

```
d <- data.frame(
  t      = c(t_neg, t_pos),
  delta = c(d_neg, d_pos),
  group  = c(rep(0, 13), rep(1, 32))
)
fit <- survreg(Surv(t, delta) ~ group,
               data = d, dist = "exponential")
summary(fit)
# Convert: psi = exp(-coef(fit)["group"])
```

Consistent with two-sample result: $\hat{\psi} \approx 2.59$, 95% CI [0.98, 6.87].

For the exponential PH model with one binary explanatory variable:

$$\hat{\lambda}_1 = \hat{\lambda}_0 \cdot \hat{\psi},$$

so fitting the regression model is equivalent to fitting separate exponential models and computing the hazard ratio.

General interpretation:

- $\hat{\psi} > 1$: group 1 has higher hazard (shorter survival)
- $\hat{\psi} < 1$: group 1 has lower hazard (longer survival)
- $\hat{\psi} = 1$: no difference

The PH model *always* gives a single number summary of the group effect.

Key points:

- Weibull PH regression: $h(t \mid x) = \lambda \gamma t^{\gamma-1} e^{\beta x}$
- Hazard ratio for one-unit increase: $\psi = e^{\beta}$
- Log-likelihood maximised numerically via `survreg()`
- `survreg()` uses AFT parametrisation: $\hat{\beta}^{\text{PH}} = -\hat{\gamma} \times \text{coefficient}$
- Single binary explanatory variable: equivalent to two-sample exponential comparison

PH Model with Multiple Explanatory Variables

With p explanatory variables $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^T$:

$$h(t \mid \mathbf{x}_i) = \lambda \gamma t^{\gamma-1} \exp(\boldsymbol{\beta}^T \mathbf{x}_i).$$

Key properties:

- Each e^{β_k} is the hazard ratio for a one-unit increase in x_k , *with all other explanatory variables held fixed*
- PH holds: hazard ratios are constant in t
- Survival: $S(t \mid \mathbf{x}_i) = \exp(-\lambda t^\gamma \exp(\boldsymbol{\beta}^T \mathbf{x}_i))$

Weibull PH in R

```
library(survival)

# Fit Weibull model with multiple explanatory variables
fit <- survreg(Surv(t, delta) ~ age + treatment,
               data = mydata, dist = "weibull")

summary(fit)

# Convert to PH parametrisation
gamma_hat <- 1 / fit$scale
beta_ph   <- -gamma_hat * coef(fit)[-1] # drop intercept
psi_hat   <- exp(beta_ph)               # hazard ratios

# 95% CIs for hazard ratios (delta method)
se_aft <- sqrt(diag(vcov(fit)))[-1]
se_ph  <- gamma_hat * se_aft
ci_lo  <- exp(beta_ph - 1.96 * se_ph)
ci_hi  <- exp(beta_ph + 1.96 * se_ph)
```

After fitting, always check:

- ① **Log-cumulative hazard plot** by group: should be parallel (PH) and linear in $\log t$ (Weibull)
- ② **Overlay fitted $\hat{S}(t)$ on KM estimate**: visually assess parametric fit
- ③ **Residual analysis**: Cox–Snell residuals should be approximately $\text{Exp}(1)$ distributed

```
# Log-cumulative hazard by explanatory-variable group
km_g <- survfit(Surv(t, delta) ~ group, data = mydata)
plot(km_g, fun = "cloglog",
      col = c("blue", "red"), lwd = 2)
```


Worked Example: Nephrectomy Data

47 kidney cancer patients; explanatory variables: age group (< 60 , $60-70$, > 70) and nephrectomy (yes/no).

```
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
               header = TRUE)
h$age <- factor(h$age,
               labels = c("<60", "60-70", ">70"))
h$nephrectomy <- factor(h$nephrectomy,
                       labels = c("no", "yes"))

fit <- survreg(Surv(time, status) ~
               age + nephrectomy,
               data = h, dist = "weibull")
```

Estimated hazard ratios: age > 70 vs < 60 : $\hat{\psi} \approx 6$; nephrectomy: $\hat{\psi} \approx 0.3$.

Summary: Multiple Explanatory Variables

Key points:

- Weibull PH with multiple explanatory variables: $h(t \mid \mathbf{x}) = \lambda \gamma t^{\gamma-1} e^{\beta^T \mathbf{x}}$
- e^{β_k} : hazard ratio for x_k , adjusting for all other explanatory variables
- Convert `survreg()` output: $\hat{\beta}^{\text{PH}} = -\hat{\gamma} \cdot \hat{\beta}^{\text{AFT}}$
- Check fit: parallel lines in log-cumulative hazard; overlay on KM
- Nephrectomy reduces hazard by $\approx 70\%$; age > 70 increases 6-fold

Next lecture: The Cox model — semi-parametric regression that avoids specifying $h_0(t)$.

Lecture 29: The Cox Model — Motivation and Partial Likelihood

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Why Semi-Parametric?

Problem with parametric models:

- Must specify $h_0(t)$ (Weibull, exponential, ...)
- Misspecified baseline $\Rightarrow$ biased $\hat{\beta}$
- Hard to verify the correct form in practice

Cox (1972) solution: treat $h_0(t)$ as an unspecified nuisance.

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x}_i), \quad h_0(t) \geq 0 \text{ unspecified.}$$

Estimate $\boldsymbol{\beta}$ without ever estimating $h_0(t)$. The explanatory-variable effects are the quantity of interest.

The Partial Likelihood

Let $t_{(1)} < \dots < t_{(r)}$ be ordered event times. At each $t_{(j)}$, define the **risk set**:

$$\mathcal{R}(t_{(j)}) = \{\text{individuals at risk just before } t_{(j)}\}.$$

Given one death occurs at $t_{(j)}$, the probability it is individual $i_{(j)}$ is:

$$\frac{h_0(t_{(j)}) \exp(\boldsymbol{\beta}^T \mathbf{x}_{i_{(j)}})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} h_0(t_{(j)}) \exp(\boldsymbol{\beta}^T \mathbf{x}_{\ell})} = \frac{\exp(\boldsymbol{\beta}^T \mathbf{x}_{i_{(j)}})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\boldsymbol{\beta}^T \mathbf{x}_{\ell})}.$$

$h_0(t)$ **cancels!**

$$L(\boldsymbol{\beta}) = \prod_{j=1}^r \frac{\exp(\boldsymbol{\beta}^T \mathbf{x}_{(j)})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\boldsymbol{\beta}^T \mathbf{x}_{\ell})}$$

Partial log-likelihood:

$$\ell_p(\boldsymbol{\beta}) = \sum_{i=1}^n \delta_i \left\{ \boldsymbol{\beta}^T \mathbf{x}_i - \log \sum_{\ell \in \mathcal{R}(t_i)} \exp(\boldsymbol{\beta}^T \mathbf{x}_{\ell}) \right\}.$$

The **maximum partial likelihood estimator (MPLE)** $\hat{\boldsymbol{\beta}}$ maximises $\ell_p(\boldsymbol{\beta})$.

Asymptotic properties: same as MLE — consistent, asymptotically normal.

Formal justification that $\hat{\beta}$ has MLE-like properties is in:

Supplementary/proof_partial_likelihood.pdf

Key result:

$$\hat{\beta} \sim N_p\left(\beta, \mathcal{I}(\hat{\beta})^{-1}\right),$$

where $\mathcal{I}(\hat{\beta}) = -\partial^2 \ell_p / \partial \beta \partial \beta^T$.

This justifies Wald CIs and likelihood ratio tests based on ℓ_p .

Handling Tied Event Times

Exact partial likelihood is intractable with ties. **Breslow's approximation:**

$$L(\boldsymbol{\beta}) \approx \prod_{j=1}^r \frac{\exp(\boldsymbol{\beta}^T \mathbf{s}_{(j)})}{\left(\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\boldsymbol{\beta}^T \mathbf{x}_{\ell}) \right)^{d_j}},$$

where d_j = deaths at $t_{(j)}$ and $\mathbf{s}_{(j)} = \sum_{k: t_k=t_{(j)}, \delta_k=1} \mathbf{x}_k$.

Default in `coxph(ties = "breslow")`. Accurate when ties are rare.

Cox Model in R

```
library(survival)

# Fit Cox model
cox_fit <- coxph(Surv(t, delta) ~ x1 + x2,
                 data = mydata,
                 ties = "breslow")

summary(cox_fit)

# Output: coefficients, exp(coef) = psi, 95% CIs
# Three overall tests: Wald, LRT, Score

# Nephrectomy example
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
                header = TRUE)

cox_neph <- coxph(Surv(time, status) ~
                  factor(age) + factor(nephrectomy),
                  data = h, ties = "breslow")

summary(cox_neph)
```

Key points:

- Cox model: $h(t | \mathbf{x}) = h_0(t)e^{\beta^T \mathbf{x}}$ with $h_0(t)$ unspecified
- Partial likelihood: $h_0(t)$ cancels at each event time
- MPLE $\hat{\beta}$: consistent and asymptotically normal
- Ties: Breslow's approximation (default in `coxph()`)
- Interpretation same as parametric: $e^{\hat{\beta}_k} = \text{hazard ratio}$
- More robust than parametric: no baseline hazard assumption

Next lecture: Cox model estimation, tied observations, and variance estimation.

Lecture 30: The Cox Model — Estimation, Inference and R

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Course notes: Chapter 23

Score Equations

The partial log-likelihood score:

$$\frac{\partial \ell_p}{\partial \beta_k} = \sum_{i=1}^n \delta_i \{x_{ik} - \bar{x}_k(t_i)\} = 0,$$

where $\bar{x}_k(t)$ is the risk-set-weighted mean of x_k :

$$\bar{x}_k(t) = \frac{\sum_{\ell \in \mathcal{R}(t)} x_{\ell k} \exp(\boldsymbol{\beta}^T \mathbf{x}_\ell)}{\sum_{\ell \in \mathcal{R}(t)} \exp(\boldsymbol{\beta}^T \mathbf{x}_\ell)}.$$

Solved by Newton–Raphson. Convergence is typically fast (< 20 iterations).

Variance Estimation

The observed information matrix:

$$\mathcal{I}(\hat{\beta}) = -\frac{\partial^2 \ell_p}{\partial \beta \partial \beta^T} \Big|_{\beta=\hat{\beta}}.$$

Asymptotic variance-covariance:

$$\widehat{\text{Var}}(\hat{\beta}) = \mathcal{I}(\hat{\beta})^{-1}.$$

Confidence interval for β_k :

$$\hat{\beta}_k \pm z_{1-\alpha/2} \widehat{\text{sd}}(\hat{\beta}_k).$$

Confidence interval for hazard ratio $\psi_k = e^{\beta_k}$:

$$\exp(\hat{\beta}_k \pm z_{1-\alpha/2} \widehat{\text{sd}}(\hat{\beta}_k)).$$

Manual Example: Coronavirus Recovery

6 patients per group. Group I: lung condition ($x = 0$); Group II: no lung condition ($x = 1$).

Ordered event times: 6, 16, 30, 32, 41, 42. At each, construct the partial likelihood term.

By symmetry of the data, the MPLE turns out to be:

$$\hat{\beta} \approx -\log 2 \approx -0.693, \quad \hat{\psi} = e^{\hat{\beta}} = 0.5.$$

Interpretation: lung-condition patients recover at **half the rate** of those without.

95% CI for ψ : [0.229, 1.093] (CI includes 1 — not significant at 5%).

Three Overall Tests

`summary(coxph())` reports three tests for $H_0 : \beta = 0$:

Test	Statistic	Distribution
Likelihood ratio	$-2(\hat{\ell}_0 - \hat{\ell}_1)$	χ_p^2
Wald	$\hat{\beta}^T \mathcal{I}(\hat{\beta}) \hat{\beta}$	χ_p^2
Score (logrank)	$U(0)^T \mathcal{I}(0)^{-1} U(0)$	χ_p^2

All three are asymptotically equivalent. LRT is generally most reliable in small samples.

Nephrectomy: Cox Model Output

```
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
               header = TRUE)
h$age <- factor(h$age, labels=c("<60", "60-70", ">70"))
h$nephrectomy <- factor(h$nephrectomy,
                       labels=c("no", "yes"))
fit <- coxph(Surv(time, status) ~
             age + nephrectomy,
             data=h, ties="breslow")
summary(fit)
```

Key output: age > 70: $\hat{\psi} \approx 6.2$ (CI [1.6, 23.6]); nephrectomy: $\hat{\psi} \approx 0.32$ (CI [0.12, 0.86]).

Key points:

- Score equations: $\hat{\beta}$ maximises partial log-likelihood via Newton–Raphson
- Variance: $\mathcal{I}(\hat{\beta})^{-1}$ from observed information matrix
- CI for hazard ratio: $\exp(\hat{\beta}_k \pm z \widehat{\text{sd}}(\hat{\beta}_k))$
- Three overall tests: LRT, Wald, Score — all asymptotically equivalent
- Nephrectomy reduces hazard $\approx 68\%$; age > 70 increases hazard 6-fold

Basic Syntax

```
library(survival)

# Fit Cox model
fit <- coxph(Surv(time, status) ~ x1 + x2 + x3,
             data = mydata, ties = "breslow")

# Summary output
summary(fit)

# Reports: coef, exp(coef), se(coef),
#           Wald z, p-value
#           exp(coef) 95% CI
#           Overall: LRT, Wald, Score tests

# Coefficients and CIs
coef(fit)
confint(fit)           # CI for beta
exp(cbind(coef(fit),   # CI for hazard ratios
          confint(fit)))
```

Interpreting summary() Output

Key lines in `summary(coxph())`:

- `coef`: $\hat{\beta}_k$ (log hazard ratio)
- `exp(coef)`: $\hat{\psi}_k = e^{\hat{\beta}_k}$ (hazard ratio)
- `se(coef)`: standard error of $\hat{\beta}_k$
- `Pr(>|z|)`: Wald p -value for $H_0 : \beta_k = 0$
- `lower .95, upper .95`: 95% CI for $\hat{\psi}_k$

```
# Example output (schematic):  
#           coef exp(coef) se(coef)  z      Pr(>|z|)  
# age60-70 -0.07  0.94      0.50   -0.13  0.90  
# age>70    1.82  6.18      0.69    2.65  0.008 **  
# nephro.yes -1.14  0.32      0.50   -2.29  0.022 *
```

Prognostic Index and Predicted Survival

The **prognostic index** (linear predictor):

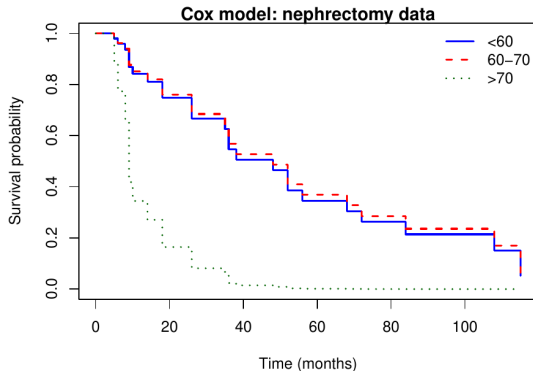
$$\eta_i = \hat{\beta}^T \mathbf{x}_i.$$

```
# Prognostic index for each observation
PI <- predict(fit, type = "lp")

# Predicted survival at specified times
# (requires baseline hazard estimate)
fit_surv <- survfit(fit,
  newdata = data.frame(age = "<60",
    nephrectomy = "yes"))
plot(fit_surv, col = "blue",
  xlab = "Time", ylab = "S(t)")
```

`survfit(coxph)` uses the Breslow estimator of $H_0(t)$ to construct $\hat{S}(t \mid \mathbf{x}_{\text{new}}) = \exp(-\hat{H}_0(t)e^{\hat{\beta}^T \mathbf{x}_{\text{new}}})$.

Example: Predicted Survival by Age Group



Baseline survival from the Cox model, by age group. The curves are proportional to each other — the PH assumption built into the model.

Comparing Nested Cox Models

```
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
               header=TRUE)
h$age <- factor(h$age, labels=c("<60", "60-70", ">70"))
h$nephrectomy <- factor(h$nephrectomy,
                       labels=c("no", "yes"))

M0 <- coxph(Surv(time, status) ~ 1, data=h)
M1 <- coxph(Surv(time, status) ~ age, data=h)
M3 <- coxph(Surv(time, status) ~ age + nephrectomy,
           data=h)

# LRT: age
-2*(M0$loglik[2] - M1$loglik[2]) # chi^2, df=2

# LRT: nephrectomy | age
-2*(M1$loglik[2] - M3$loglik[2]) # chi^2, df=1
```


Summary: The Cox Model in R

Key points:

- `coxph(Surv(t,d) ~ x, data, ties="breslow")`
- `summary()`: coefficients, hazard ratios, CIs, p-values
- `exp(coef(fit))`: hazard ratios; `confint(fit)`: log-scale CIs
- Prognostic index: $\hat{\beta}^T \mathbf{x}$ via `predict(fit, type="lp")`
- Predicted survival: `survfit(fit, newdata=...)`
- LRT: $-2(\hat{\ell}_0 - \hat{\ell}_1)$ via `loglik[2]` slot

Next lecture: Cox model diagnostics — checking the PH assumption.

Lecture 31: Cox Model Diagnostics and Checking PH

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Course notes: Chapter 24

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Diagnostic Objectives

After fitting a Cox model, check:

- ➊ **Proportional hazards:** is the hazard ratio constant over time?
- ➋ **Linearity:** do continuous explanatory variables enter log-linearly?
- ➌ **Influential observations:** are any observations unduly influential?

Tools:

- Log-cumulative hazard plots (graphical PH check)
- Schoenfeld residuals and `cox.zph()` (formal PH test)
- Martingale residuals (linearity check)
- Deviance residuals (outlier detection)

Schoenfeld Residuals

At each event time $t_{(j)}$, the Schoenfeld residual for explanatory variable k is:

$$r_{jk} = x_{(j)k} - \bar{x}_k(t_{(j)}),$$

where $\bar{x}_k(t_{(j)})$ is the risk-set-weighted mean.

- Under PH: r_{jk} should be centred at 0 with no trend in time
- Systematic *trend* in r_{jk} vs time $\Rightarrow$ PH violation

Formal test: fit a smooth of r_{jk} vs $t_{(j)}$ and test the slope.

In R: `cox.zph()` implements this test.

Schoenfeld Residuals in R

```
library(survival)
fit <- coxph(Surv(t, delta) ~ x1 + x2,
             data = mydata)

# Test PH assumption
ph_test <- cox.zph(fit)
print(ph_test)
# Output: test statistic, p-value per explanatory variable
# + global test
# p < 0.05 => PH may be violated

# Plot Schoenfeld residuals
plot(ph_test) # one plot per explanatory variable
# Smooth line should be approximately flat
```

Martingale Residuals

$$M_i = \delta_i - \hat{H}_0(t_i) \exp(\hat{\beta}^T \mathbf{x}_i).$$

- Range: $(-\infty, 1]$; sum to 0
- Plot vs continuous explanatory variable to check linearity
- Non-linear trend $\Rightarrow$ consider transformation or smooth term

```
# Martingale residuals
mart <- residuals(fit, type = "martingale")

# Check linearity of continuous explanatory variable 'age'
plot(mydata$age, mart,
     xlab = "Age", ylab = "Martingale residual")
lines(lowess(mydata$age, mart), col="red", lwd=2)
abline(h = 0, lty = 2)
```

$$d_i = \text{sign}(M_i) \sqrt{-2(M_i + \delta_i \log(\delta_i - M_i))}.$$

- More symmetric around 0 than martingale residuals
- $|d_i| > 3$: potentially influential or outlying

```
dev <- residuals(fit, type = "deviance")
plot(seq_along(dev), dev, type = "h",
     xlab = "Observation index",
     ylab = "Deviance residual")
abline(h = c(-2, 0, 2), lty = c(2,1,2))
```


Key points:

- PH check: log-cumulative hazard plots (parallel lines) + `cox.zph()`
- Schoenfeld residuals: no trend with time $\Rightarrow$ PH satisfied
- `cox.zph()`: formal test; $p < 0.05$ suggests PH violation
- Martingale residuals: check linearity of continuous explanatory variables
- Deviance residuals: identify influential observations ($|d_i| > 3$)
- Small p -value in `cox.zph()` $\Rightarrow$ consider stratification or time-varying coefficients

Next lecture: Model selection — parametric vs Cox, LRT.

Lecture 32: Model Selection — Parametric vs Cox

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Course notes: Chapter 25

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Why Model Selection Matters

After fitting survival models, we must choose between:

- Competing parametric models (exponential vs Weibull vs Gompertz)
- Parametric vs semi-parametric (Cox) models
- Nested models: include/exclude explanatory variables

Two main tools:

- ① **Likelihood ratio test (LRT):** for nested models
- ② **Graphical checks:** for non-nested comparisons and overall fit — overlay $\hat{S}(t)$ on KM, Cox–Snell residuals

Likelihood Ratio Test for Nested Models

For nested models $M_0 \subset M_1$ (both parametric, or both Cox):

$$T = -2(\hat{\ell}_0 - \hat{\ell}_1) \sim \chi^2_{q-p},$$

where $p = \dim(M_0)$, $q = \dim(M_1)$.

Comparison	χ^2 d.f.	Example
Add 1 explanatory variable	1	nephrectomy age
Add age group	2	age (3 levels) null
Exp vs Weibull	1	$\gamma = 1$ constrained

Note: LRT compares log-likelihoods of the *same type*: partial likelihoods for Cox models, full likelihoods for parametric models.

LRT in R: Cox Models

```
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
               header = TRUE)
h$age <- factor(h$age,
               labels = c("<60", "60-70", ">70"))
h$nephrectomy <- factor(h$nephrectomy,
                       labels = c("no", "yes"))

M0 <- coxph(Surv(time, status) ~ 1, data=h)
M1 <- coxph(Surv(time, status) ~ age, data=h)
M3 <- coxph(Surv(time, status) ~
            age + nephrectomy, data=h)

# LRT: effect of age (df=2)
T_age <- -2*(M0$loglik[2] - M1$loglik[2])
pchisq(T_age, df=2, lower.tail=FALSE)

# LRT: nephrectomy given age (df=1)
T_neph <- -2*(M1$loglik[2] - M3$loglik[2])
pchisq(T_neph, df=1, lower.tail=FALSE)
```

LRT in R: Parametric Models

```
# Fit competing parametric models
fit_exp <- survreg(Surv(time, status) ~
                  age + nephrectomy,
                  data = h, dist = "exponential")
fit_wei <- survreg(Surv(time, status) ~
                  age + nephrectomy,
                  data = h, dist = "weibull")

# Exponential is nested in Weibull (gamma=1)
# LRT test (change in deviance):
T_ew <- -2*(logLik(fit_exp) - logLik(fit_wei))
pchisq(T_ew, df=1, lower.tail=FALSE)
```

Parametric vs Cox: When to Use Which?

	Parametric	Cox
Baseline hazard	Specified	Unspecified
Robustness	Lower	Higher
Efficiency	Higher (if correct)	Lower
Prediction	Full $\hat{S}(t)$	Needs $\hat{H}_0(t)$
PH required	Yes (Weibull)	Yes

Practice: fit both; if hazard ratios agree and KM overlay looks good for parametric, use it for efficiency. If not, prefer Cox.

Overlay $\hat{S}(t)$ on KM:

- Good agreement $\Rightarrow$ parametric model adequate
- Systematic divergence $\Rightarrow$ model misspecified

Cox–Snell residuals: if model is correct, $\hat{r}_i = \hat{H}(t_i | \mathbf{x}_i)$ should be approximately $\text{Exp}(1)$.

- Plot KM of residuals: should follow $\exp(-r)$
- Plot $-\log(\hat{S}(r))$ vs r : should be approximately straight line through origin with slope 1

Key points:

- LRT: $-2(\hat{\ell}_0 - \hat{\ell}_1) \sim \chi^2_{q-p}$ for nested models
- Non-nested or Cox-vs-parametric comparisons: use graphical checks, not likelihood-based criteria
- If parametric fits well (KM overlay, Cox–Snell residuals), prefer it: higher efficiency
- If uncertain about baseline hazard form, prefer Cox: more robust
- Cox–Snell residuals: formal goodness-of-fit for parametric models

Next lecture: Extended case study — full pipeline from EDA to final model.

Lecture 33: Extended Case Study

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Course notes: Chapter 26

MATH3701 Statistical Modelling notes

Case Study: Hypernephroma

47 kidney cancer patients. Response: time to death (months). Explanatory variables:

- **age:** < 60, 60–70, > 70 (3 levels)
- **nephrectomy:** no/yes (surgical kidney removal)

Full analysis pipeline:

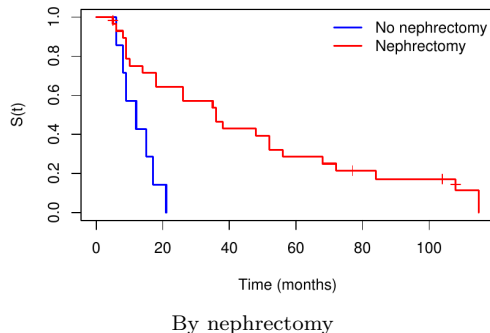
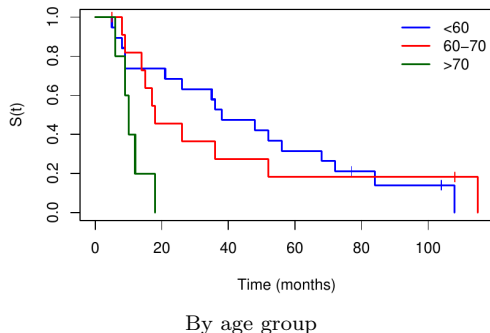
- ① EDA: Kaplan–Meier curves by group
- ② Check PH: log-cumulative hazard plots
- ③ Fit Cox model; compare nested models via LRT
- ④ Diagnostics: Schoenfeld residuals, `cox.zph()`
- ⑤ Sensitivity: compare with Weibull parametric model
- ⑥ Interpret final model

Step 1: Data Loading and EDA

```
h <- read.table("https://www.richardpmann.com/MATH3701/Datasets/hypernephroma.dat",
               header = TRUE)
h$age <- factor(h$age,
               labels = c("<60", "60-70", ">70"))
h$nephrectomy <- factor(h$nephrectomy,
                       labels = c("no", "yes"))

# KM by age group
km_age <- survfit(Surv(time, status) ~ age,
                 data = h)
plot(km_age, col = c("blue", "red", "green"),
     lwd = 2, xlab = "Time (months)",
     ylab = "S(t)", main = "KM by Age Group")
legend("topright",
      legend = c("<60", "60-70", ">70"),
      col = c("blue", "red", "green"), lwd=2)
```

Step 1: EDA Output



Age > 70 has clearly worse survival; patients who had nephrectomy appear to survive longer on average.

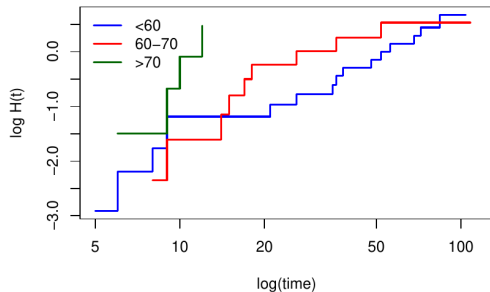
Step 2: PH Check

```
# Log-cumulative hazard plot by age
plot(km_age, fun = "cloglog",
     col = c("blue","red","green"), lwd = 2,
     xlab = "log(time)", ylab = "log H(t)")
# Roughly parallel lines => PH plausible

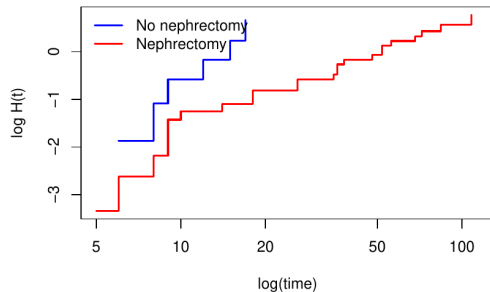
# By nephrectomy
km_neph <- survfit(
  Surv(time, status) ~ nephrectomy, data=h)
plot(km_neph, fun = "cloglog",
     col = c("blue","red"), lwd = 2,
     xlab = "log(time)", ylab = "log H(t)")
```

Parallel lines $\Rightarrow$ PH assumption plausible for both explanatory variables.

Step 2: PH Check Output



By age group



By nephrectomy

Both plots show approximately parallel lines, supporting the PH assumption for both explanatory variables.

Step 3: Cox Model Fitting

```
M0 <- coxph(Surv(time,status) ~ 1,      data=h)
M1 <- coxph(Surv(time,status) ~ age,    data=h)
M3 <- coxph(Surv(time,status) ~
            age + nephrectomy,          data=h)

# LRT: age (df=2)
T1 <- -2*(M0$loglik[2] - M1$loglik[2])
pchisq(T1, df=2, lower.tail=FALSE)    # p ~ 0.04

# LRT: nephrectomy | age (df=1)
T2 <- -2*(M1$loglik[2] - M3$loglik[2])
pchisq(T2, df=1, lower.tail=FALSE)    # p ~ 0.02

summary(M3)
```

Step 3: LRT Results

Comparison	T	d.f.	p
Null $\rightarrow$ age	6.1	2	0.047
age $\rightarrow$ age + nephrectomy	5.3	1	0.021

Both explanatory variables are significant at the 5% level.

Estimated hazard ratios (final model):

- Age 60–70 vs < 60 : $\hat{\psi} \approx 0.94$ (CI: [0.25, 3.6])
- Age > 70 vs < 60 : $\hat{\psi} \approx 6.2$ (CI: [1.6, 24])
- Nephrectomy yes vs no: $\hat{\psi} \approx 0.32$ (CI: [0.12, 0.86])

Step 4: Diagnostics

```
# PH test for final model
ph_test <- cox.zph(M3)
print(ph_test)
# p-values: age ~0.4, nephrectomy ~0.7,
#           global ~0.5 => PH not rejected

plot(ph_test) # flat Schoenfeld residuals

# Deviance residuals
dev <- residuals(M3, type = "deviance")
plot(seq_along(dev), dev, type = "h",
     xlab = "Index",
     ylab = "Deviance residual")
abline(h = c(-2, 0, 2), lty = c(2,1,2))
# No |d_i| > 3: no extreme outliers
```

Step 5: Weibull Sensitivity Check

```
# Fit Weibull for comparison
fit_wei <- survreg(Surv(time,status)~age+nephrectomy,
                  data=h, dist="weibull")
gamma_hat <- 1 / fit_wei$scale
beta_ph <- -gamma_hat * coef(fit_wei)[-1]
exp(beta_ph) # age>70: ~6.0; nephrectomy: ~0.30

# Is Weibull's shape needed? Exponential (shape=1) is
# nested in Weibull -- compare via deviance change
fit_exp <- survreg(Surv(time,status)~age+nephrectomy,
                  data=h, dist="exponential")
dev_wei <- -2*logLik(fit_wei); dev_exp <- -2*logLik(fit_exp)
dev_exp - dev_wei # ~ chi^2_1; cf. qchisq(.95,1)=3.84
```

Weibull and Cox give very similar hazard ratios $\Rightarrow$ results robust to model choice.

Step 6: Final Interpretation

Final model: Cox proportional hazards with age and nephrectomy.

- **Age > 70:** hazard $\approx 6\times$ higher than < 60 ($p < 0.01$)
 $\Rightarrow$ older patients die substantially faster from kidney cancer
- **Age 60–70:** no significant difference from < 60 ($p \approx 0.9$)
- **Nephrectomy:** hazard reduced by $\approx 68\%$ ($p \approx 0.02$)
 $\Rightarrow$ surgical removal associated with significantly longer survival

Diagnostics: PH holds (`cox.zph` $p > 0.4$); no influential observations ($|d_i| \leq 2.5$); Weibull confirms findings.

Key points:

- Full analysis: EDA $\rightarrow$ PH check $\rightarrow$ Cox fitting $\rightarrow$ LRT $\rightarrow$ diagnostics $\rightarrow$ sensitivity
- KM and log-cumulative hazard plots guide model choice
- LRT confirms both age and nephrectomy are significant
- Diagnostics confirm PH holds; no outliers
- Weibull and Cox agree: parametric model adequate here
- Interpret hazard ratios carefully: adjusted for other explanatory variables

Next lecture: Course review and synthesis.

Lecture 34: Course Review and Synthesis

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Part	Topic	Lectures
I	Generalised Linear Models	1–12
II	Generalised Additive Models	13–19
III	Survival Analysis	20–34

Unifying theme: extending the normal linear model to handle:

- Non-normal responses (GLMs)
- Non-linear explanatory-variable effects (GAMs)
- Time-to-event responses with censoring (Survival)

Three-component structure:

- ① **Random:** $Y_i \sim$ exponential family; $\text{Var}(Y_i) = V(\mu_i)\phi$
- ② **Systematic:** linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$
- ③ **Link:** $g(\mu_i) = \eta_i$; canonical links simplify inference

Model	Distribution	Link	Use
Linear	Normal	Identity	Continuous Y
Logistic	Binomial	Logit	Proportions
Poisson	Poisson	Log	Counts

Fitting: IRLS; **Inference:** LRT, Wald, deviance residuals

Extended three-component structure:

$$g(\mu_i) = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \dots$$

where f_j are smooth functions estimated from data.

Key ideas:

- Splines: piecewise polynomials with smoothness constraints
- Interpolating splines: pass through all data points
- Smoothing splines: minimise penalised LS criterion
- λ : smoothing parameter; selected by GCV
- GAMs: extend GLMs using penalised deviance; `mgcv::gam()`

Effective degrees of freedom: $\text{EDF} = \text{tr}(\mathbf{S}_\lambda)$ measures model complexity.

Key functions:

$$S(t) = \exp(-H(t)), \quad h(t) = \frac{f(t)}{S(t)}, \quad H(t) = \int_0^t h(s) ds.$$

Method	Baseline	Explanatory Variables
KM	Non-parametric	No
Parametric (Weibull)	Specified	Via PH structure
Cox model	Unspecified	Via PH structure

PH model: $h(t \mid \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x})$; hazard ratio $\psi_k = e^{\beta_k}$.

All three parts share:

- A **linear predictor** $\eta = \mathbf{x}^T \boldsymbol{\beta}$ (or smooth extension $\sum f_k$)
- Maximum likelihood estimation (full or partial)
- Inference via LRT, Wald, and information-based CIs
- Model checking through residuals and graphical diagnostics
- Model selection via likelihood-based tests

Key distinction: survival analysis adds the complication of censored observations, requiring the censored likelihood and semi-parametric methods.

- Derive the variance function $V(\mu)$ from the exponential family form
- Write down and interpret the score equations
- Explain IRLS: what is the working response and working weights?
- Interpret logistic and Poisson regression coefficients
- Conduct and interpret a LRT for nested GLMs
- Define Pearson and deviance residuals; identify outliers
- Interpret overdispersion; quasi-likelihood approach

- Define a spline: knots, polynomial pieces, smoothness constraints
- State the natural spline optimality proposition
- Write down the penalised LS criterion and explain the role of λ
- Derive or state the solution to the smoothing spline problem
- Explain OCV and GCV; compute GCV from the smoothing matrix
- Define $\text{EDF} = \text{tr}(\mathbf{S}_\lambda)$; interpret its value
- Write down the penalised deviance for a GAM

Exam Pointers: Survival Analysis

- Derive the censored likelihood and the MLE for exponential data
- State the Kaplan–Meier formula; compute by hand for small examples
- Define the PH property; verify it for Weibull
- Interpret hazard ratios from parametric and Cox models
- Derive the partial likelihood; explain why $h_0(t)$ cancels
- State and apply the Breslow approximation for tied events
- Describe Schoenfeld, martingale, and deviance residuals and their use
- Compare models using LRT

MATH3701 in three sentences:

GLMs extend the linear model to handle non-normal responses by linking a linear predictor to the mean via a canonical link function and estimating via IRLS.

GAMs replace the linear predictor with smooth additive functions estimated via penalised likelihood, with the smoothing parameter chosen by cross-validation.

Survival analysis handles time-to-event data under censoring using either parametric models (Weibull) or the semi-parametric Cox model, both using the proportional hazards structure.

Good luck in the examination!